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Is there evidence of children outdoor active mobility behaviour and neighbourhood's safety? a systematic review

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ABSTRACT

Objective: This review provides a synthesis of evidence regarding the impact of neighbourhood road and personal safety on diverse forms of outdoor active mobility behaviour and health indices for primary-school-aged children.

Design: Systematic review

Data sources: Google Scholar, PubMed, Scopus, Science Direct, SpringerLink and Web of Science were searched from inception until July 2020.

Data extraction and synthesis: Study selection and quality assessments were conducted independently by two reviewers. Data were extracted and organised into a socio-ecological framework. Methodological evaluation to derive studies' quality was adapted and used with a vote-counting technique to quantify evidence strength.

Results: Informed by the synthesis of 29 studies, active behaviour among young people has strong correlations with child's sex/gender, age, ethnic background, land-use diversity, an inverse association with car ownership and proximity, independent of safety. Parental perceived personal safety is strongly correlated with children's active mobility but mostly affecting active travel type. There were limited findings of parental/child concern about 'stranger danger' as well as perceived personal safety among children. A moderate yet consistent correlation between parents' or children's perceived road danger and varying types of child's active behaviour was also found. However, these associations penetrate through variation in child's individual, family and neighbourhood characteristics and across time. There were limited and inconsistent findings that explored the correlation between active behaviour and measured safety.

Conclusion: This review depicted the harm caused by safety concerns restricting children's active behaviours. There is an urgent need for a collaborative action across all socio-ecological levels to reverse the decline in child's active behaviour. We drew upon findings to outline future implications. Greater understanding of children's own fears, and hence engaging youngsters in the design process of places may lead to more child-friendly cities. The adapted methodological quality assessment can guide future studies with similar constructions.

Keywords: Children's outdoor active mobility behaviour, Parents' perceived safety, Children's perceived safety, Measured safety, Neighbourhood safety, Objective measures, Socio-ecological framework.

Article Summary: Strengths and limitations of this study

- This systematic review is original in conducting a methodological quality appraisal by integrating studies' reporting and relevant measurement methods to derive an understanding of the strength in the evidence underpinning the review.
- This review is novel in assessing the impact of safety on children's varied types of active mobility behaviour of evidence and represents the most comprehensive synthesis of existing literature.
- Peer-reviewed studies published in the English language were included, and meta-analysis review was not possible due to the heterogeneity in measurement methods and studies outputs.

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INTRODUCTION

The harm caused by restricting children's active mobility behaviours may expand beyond the immediate effects on physical health,[1] to overall social skills[2] and healthy brain development.[3] With more children living in urban settings than ever before,[4] neighbourhood safety concerns as they are perceived (by parents or children) or measured (actual) are seen to have mixed results in shaping children's outdoor active behaviour.[5-8] Safety is a complex concept,[9] yet involves consideration of harm from strangers ('stranger danger) as well as personal and road dangers.[10 11] A growing body of work suggests that the neighbourhood is an essential venue for youth to accumulate the recommended daily levels of physical activity through various forms of structured or unstructured active behaviour.[9 12] However, parents or children perceiving risk in their surroundings may result in children having lower confidence in exploring their surroundings[12] and contributing to their community.[13]

Previously found review in specific to assess the impact of neighbourhood safety has addressed adult populations.[14] Limited works found synthesising influences of safety on children's active mobility.[9] Available reviews were either non-systematic,[9] specific to one form of active mobility behaviour where safety was combinedly examined with other aspects of environmental correlates, for school travel[10 15-18] or independent mobility,[19] or have combined evidence for active travel for a broad range of child age groups (three or five to 18 years),[8 20] which may contribute to the mixed findings among reviews.[8 21] Thus, a gap remains in the evidence base regarding a deeper understanding of the impact of both measured and perceived safety on children's daily active mobility behaviour. At the same time, persistent variances in measurement methods and output add complexity when synthesising findings across studies.[15 17] Due to that, alternatives to narrative assessments, dismantling safety influences and understanding the strength of associations in the evidence is needed. Thence, the aims of this review are (1) to systematically synthesise evidence of associations between a neighbourhood's measured and perceived safety and primary-school-aged children's outdoor active mobility behaviour (COAMB), and (2) to develop a context-specific methodological quality evaluation.

Inevitably, there is evidence in prior research linking children's daily active play, exercise and travel to school and other leisure destinations with profound long-term positive outcomes.[22] Evidently, active school travel behaviour is a significant contributor to overall child behaviour.[23] However, beyond school travel, other forms of active behaviour incorporated into children's daily routines[24] are vital in reaching recommended levels of daily physical activity.[25] Thus, building on more recent practices[26] when examining neighbourhood exposures, we synthesise the impact of neighbourhood safety on various forms of COAMB in terms of active play, active travel to destinations and exercise for their crucial preventive effects on obesity among children.[22]

METHOD

A rigorous systematic search of existing literature was conducted and reported based on the methodology guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis extension for Scoping Review (PRISMA-ScR) checklist.[27] Though the search was planned in advance, no protocol was registered (see Online Supplemental File 1).

Search strategy

To fulfil the first objective of this review study, a search strategy was designed to identify all relevant studies. Searches were conducted in six electronic databases: *Google Scholar*, *PubMed*, *Scopus*, *Science Direct*, *SpringerLink* and *Web of Science*. All databases were searched from the review inception until July 2020. Search terms were explicitly developed to identify all studies linked with the objectives of this study. The terms pertaining to this review were grouped by (1) target population, (2) AND active

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behaviour, AND (3) neighbourhood safety, OR (4) moderators and mediators, OR (5) spatiotemporal aspects (GPS, GIS). The full search thesauruses and strings are available in Online Supplemental File 2.

Eligibility criteria

Studies were included for review if they were peer-reviewed, published in English, reported participants aged between five and 12 years (primary school children), addressed neighbourhood safety (personal and road safety) either as perceived by parents and/or children or measured (actual) and reported or measured children's active mobility outside school hours. Included publications were not restricted to a single geographic location or a specific time. The following types of studies were excluded for failing to meet the criteria: conference abstracts, reports, PhD or master theses, protocols and intervention studies (e.g., on traffic calming).

Study selection process

Using search terms developed for this review, the study selection process began with screening titles to identify and retrieve only those works that clearly met the inclusion criteria and were therefore potentially relevant. Next, retrieved studies' abstracts were scanned against inclusion and exclusion criteria. Following this, the full texts of the articles were examined. Included studies were those that were reviewed in full after matching inclusion criteria. Additional papers were identified when searching the reference lists of included studies. Two people independently screened paper abstracts and full texts, and disagreement concerning inclusion was resolved through a third person.

Data extraction process

Data extracted to an Excel sheet included author(s) and year of publication, year of data collection, location of study, sample size (separated by gender) and age/grade. In addition, data was collected on examined neighbourhoods' safety (perceived or measured), methods of measuring or reporting children's active behaviour with outcome of measures (e.g., body mass index [BMI] or medium-to-vigorous physical activity [MVPA]), methods of measuring or assessing child's neighbourhood, variables accounted for analysis including cofounders (e.g. sex/gender/family characteristics) and finally the summary of findings. The output was a systematic narrative summary table of the 29 relevant papers (Table 1 in Online Supplemental File 2). The screening and selection processes are visualised in the PRISM diagram in Figure 1, and the corresponding PRISMA_Scr checklist was completed during the review (see Online Supplemental File 1).

Patient and public involvement

No patient involved

Data synthesis

Findings of the association between COAMB and safety were compared by creating another table (Table 2 in Online Supplemental File 2) and ordering evidence according to a socio-ecological framework. In this framework, exposures in active living are organised into multiple levels of influences.[28] Additionally, we created another table to group studies by similarity of examined forms of active behaviour, thereby offering an opportunity to assess the impact of safety by the subgrouping of COAMB shapes (Table 3 in Online Supplemental File 2). The subgrouping corresponds to child's main types of structured or unstructured active behaviour addressed in literature excluding during school hours.[29]

Methodological assessments of the quality of individual studies

A rigour methodological quality assessment of studies was carried out, and each work was appraised independently and critically by two reviewers (RZ, BJ). Any rating discrepancies were discussed, and a shared decision was reached in required cases. Review of previous research assessed quality of studies for children active behaviour [15 19 23 24, 19 24] resulted in the adoption of 13 quality assessment criteria. The list was then expanded with seven methodological quality criteria to increase its suitability and capture actual methodological measures that were previously overlooked but likely to affect results. For example, criteria were added regarding measurement method for COAMB (reported or objectively

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measured), temporality elements (change in time of day, weekdays or weekends), the delineation of the neighbourhood area (assessed or measured) and accounting for confounders (age, gender, ethnicity and family characteristics), which are fundamental points in children's active behaviour evidence-based studies. The criteria checklist comprised six sections under which every question was scored and a total overall score out of eight points assigned. The scoring chart is presented below in Table 1. In the event of employing composite technique of measures in Part D (items 13 and 14) such as an accelerometer and GPS, the study accumulates 0.5 of points than when only used one tool that collects 0.25 point. In the instance of specific criterion that was not applicable—for example, BMI with no temporal and spatial characteristics—the criterion was discounted from the sum of the points to derive the total score. The quality score was then converted into a percentage by dividing each study quality score by the highest score possible (adjusted for criterion measured); the resulting figure was then multiplied by 100 to obtain a quality percentage.[24] A study quality percentage score of ≥ 66.7% was considered robust, while a score between ≥ 50 and < 66.6% was moderate and < 50% low (see Table 2 in Online Supplemental File 2). The strength in the evidence was then determined from the proportion of high-quality studies showing concurring and significant findings.[24]

Table 1. Criteria for quality assessment and scoring for each criterion from previous published systematic reviews

Study	Criteria	Description	Total quality score
Characteristics quality assessment	Part A	1. **Study objectives**: are objectives clear? 2. **Design**: Was the study design appropriate for research undertaken? 3. **Target population:** has target population defined? 4. **Random sampling:** Was a random sampling of the target population taken or was sampling appropriate for the study design?	(0.25 point for each sub item) Total = 1
	Part B	5. **Study participant number** Was the number of participants at each stage reported. 6. **Participants inclusion/exclusion:** Were criteria for inclusion and exclusion of the participants used? 7. **Study population:** Was the study population sufficiently described, i.e., sample size, gender, age, indicators of socio-economic status? 8. **Participant recruitment:** Was participant recruitment described or referred to?	0.25 for each point Total = 1
	Part C	9. **Response rate:** Was the response rate 60% or more? 10. **Data collection**: Did describe data collection, i.e., by mail, by interview, objective measure? 11. **Data sources:** Did the study describe source of data, i.e., questionnaire, survey, focus group, accelerometer, GPS? 12. **Missing data:** Were numbers/percentages of the participants with missing active behaviour data reported and did at least 80% of enrolled participants provide complete data to include?	0.25 for each point Total = 1
Methodological measures and analysis assessment	Part D Measures	13. **Objective measures:** Did the study objectively measure active mobility, i.e., activity tracking. Spatial technology or web application? 14. **Subjective measure:** Were measures reliable and valid?	GPS = 0.25 (accelerometer) = 0.25 questionnaire = 0.25 Travel diary = 0.25 N/A Total = between 0.25 -1
		15. **Temporal measures:** was measured active behaviour related to temporal characteristics, i.e., weekends vs weekdays or before and after school?	Yes = 0.5 No = 0 N/A Total = 0.5
		16. **Measures perceived safety characteristics:** Did the study account for spatial or temporal features of the safety feelings? (weekdays vs weekends or daytime (before and after school)) Or was measured safety conducted using geocoding?	Yes = 0.5 No = 0
		17. **Area of exposures:** Did the study delineate exposed area "neighbourhood" objectively or evaluate arbitrarily?	Objectively = 0.5 Arbitrary = 0 N/A
	Part E Analysis	18. **Analysis: W**as it clear what was done to determine statistical significance, e.g., p-value, confidence interval, OR was a spatial statistical method?	Yes = 0.5 No = 0
		19. **Output evidence:** Did the analysis account for spatiotemporal behaviour of evidence?	Yes = 1 No = 0
	Part F variables	20. **Cofounders:** Did the study account for cofounders of age, sex, ethnicity and family characteristics?	Each is 0.25 Total = 1

Evidence synthesis and strength

The quantitative summary of evidence was collated by the socio-ecological framework to overcome heterogeneity across studies. [15] Relevant findings were classified as showing statistically significant positive (+) or negative (-) associations or 0 for a non-significant association (Table 2 in Supplemental File 2). To enable synthesis, a reverse coding of the original direction of associations (i.e., positive to negative or vice versa) was performed as necessary, depending on the wording used in each work.[15] For

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example, low perceived safety association with decreased active mobility behaviour was recoded to high perceived safety associated with increased active mobility behaviour. A minimum of five studies with significant or non-significant findings was sufficient to generate a consensus.[15 17] When less than five studies (four studies) were available for synthesis, evidence was rated as limited,[24] and three studies were rated insufficient. For the purpose of this review, we adapted earlier practices[24] to derive the strength in the evidence from the proportion evidence with robust quality to the total association that agreed in the direction of significance (i.e., positive, negative or non-significant). However, we rescaled the minimum to assign the strength levels in the evidence using the proportion of studies found with robust quality in this review to those found in earlier review.[24] The output evidence was called strong when the threshold of ≥ 27% in the total synthesised evidence agreeing in the direction is accumulated in minimum of five studies. Thus when ≥ 27% of the studies that had a consensus in the direction were of robust quality, the fair quality ranged from ≥ 20%–< 27%, and weak were < 20%.

Study selection

Following the title screening, a total of 231 papers across the six databases were initially identified as potentially relevant. After removing 128 duplicates, the abstract review resulted in 64 articles underwent a full-text review. Seven studies were identified from a manual search of individual reference lists resulting in a total of 29 studies meeting inclusion criteria. The selection process is visualised in Figure 1.

RESULTS

Study characteristics

The characteristics of the 29 studies—including settings, measurement methods and evidence output—are summarised in (Table 1in Online Supplemental File 2). Aside from three longitudinal studies[11 30 31], 25 of the 29 studies (86%) had a cross-sectional design. Twelve of the 29 studies (41%) featured analysis using data from other larger scale or government projects (e.g., the Children Living in Active Neighbourhoods [CLAN] study,[32 33] the Built Environment and Active Transport [BEAT] project[34] and the Spatio-Temporal Exposure and Activity Monitoring [STEAM] project[35]). Populations varied across studies, with both parents and children represented. The sample sizes ranged from 35 children[36] to 31,000 households.[37] Studies had alternatively used age groups or grades to refer to children's age, but the mean ages were within the specified range of five–12 years. Included works have examined associations between forms of COAMB and perceived safety (21 of 29 studies; 73%), measured safety (five of 29; 17%) or both measured and perceived safety (three of 29; 10%). Collectively, the publications spanned Asia, Europe, North America and Oceania (Diagram 1 in Online Supplemental File 2). Aside from two Iranian studies[38 39] and one from Malaysia,[40] the remaining manuscripts (26 of 29; 90%) addressed populations in developed countries, with five each from the United States[31 41-44] and the United Kingdom;[11 45-47] four each from Canada[30 34 48 49] and Australia;[32 33 50 51] two each from New Zealand[52 53] and the Netherlands;[54 55] and one each from Portugal,[56] Finland[36] and Austria.[57] Except for one study,[58] all studies were published between 2007 and 2020. Two analysis used BMI[31 46] as a measure of children's active health levels. Subjective assessment was the most common method of measures of children's active behaviour. Ten of 29 studies (34%) employed objective measures such as accelerometers,[32 33 43] pedometers,[51] GPS[36] alone or in combination with an accelerometer,[34 49 52] or GIS loggers.[30] In view of the heterogeneity in the measurement methods, disparity in studies units of outcome were identified across both reported and measured COAMB (Diagram 2 in Online Supplemental File 2). Of 24 studies examined personal safety, majority assessed safety as perceived by parents including seven studies featured safety perception among children.[39 45 47 48 51 58] Aside from five studies[11 33 49 52 55] examining objective measures of road safety, 12 studies assessed roads danger as perceived primarily by parents. Findings in 25 of the 29 studies (86%) were reported separately for boys and girls.

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Methodological quality assessments and strengths in evidence

We found nine of 29 studies (31%) to be of robust quality, 16 studies (55%) to be fair, and four (14%) to be weak. With regard to subgrouping of studies by active mobility types, we recognised active travel behaviour (independent walking or cycling to local destinations such as shops, friend's houses, playgrounds or parks) in 10 of 29 (34%) studies, active school travel in seven of 29 studies (24%), active play, jointly examined active travel and exercise in seven of 29 studies (24%), solely active play in three (10%) and two (8%) assessed BMI (Figure 2).

Synthesis of the evidence

The summary of statistically significant and non-significant findings in the 29 studies examining influences on COAMB is shown in Figures 3-6 below.

Independent from safety, we found that forms of COAMB have a strong sex/gender variation association (10 out of 15 studies; 40% robust quality), including three studies depicting temporal weekdays and weekends[34] variation and another two varied by time segment of the day (after dark).[50 51] We found a strong age/grade association (six of seven studies; 33.3% robust quality) to variation in children active behaviour. A weak strength but consistent negative association (eight studies; 12.5% robust studies) of car ownership (the most consistently addressed family characteristics variable), and a strong association of ethnic background in five of six studies (40% robust quality), with one study depicting temporal behaviour change by day type (weekends to weekdays,[52] were depicted (Figure 3).

On the neighbourhood level, we found reported or measured COAMB to have a strong negative association between safety concerns measured by parents' perception (17 of 18 studies; 29.4% robust quality) and varied forms of reported or measured COAMB. (Figure 4). Limited and inconsistent association in four studies reported findings on 'stranger danger' among parents inclusive non-significant,[53 56] negative association[43] with one showing a temporal association (weekdays).[34] Parents' concern over road safety (11 studies; 25% robust quality) was moderately yet consistently negatively associated with COAMB, including one study reporting variation by gender (significant for girls).[51]

In the works that apprehend safety among children, aside from one study,[48] remaining works were quantified as fair or weak levels of quality hence unlikely to yield strong evidence. Nevertheless, four studies[39 47 48 51] were found personal safety to have negative and consistent association with COAMB, including variation by sex/gender[47]. While insufficiently addressed, perceived stranger danger among children demonstrated a consistently negative association with COAMB.[45 50 58] Road safety was consistently and negatively associated in five studies but varied by gender,[47 58] with one study was particularly notable for boys from low socio-economic status SES backgrounds[50] (Figure 4).

Despite the fact that about 50% of robust quality studies were located in those examined actual measures of safety, the synthesis of evidence showed inconsistency in the association of COAMB with measured personal[30 31 43 44] and road safety.[9 11 49 55] Thus, whilst unlikely that definitive evidence can be drawn, evidence saw a general indication of negative impact of measured crime on children active health. A significant association in the opposite direction[30] for personal safety and the disagreed in the direction for road safety,[59] with another lacked the association[33] depict the inconsistencies in the evidence (Figure 5). Other characteristics were explored in studies, on the neighbourhood level, that may contingently ally with safety. For example, distance travelled by children, whether perceived or measured (10 studies; 40% robust quality), has a strong, but inverse association. Additionally, land-use diversity (six studies; 66.6% robust studies) exhibited strong association with COAMB (Figure 6). Other variables displayed consistently positive associations yet are inadequately examined (see Table 2 in Online Supplemental File 2.

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DISCUSSION

This systematic amalgamation, synthesis influences found in 29 studies on COAMB in specific to neighbourhood perceived and measured safety. Overall, we found that the association of parents' or children reported COAMB and perceived safety was more likely to present consistency in the direction than when active behaviour and/or safety exposures were objectively measured. Most importantly, is the notable dearth of understanding safety concerns in the eyes of young people. It is also worthy to note that discounting five studies, remaining included works have fallen short of considering the time element in the measures or the output. Previous literature has demonstrated COAMB[60] or perceived safety[61] to differ by day type (weekend to weekday) or by time during the day. Nevertheless, this review comprehensibly indicates that COAMB is multifaceted in relation to the socio-ecological domains of individual, family and neighbourhood and across various types of COAMB including active travel to school or other local destinations, active play and exercises (see Diagrams 3, 4, 5 and 6 in Online Supplemental File 2).

Our first finding to demonstrate in this review is that influences across the socio-ecological levels, including neighbourhood safety modifying COAMB, are ubiquitous across countries. The review identified studies conducted in Europe, North America, Asia and Oceania. While different regions or countries are likely to have somewhat different impacts, the findings did not seem to vary between countries with disparate geographical locations. Across studies, the varying of COAMB by sex/gender as boys were shown to range over a larger space, travel greater distances and have higher levels of physical activity. The previous finding, however, disagreed with two previous reviews on independent mobility[62] and active school travel,[17] yet was in agreement with another two reviews on physical activity.[7 63] Despite that, it is unlikely to escape validating our observation in (Diagram 3 in Online Supplemental File 2) where less evidence was among active travel types than in those in active play and physical activity. The association to age across all types of active behaviour found in the current paper disagree with earlier two reviews[7 63] may be explained by the broad children's age categories (children aged three to 12 years[63] or three to 18 years[7]) and by investigating physical activity type of active behaviour. Earlier longitudinal study suggested that some forms of active travel (such as school travel) become more prevalent with age starting from six years,[64] which may depict the consistency with a review on school active travel.[17] Though this review didn't resolve the debate due to inconsistencies in the subgroupings used for race/ethnic background; there is enough indication of ethnic differences association with active health behaviour differences. The inverse association found between car ownership and COAMB affects primarily active travel types to local destinations and school coincides with previous reviews.[10 15 17]

The second critical finding of this review is that children living in neighbourhoods of low safety are denied the opportunity for active mobility behaviour but more effects on active travel (Diagram 4 in Online Supplemental File 2). The relation of personal safety concerns among parents to COAMB favoured the notion of the significant parental role in impacting primary-school-aged children's active health behaviour.[10] However, this finding contrasts with previous reviews on active travel[8] (inconsistent findings), and on physical activity[7] (lack of effects). It is worth noting that both reviews addressed a broad range of children ages (five–18 years old and three–18 years old) that may contribute to this outcome. Nevertheless, our findings concurred with a meta-analysis on adult population[14] and a longitudinal national study in the US on children.[65] Despite all, this review saw that the evident impact of personal safety on COAMB features more nuanced gender variation (more concern for girls),[32 36 51] temporal variations (on weekdays only[52] or on weekends[32]), and intersections between gender and temporality when after dark and girls.[36]

The novel finding from this review is the apparent variation in the impact of between personal safety and road danger on the various types of COAMB (Diagram 4 in online supplemental file 2). Parents' concern over personal safety is seen mostly gathering in active travel to school and to local destinations

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inclusive. However, parents' concerns over road danger were shown to affect all types of COAMB. The finding suggests that perceived risk on personal safety restricts children actively travel and consequently limits their active space which confirms with earlier study[12] revealing shrinking children space with the rise of safety fears. Despite a dearth of evidence on 'stranger danger', current review finding suggests that the vulnerability of children (particularly for girls) is more significant than parents perceived harm from stranger danger. Insufficient evidence indicates that high score of perceived social norms among parents[53 54] or children[47] may encourage children active behaviour.

The inconsistencies yielded of the associations between COAMB and objectively measured safety (actual) compromise the ability to draw a conclusion and seeks further research. Despite that, there is indication of a negative impact on children active behaviour and health indices. Several possible explanations for the inconsistencies in the findings can be briefed. For example, the different study designs as a longitudinal study that found an association[66] disagreed with the remaining cross-sectional direction, where causality could not be inferred.[9 63] Additionally, the variation in measurement methods among studies showed that accounting for actual crime and its categories varied across studies. Furthermore, the methods adopted to delineate child's neighbourhood and derive safety exposures varied across studies that may lead to variation in the output results.[67]

The dense populations offering the sense of safety characterising residential and commercial areas could explain the increase of active behaviour, which agrees with similar findings portrayed in an earlier review.[19] The notable inverse relation to perceived or measured distance travelled affects children's active travel types, including the journey to school as well as to the other destinations in the neighbourhood. This finding agrees with earlier reviews on school travel[15 17] and active travel.[8] See Diagram 4 in Online Supplemental File 2.

Finally, our synthesis of evidence using a tailored methodological quality appraisal was validated in this review to derive evidence strength and has identified the inconsistencies found in studies employed objective measures. Thus, we hope our methodology is informative for researchers evaluating similar constructs and can help to more effectively standardise measurement methods and quality assessments going forward.

Implications

The portrayed significant association of COAMB across all socio-ecological levels indicates that multiple factors govern children's active behaviour. Consequently, successful implementation of any suggested intervention strategies to reverse the current curve in declining levels of children's active behaviour[25 68] will necessitate collaboration across all levels: children, family, school, community-based organisations, city planners and policymakers. Thus, disentangling influences across the aforementioned domains aid to recommend effective intervention and planning.

To begin, strategies that are gender and age-specific are warranted for intervention and design planning policies. Incorporating these differences when redesigning existing neighbourhood spaces and/or in new urban areas may reduce current inequities in children's active behaviour. Ignoring gendered primary-school-aged children's active behaviour may have a prolonged lifelong implication of having one-sided adult society (male) regarding physical activity. Similarly, the evidence of COAMB associations with ethnic differences suggests that a subgrouping of population is constantly inactive; thus, interventions for vulnerable groups of the population are warranted. Parental involvement in educational programmes to promote children's free play and active travel or children's encouragement in school and community programmes to participate in active travel and leisure active play could be useful. Earlier evidence also supports the positive influences of active parents[56] or positive peer influence.[69]

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The evident impact of safety on COAMB could imply that current cities' and roads' design are unwelcoming and unsafe for kids. To alleviate the impact of perceived safety across all shapes of COAMB, we vow for revitalise existing design and propose new strategies. Coupled with that research should delve into a more specific understanding of determinants[70] and measurable parents' concerns of personal safety,[61 71 72] influences of SES background,[73] association to spatial location[74], and account for time.[75] In tandem with this, raised voices are calling to engage children in the design yet merging with parents, caregivers, and other relevant stakeholders to sharpen child-specific designs may remain the key for the suggested child-friendly city model[76]. Available evidence further suggests that current streets' design are unsafe and not appealing by children or by their parents. Prompt policy interventions on infrastructures to address heavy traffic, fast drivers, high-speed roads and lack of road signals, traffic lights, availability of sidewalks and cycling infrastructures could ensure safe crossings and may alleviate parents' and children's perceived concerns over road danger. Indeed, an improved understanding of measured pedestrian safety impacts may benefit future road design in cities. The association of travelled distance and land use diversity to COAMB clearly call for a redesign of fear-related physical features that promote personal safety. For example, the 'activity-friendly neighbourhood' indorsed by Lee and Moudon[77] recommends the proximity of safe places that provide environment and public play opportunities to children's homes. Building on findings in our review, we put forward, a prompt consideration for upgrading local features to promote walkability, desirable environmental features (e.g., schools, shops, libraries), and active playing in local parks or exercise. Furthermore, parents' or children's perceptions of risks may be mitigated by features such as well-lit sidewalks and reduction of graffiti, litter or roaming dogs in existing cities designs.

What's more is the fast pace of technological advancement, offering real-time GPS localities of mobility yet embedded in mobile phones or smartwatches. Thus, it may seem favoured by parents to equip children with such technologies offering a sense of security. Along with that, web-based applications are emerged where instant updates on locations of personal safety incidents are surfacing[61]. By the same token, GPSs-enabled watches to alarm parents when a child leaves a safe zone are surfacing. Offering these technologies may encourage parents to grant children greater independence to explore and expand their space, thus promoting active behaviour. At that moment, research may explore active behavioural changes among children when equipped with such technologies.

Overall, a real shift in children's physical active behaviour in the safety context can be achieved, but it requires collaboration across the socio-ecological domains for successful intervention programmes and policy planning. Failure to couple changes on the levels of the individual, family, and neighbourhood to promote an active child behaviour culture may result in city facilities governments have invested in, remaining barren because of safety concerns.

Strengths and limitations

This review has several limitations. First, only English-language and peer-reviewed studies were included. Second, despite every effort to identify all relevant studies through the variety of terms used, some relevant studies may have been omitted due to multiple synonyms used in research for the same word. Third, observed heterogeneity in study measures prevented a straightforward meta-analysis; thus, we conducted quantitative vote counting to overcome existing disparities. Finally, the majority of studies were of fair or weak quality. Despite these limitations, this review has important strengths. This work synthesised findings and derived a quantitative understanding of evidence strength by combining quality of studies reporting and critical contextual assessment of measurement methods. This paper has comprehensively addressed impact of safety in specific to primary-school-aged children, eliminating inconsistencies due to age groups. Another strength is being the pioneer in representing the impact of

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neighbourhood safety exposures across varied types of COAMB. Finally, no restriction was attempted to confine this work regarding either date of publication or study region.

CONCLUSION

This review synthesised shreds of evidence across the socio-ecological domains in particular to neighbourhood safety (personal and road danger) exposures and modified types of COAMB to design relevant interventions. Expanding from previous reviews, a methodological quality assessment was undertaken that may aid future reviews with similar constructions in overcoming the heterogeneity found across studies. It is our hope that future studies pay more attention to these measures to increase reliability in evidence-based research. Further studies are warranted to better understand the effect of measured safety on COAMB, and to explore the varying impact between measured and perceived safety. Independent from safety, gender variation, ethnicity, access to cars, distance travelled and modified COAMB is ubiquitous across countries.

Every child has the right to feel safe at all times. However, our review revealed that perceived safety—among parents or children—indeed modifies children's various active travel shapes. Perceived risk on personal safety restricts child's active behaviour significantly, with evidence crowds in active travel journies to various destinations, including school as well as to local destinations. The effect of danger perceived from traffic spread over all types of active travel, active play and exercise. In either case, the strength of the associations penetrates gender, age and ethnicity variations; distance travelled (inversely); and temporal distinctions between weekends/weekdays or different times of the day.

This review has illustrated the urgent need for collaborative actions across all socio-ecological levels to reverse a decline in children's active behaviour. Although taking steps to improve the sense of safety in neighbourhoods and reduce crime requires no further justification on its own, there is enough evidence that improving roads and personal safety perceptions may contribute to parents and their children embracing an active mobility culture in its various forms.

Figures Legend

Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses for Scoping Review, PRISMA_ScR for the screening and selection process.

Figure 2. Studies re-grouped by the type of active behaviour with output level from the methodological quality assessment. The x-axis represents re-categorised active behaviour in association with neighbourhood safety. y-axis denotes the number of studies accumulated strength type of robust when summing total score (≥ 66.66%) against 20 criteria, moderate when scored ≥ 50–< 66.6% and weak when an accumulated point was < 50%.

Figure 3. Association between individual child and family characteristics and active mobility behaviour. The x-axis represents examined variables that accumulated required number of studies to synthesise evidence in the categories of gender, age, access to access to car and ethnicity. y-axis denotes the number of studies that examined the association and output of accumulated significance or non-significance. Note: #T = number of studies that found temporal characteristics (changes between weekdays and weekends or segments of the day (before and after school) in the association, #D = number pf studies found variation by distance travelled.

Figure 4. Association of perceived safety (parents and children) of personal and road danger to children reported and measured active mobility behaviour. The x-axis represents examined variables that accumulated required number of studies to synthesise evidence in categories of perceived personal, road safety and 'stranger danger' by parents or children. y-axis denotes the number of studies that accumulated significant. Note: #T = number of studies examined temporal behaviour, #G (D)= # of studies varied in association by gender including one study by distance, #A = number pf studies varied by child's age, #SES = number of studies varied by the socio-economic status.

Figure 5. Association of measured (actual) safety to children and measured outdoor active mobility behaviour. The x-axis represents category of the measured safety. y-axis denotes accumulated number of studies, examining the association per level of quality. -# = number of studies in the opposite direction.

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Figure 6. Relevant attributes of the physical environments in the safety context in association with child reported or measured active behaviour. The x-axis represents category of the attributes of the physical environment. y-axis denotes accumulated number of studies, examining the association with quality (robust, fair, weak).

Supplementary information

Supplemental File 1 PRISMA checklist
Supplemental File 2: Table 1 is summary of studies' characteristics, Table 2 is findings of association by the socio-ecological framework of the included studies, Diagram 1: show studies distribution by country of origin, Diagram 2: show disparity of studies measurement methods and in unit of outcome, Diagrams 3 – 6, Details of search terms used. Table 3: display categories of studies by active mobility behaviour type, and Table 4 show each study quality score.

Declarations

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Authors Contributors: RZ carried concept/design, data retrieve, screening of the literature for inclusion, rating of the methodological quality, synthesised the findings, drafted and finalised the manuscript. RN, OG and AD reviewed the manuscript and approved the final outcome.

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Availability of data and materials: All data generated for synthesising methods during the systematic review are included in the published article, and its supplementary information files 1, 2 & 3.

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References

1. WHO. Global Status Report on Non-Communicable Diseases 2010. 2010:176.
2. Pellis SM, Pellis VC, Bell HC. The Function of Play in the Development of the Social Brain. *American Journal of Play* 2010;v2 (n3):p278-96
3. Council IoMNR. From Neurons to Neighborhoods: The Science of Early Childhood Development. Washington, DC: The National Academies Press 2000.
4. Schwab K. Play access in cities is critical to children's 2020 [cited 2020 3/11/2020]. Accessed 9/10/2020 2020.
5. Garofalo J. The Fear of Crime: Causes and Consequences. *THE JOURNAL OF CRIMINAL LAW & CRIMINOLOGY* 1981;72(2):839.
6. Lorenc T, Petticrew M, Whitehead M, et al. Crime, fear of crime and mental health: synthesis of theory and systematic reviews of interventions and qualitative evidence. : Public Health Research 2014.
7. Davison K, Lawson C. Do attributes in the physical environment influence children's physical activity? A review of the literature. *International Journal of Behavioral Nutrition and Physical Activity* 2006;3(1):19. doi: 10.1186/1479-5868-3-19
8. Panter JR, Jones AP, van Sluijs EMF. Environmental determinants of active travel in youth: A review and framework for future research. *International Journal of Behavioral Nutrition and Physical Activity* 2008;5(1):34. doi: 10.1186/1479-5868-5-34
9. Carver A, Timperio A, Crawford D. Playing it safe: The influence of neighbourhood safety on children's physical activity—A review. *Health & Place* 2008;14(2):217-27. doi: https://doi.org/10.1016/j.healthplace.2007.06.004
10. McMillan TE. Urban Form and a Child's Trip to School: The Current Literature and a Framework for Future Research. *Journal of Planning Literature* 2005;19(4):440-56. doi: 10.1177/0885412204274173
11. Carver A, Panter JR, Jones AP, et al. Independent mobility on the journey to school: A joint cross-sectional and prospective exploration of social and physical environmental influences. *Journal of Transport & Health* 2014;1(1):25-32. doi: https://doi.org/10.1016/j.jth.2013.12.003
12. Jack G. Place matters: The significance of place attachments for children's well-being. *British Journal of Social Work* 2010;40(3):755-71. doi: https://doi.org/10.1093/bjsw/bcn142
13. Donovan J. Designing the compassionate city to overcome built-in biases and help us live better La Trobe University: Routledge; 2018 [Available from: https://theconversation.com/designing-the-compassionate-city-to-overcome-built-in-biases-and-help-us-live-better-92726 accessed 1/11/2020 2020.
14. Rees-Punia E, Hathaway ED, Gay JL. Crime, perceived safety, and physical activity: A meta-analysis. *Preventive Medicine* 2017 doi: https://doi.org/10.1016/j.ypmed.2017.11.017
15. Ikeda E, Hinckson E, Witten K, et al. Associations of children's active school travel with perceptions of the physical environment and characteristics of the social environment: A systematic review. *Health & Place* 2018;54:118-31. doi: https://doi.org/10.1016/j.healthplace.2018.09.009
16. Wong BY-M, Faulkner G, Buliung R. GIS measured environmental correlates of active school transport: A systematic review of 14 studies. *International Journal of Behavioral Nutrition and Physical Activity* 2011;8(1):39. doi: 10.1186/1479-5868-8-39
17. Rothman L, Macpherson AK, Ross T, et al. The decline in active school transportation (AST): A systematic review of the factors related to AST and changes in school transport over time in North America. *Preventive Medicine* 2018;111:314-22. doi: https://doi.org/10.1016/j.ypmed.2017.11.018
18. Pang B, Kubacki K, Rundle-Thiele S. Promoting active travel to school: a systematic review (2010–2016). *BMC Public Health* 2017;17(1):638. doi: 10.1186/s12889-017-4648-2
19. Marzi I, Reimers A. Children's Independent Mobility: Current Knowledge, Future Directions, and Public Health Implications. 2018;2018:2441. doi: 10.3390/ijerph15112441
20. Pont K, Ziviani J, Wadley D, et al. Environmental correlates of children's active transportation: A systematic literature review. *Health & Place* 2009;15(3):849-62. doi: https://doi.org/10.1016/j.healthplace.2009.02.002

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

21. Davison KK, Werder JL, Lawson CT. Children's active commuting to school: current knowledge and future directions. *Preventing chronic disease* 2008;5(3):A100-A00.
22. Steinbeck KS. The importance of physical activity in the prevention of overweight and obesity in childhood: a review and an opinion. *Obesity Reviews* 2001;2(2):117-30. doi: 10.1046/j.1467-789x.2001.00033.x
23. Lubans DR, Boreham CA, Kelly P, et al. The relationship between active travel to school and health-related fitness in children and adolescents: A systematic review. *The International Journal of Behavioral Nutrition and Physical Activity* 2011; 8(5) doi: http://dx.doi.org.dbgw.lis.curtin.edu.au/10.1186/1479-5868-8-5
24. Schoeppe S, Duncan MJ, Badland H, et al. Associations of children's independent mobility and active travel with physical activity, sedentary behaviour and weight status: A systematic review. *Journal of Science and Medicine in Sport* 2013;16(4):312-19. doi: https://doi.org/10.1016/j.jsams.2012.11.001
25. ABS. Australian Health Survey: Physical Activity, 2011-12 In: Australian Bureau of Statistics, ed. Canberra: Australian Bureau of Statistics, 2013.
26. Leung KYK, Loo BPY. Association of children's mobility and wellbeing: A case study in Hong Kong. *Travel Behaviour and Society* 2017;9:95-104. doi: https://doi.org/10.1016/j.tbs.2017.07.004
27. Tricco AC, Lillie E, Zarin W, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Annals of internal medicine* 2018;169(7):467-73. doi: 10.7326/m18-0850 [published Online First: 2018/09/05]
28. Mehtälä MAK, Sääkslahti AK, Inkinen ME, et al. A socio-ecological approach to physical activity interventions in childcare: a systematic review. *International Journal of Behavioral Nutrition and Physical Activity* 2014;11(1):22. doi: 10.1186/1479-5868-11-22
29. Borghese MM, Janssen I. Development of a measurement approach to assess time children participate in organized sport, active travel, outdoor active play, and curriculum-based physical activity. *BMC Public Health* 2018;18(1):396. doi: 10.1186/s12889-018-5268-1
30. Vonderwalde M, Cox J, Williams GC, et al. Objectively measured crime and active transportation among 10–13year olds *Preventive Medicine Reports (PMR)* 2019;13 (2019) (48-51)
31. Suminski RR, Robson SM, May LL, et al. Actual neighborhood-level crime predicts body mass index z-score changes in a multi-racial/ethnic sample of children. *Preventive Medicine Reports* 2018;12:164-69. doi: https://doi.org/10.1016/j.pmedr.2018.09.012
32. Carver A, Timperio A, Hesketh K, et al. Are children and adolescents less active if parents restrict their physical activity and active transport due to perceived risk? *Social Science & Medicine* 2010;70(11):1799-805. doi: https://doi.org/10.1016/j.socscimed.2010.02.010
33. Carver A, Timperio AF, Crawford DA. Neighborhood road environments and physical activity among youth: the CLAN study. *J Urban Health* 2008;85(4):532-44. doi: 10.1007/s11524-008-9284-9 [published Online First: 2008/04/25]
34. Faulkner G, Mitra R, Buliung R, et al. Children's outdoor playtime, physical activity, and parental perceptions of the neighbourhood environment. *International Journal of Play* 2015;4(1):84-97.
35. Loebach J, Gilliland J. Neighbourhood play on the endangered list: examining patterns in children's local activity and mobility using GPS monitoring and qualitative GIS. *Children's Geographies* 2016;14(5):573-89. doi: 10.1080/14733285.2016.1140126
36. Fagerholm N, Broberg A. Mapping and characterising children's daily mobility in urban residential areas in Turku, Finland. *Fennia* 2011;189(2):31-46.
37. Waygood EOD, Susilo YO. Walking to school in Scotland: Do perceptions of neighbourhood quality matter? *IATSS Research* 2015;38(2):125-29. doi: https://doi.org/10.1016/j.iatssr.2014.12.002
38. Mehdizadeh M, Mamdoohi A, Nordfjaern T. Walking time to school, children's active school travel and their related factors. *Journal of Transport & Health* 2017;6:313-26. doi: https://doi.org/10.1016/j.jth.2017.01.012
39. Shokoohi R, Hanif NR, Dali MM. Children Walking to and from School in Tehran: Associations with Neighbourhood Safety, Parental Concerns and Children's Perceptions. *Procedia - Social and Behavioral Sciences* 2012;38:315-23. doi: https://doi.org/10.1016/j.sbspro.2012.03.353

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

40. Tung SEH, Ng XH, Chin YS, et al. Associations between parents' perception of neighbourhood environments and safety with physical activity of primary school children in Klang, Selangor, Malaysia. *Child: Care, Health and Development* 2016;42(4):478-85. doi: 10.1111/cch.12355
41. Oluyomi AO, Lee C, Nehme E, et al. Parental safety concerns and active school commute: correlate across multiple domains in the home-to-school journey. *International Journal of Behavioral Nutrition and Physical Activity* 2014;11(1):32. doi: 10.1186/1479-5868-11-32
42. Roberts JD, Knight B, Ray R, et al. Parental perceived built environment measures and active play in Washington DC metropolitan children. *Preventive Medicine Reports* 2016;3:373-78. doi: https://doi.org/10.1016/j.pmedr.2016.04.001
43. Stephanie H. Kneeshaw-Price, Brian E. Saelens, James F. Sallis, et al. Neighborhood Crime-Related Safety and Its Relation to Children's Physical Activity. *Journal of Urban Health: Bulletin of the New York Academy of Medicine* 2015;92(3) doi: 10.1007/s11524-015-9949-0
44. Zhu X, Lee C. Walkability and Safety Around Elementary Schools Economic and Ethnic Disparities *American journal of preventive medicine* 2008;34(4) doi: 10.1016/j.amepre.2008.01.024
45. Alton D, Adab P, Roberts L, et al. Relationship between walking levels and perceptions of the local neighbourhood environment. *BMJ* 2007;92(1): 29–33. doi: http://doi.org/10.1136/adc.2006.100826
46. Noonan R, Boddy L, Knowles Z, et al. Cross-sectional associations between high-deprivation home and neighbourhood environments, and health-related variables among Liverpool children. *BMJ Open* 2016;6(e008693)
47. Page AS, Cooper AR, Griew P, et al. Independent mobility, perceptions of the built environment and children's participation in play, active travel and structured exercise and sport: the PEACH Project. *International Journal of Behavioral Nutrition and Physical Activity* 2010;7(1):17. doi: 10.1186/1479-5868-7-17
48. Loebach JE, Gilliland JA. Free-Range Kids? Using GPS-derived activity spaces to examine children's neighborhood activity and mobility. *Environment and Behavior* 2016;48(3):421-53.
49. Nguyen A, Borghese MM, Janssen I. Pedestrian traffic safety and outdoor active play among 10–13 year olds living in a mid-sized city. *Preventive Medicine Reports* 2018;10:304-09. doi: https://doi.org/10.1016/j.pmedr.2018.04.010
50. Timperio A, Crawford D, Telford A, et al. Perceptions about the local neighborhood and walking and cycling among children. *Preventive Medicine* 2004;38(1):39-47. doi: https://doi.org/10.1016/j.ypmed.2003.09.026
51. Villanueva K, Giles-Corti B, Bulsara M, et al. How far do children travel from their homes? Exploring children's activity spaces in their neighborhood. *Health & Place* 2012;18(2):263-73. doi: https://doi.org/10.1016/j.healthplace.2011.09.019
52. Oliver M, Mavoa S, Badland H, et al. Associations Between the Neighbourhood Built Environment and Out of School Physical Activity and Active Travel: An Examination from the Kids in the City Study. *Health & Place* 2015;36(2015):57–64. doi: http://dx.doi.org/10.1016/j.healthplace.2015.09.005
53. Lin E-Y, Witten K, Oliver M, et al. Social and built-environment factors related to children's independent mobility: The importance of neighbourhood cohesion and connectedness. *Health & place* 2017;107(113)
54. van den Berg P, Waygood EOD, van de Craats I, et al. Factors affecting parental safety perception, satisfaction with school travel and mood in primary school children in the Netherlands. *Journal of Transport & Health* 2020;16:100837. doi: https://doi.org/10.1016/j.jth.2020.100837
55. Helbich M, Emmichoven MJ, Dijst MJ, et al. Natural and built environmental exposures on children's active school travel: A Dutch global positioning system-based cross-sectional study. *Health Place* 2016;39:101-9. doi: 10.1016/j.healthplace.2016.03.003 [published Online First: 2016/03/25]
56. Santos MP, Pizarro AN, Mota J, et al. Parental physical activity, safety perceptions and children's independent mobility. *BMC public health* 2013;13:584-84. doi: 10.1186/1471-2458-13-584
57. Stark J, Frühwirth J, Aschauer F. Exploring independent and active mobility in primary school children in Vienna. *Journal of Transport Geography* 2018;68:31-41. doi: https://doi.org/10.1016/j.jtrangeo.2018.02.007

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

58. Davis A, Jones L. Environmental constraints on health: listening to children's views. *Health Education Journal* 1996;55(4):363-74.
59. Fjørtoft I. Exploring Children's Environments

GPSNHRN the Global Positioning Systems in Health Research Network 2011
60. Brooke HL, Corder K, Atkin AJ, et al. A Systematic Literature Review with Meta-Analyses of Within- and Between-Day Differences in Objectively Measured Physical Activity in School-Aged Children. *Sports Medicine* 2014;44(10):1427-38. doi: 10.1007/s40279-014-0215-5
61. Solymosi R, Bowers K, Fujiyama T. Mapping fear of crime as a context-dependent everyday experience that varies in space and time. *Legal and Criminological Psychology* 2015;20(2):193. doi: http://dx.doi.org/10.1111/lcrp.12076
62. Ding D, Sallis JF, Kerr J, et al. Neighborhood Environment and Physical Activity Among Youth: A Review. *American Journal of Preventive Medicine* 2011;41(4):442-55. doi: https://doi.org/10.1016/j.amepre.2011.06.036
63. SALLIS JF, PROCHASKA JJ, TAYLOR WC. A review of correlates of physical activity of children and adolescents. *Medicine & Science in Sports & Exercise* 2000;32(5):963-75.
64. Pabayo R, Gauvin L, Barnett TA. Longitudinal Changes in Active Transportation to School in Canadian Youth Aged 6 Through 16 Years. *Pediatrics* 2011;128(2):e404-e13. doi: 10.1542/peds.2010-1612
65. Datar A, Nicosia N, Shier V. Parent perceptions of neighborhood safety and children's physical activity, sedentary behavior, and obesity: evidence from a national longitudinal study. *Am J Epidemiol* 2013;177(10):1065-73. doi: 10.1093/aje/kws353 [published Online First: 2013/04/11]
66. Vonderwalde M, Cox J, Williams GC, et al. Objectively measured crime and active transportation among 10–13 year olds. *Preventive Medicine Reports* 2019;13:48-51. doi: https://doi.org/10.1016/j.pmedr.2018.11.010
67. Zhao P, Kwan M-P, Zhou S. The Uncertain Geographic Context Problem in the Analysis of the Relationships between Obesity and the Built Environment in Guangzhou. *International Journal of Environmental Research and Public Health* 2018;15(308):1-20.
68. WHO. Equity, Social Determinants and Public Health Programmes: World Health Organization, 2010:303.
69. Salvy S-J, Roemmich JN, Bowker JC, et al. Effect of Peers and Friends on Youth Physical Activity and Motivation to be Physically Active. *Journal of Pediatric Psychology* 2008;34(2):217-25. doi: 10.1093/jpepsy/jsn071
70. Sallis JF, Simons-Morton BG, Stone EJ, et al. Determinants of physical activity and interventions in youth. *Medicine and science in sports and exercise* 1992;24(6):248-57.
71. Chataway ML, Hart TC. (Re)Assessing contemporary "fear of crime" measures within an Australian context. *Journal of Environmental Psychology* 2016;47:195-203. doi: https://doi.org/10.1016/j.jenvp.2016.06.004
72. Lorenc T, Petticrew M, Whitehead M, et al. Fear of crime and the environment: systematic review of UK qualitative evidence. *BMC Public Health* 2013;13(1):496. doi: 10.1186/1471-2458-13-496
73. Killias M, Clerici C. Different measures of vulnerability in their relation to different dimensions of fear of crime. *British journal of criminology* 2000;40(3):437-50.
74. Matthews SA. Thinking about place, spatial behavior, and spatial processes in childhood obesity. *American journal of preventive medicine* 2012;42(5):516-20.
75. Boessen A, Hipp JR, Butts CT, et al. Social fabric and fear of crime: Considering spatial location and time of day. *Social Networks* 2017;51:60-72.
76. Wood J, Bornat D, Bicquelet-Lock A. Child Friendly Planning in the UK, A Review. Heriot-Watt University: Institute for Social Policy, Housing and Equalities Research, 2019.
77. Lee C, Moudon AV. Physical Activity and Environment Research in the Health Field: Implication for Urban and Transportation Planning and Research. *Journal of Planning Literature* 2004;19(2):147-81.

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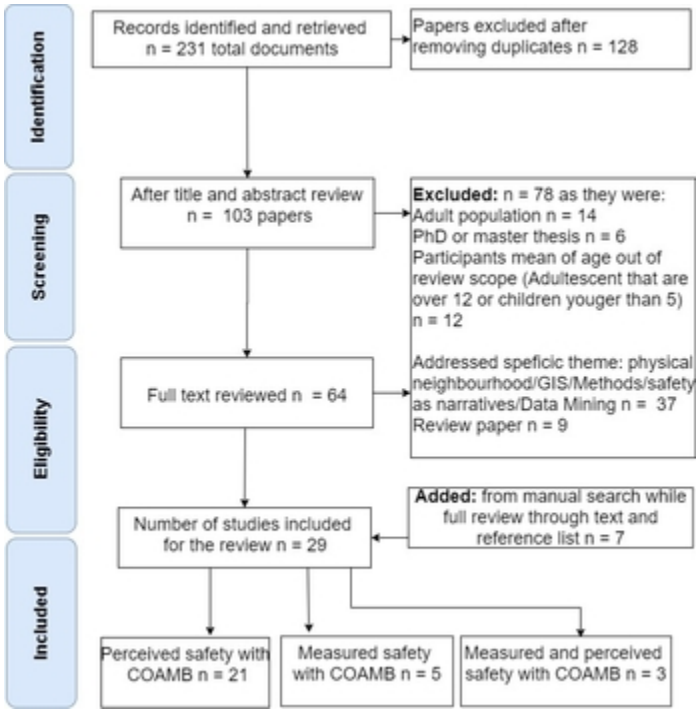


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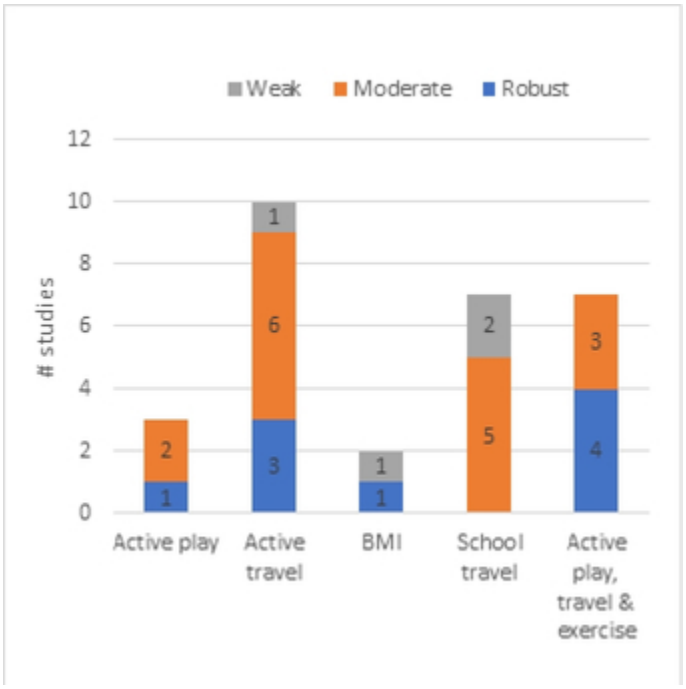


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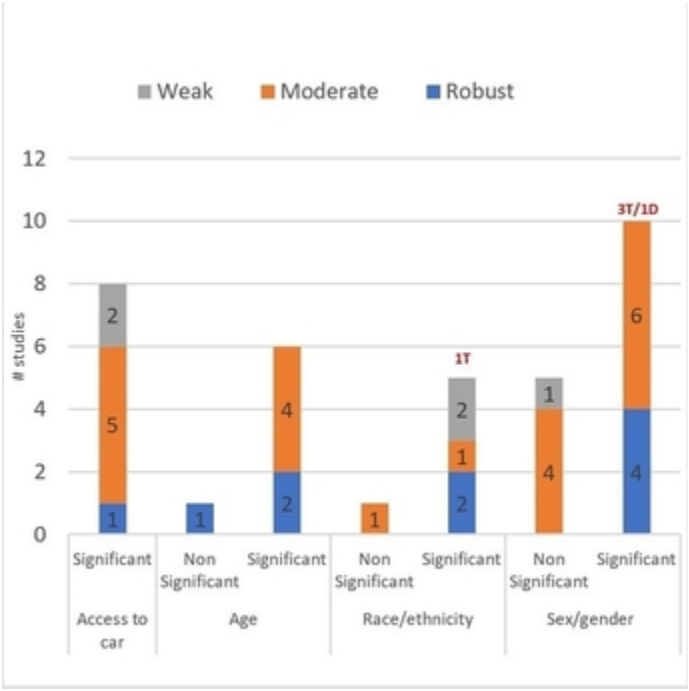


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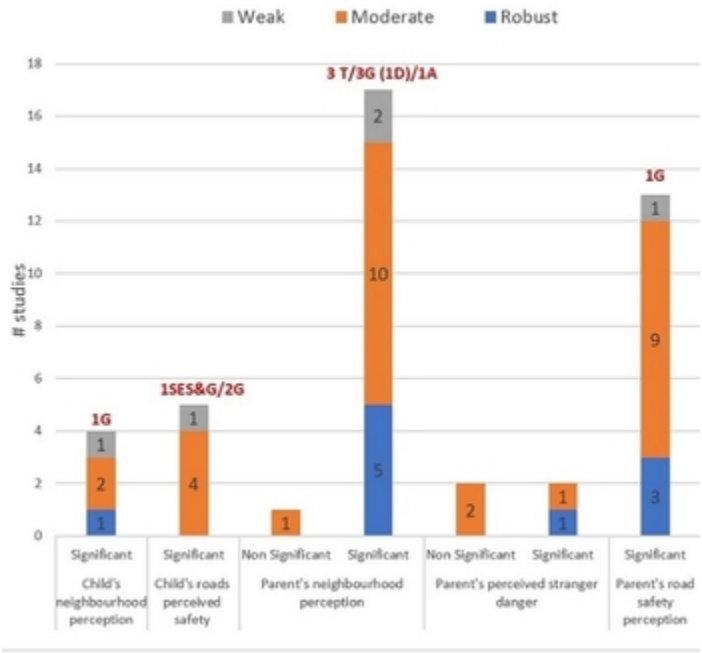


Figure 4. Association of perceived safety (parents and children) of personal and road danger to children reported and measured active mobility behaviour. The x-axis represents examined variables that accumulated required number of studies to synthesise evidence in categories of perceived personal, road safety and 'stranger danger' by parents or children. y-axis denotes the number of studies that accumulated significant. Note: #T = number of studies examined temporal behaviour, #G (D)= # of studies varied in association by gender including one study by distance, #A = number pf studies varied by child's age, #SES = number of studies varied by the socio-economic status.

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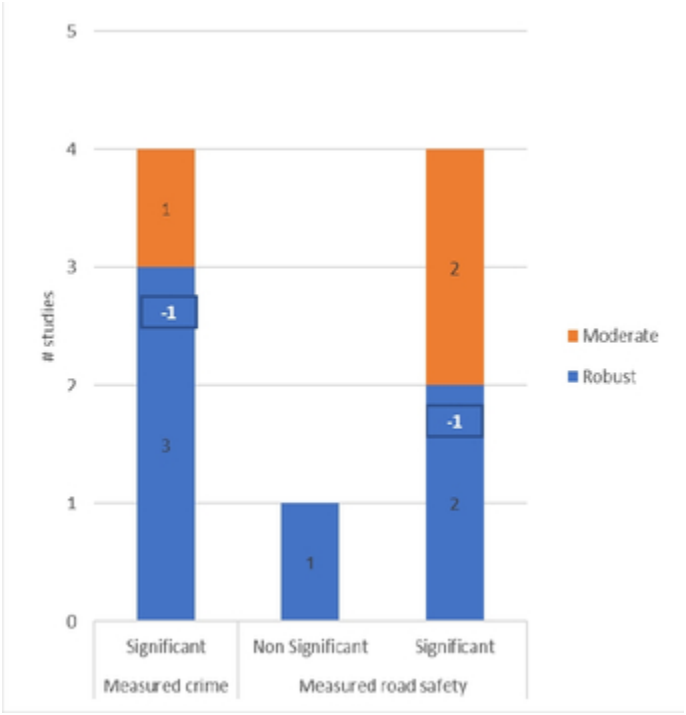


Figure 5. Association of measured (actual) safety to children and measured outdoor active mobility behaviour. The x-axis represents category of the measured safety. y-axis denotes accumulated number of studies, examining the association per level of quality. -# = number of studies in the opposite direction.

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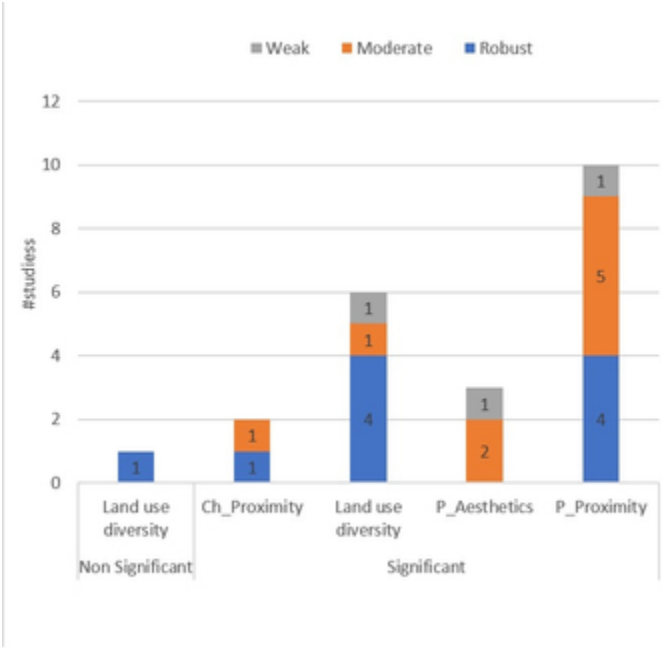


Figure 6. Relevant attributes of the physical environments in the safety context in association with child reported or measured active behaviour. The x-axis represents category of the attributes of the physical environment. y-axis denotes accumulated number of studies, examining the association with quality (robust, fair, weak).

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Table 1. Characteristics of included studies and findings of evidence of influences on children outdoor active mobility behaviour

#	Study's author(s)/year of data collection n = Participants number/ (sex/gender: number/% of M/F)/ Age (or mean age)/Grade	Country/ Study Design/ [Project]	Measures Safety (Perceived by/derived from)/ Measured	Active Behaviour Outcome / Measures	Definition of neighbourhood	Study cofounders/Variables	Findings of Associations
1	Alton, Adab, Roberts, and Barrett (2007)/-/ n= 473 (250 M and 60 F)/ 9-11 years old/	United Kingdom Cross-section	Children/questionnaire on perception of local environment + preferred travel method	Level of walking (high walkers, low walkers)/ Self-report Past 7-days	Local area	Sex/gender, race/ethnicity (Asian, black, Chinese, mix, others, white), family characteristics (cars ownership and number of rooms in primary residence) as a proxy for SES)	Child from ethnic minority walks significantly less. Perception of parental concerns over road safety and heavy traffic associated with a higher walking level. Child perception of lack of suitable leisure space in the neighbourhood and worry about a stranger is associated with less walking levels. Child perceptions of the local environment correlate with walking level.
2	(Carver, Timperio, & Crawford, 2008)/between July and December 2004/ n = 188 (44% boys) 8-9 years old	Australia Cross-section from [CLAN] longitudinal study	Parents survey for personal characteristics +frequency of children walking and cycling to 15 destinations + Measured road safety[1]	*MVPA Outside school hours Accelerometer for 8-days	800 m around participant home	Sex/gender, age, Active transport +MVPA + measured road environment	No association between children likelihood of making at least seven walking/cycling trips per week to neighbourhood destination and roads environment
3	Carver, Timperio, Hesketh, and Crawford (2010)/2001 n = 170 (51% M)/ 10-11 years old 15-17 years old (excluded) 2001-2005 longitudinal study	Australia Cross-section from [CLAN] longitudinal study	Parents questionnaire Indices for (avoidance and defensive behaviour, and perceived risk) + active transportation to 15 destination	*Active transport/MVPA outside school hours/ Accelerometer	Local area	Sex/gender, parents (avoidance, defensive and perceived risk)	The lower level of active transport and lower level of MVPA in a neighbourhood outside school hours associated with constrained behaviour exhibited by parents on both boys and girls. Reduced MVPA associated with constrained behaviour on weekends for boys. Higher constrained behaviour associated with higher MVPA for girls (limitation of study)
4	(Carver, Panter, Jones, & van Sluijs, 2014)/T1 (April-July 2007) n= 1121 (9-10) years old (43%M) T2 (April-July 2008) n= 491 (39%M)	United Kingdom Longitudinal from [SPEEDY] study	Parents perception questionnaire (social/physical environment + rules regarding their children physical activity + perception of traffic safety concerns)	Independent mobility to school/ from children questionnaire	Within 800 m pedestrian network buffer around the home (10 min walk)	Sex/gender, sociodemographic (siblings, cars ownership, parents' education) + environmental characteristics around the home and (within 100 m buffer of the shortest route to school	Car access is associated longitudinally with boys walking/cycling independently to school. The proportion of main roads in the neighbourhood and parental encouragements of walking/cycling associated longitudinally with girls walking/cycling to independently to school. Land use mix is associated positively with girls walking cycling independently to school. Boys that are allowed by their parents to play outside have higher odd to walk/cycle independently to school.
5	(Davis & Jones, 1996)/-/ n = 492 of children (not reported gender)/ 9-11-years-old 13-14 years old (excluded)	United Kingdom Cross-section	Children (focussed group discussion)	Independent mobility/ discussion	Local destination	Sex/gender, age + stranger danger and traffic danger	Impact of traffic for both genders and stranger danger, especially for girls. Car escort journey developing unhealthy habits of sedentary living with its associated risks of increased cardiovascular illness.
6	(Fagerholm & Broberg, 2011)/ Sep.-Oct. 2009 n = 35 children (18M/17F)/ 10-11 years old	Finland Cross-section	Parents questionnaire (children and parent) mobility patterns + mobility licences, perceived safety	*Active route (home to destinations)/ GPS + Travel diary for 7-days	Buffer 500-m from home	Sex/gender Land use types	Gender difference shows in the distance and speed children travel actively to the boys' advantages. Land-use type associated with different mobility patterns (increase in urban areas). A high perception of safety from parents in residential areas resulted in a high level of independent mobility. Weekdays and weekends have different mobility patterns in term of proximity (near the home on weekdays) with more time out on weekends.
7	Faulkner, Mitra, Buliung, Fusco, and Stone (2015)/ April2010 – May2011 n = 736 (47% M and 52%F) included in the analysis/ (10-12 years old), grade 5-6	Canada Cross-section from [BEAT] project	Parents questionnaire (child outdoor active play + parents' perception of personal and road the neighbourhood safety)	*Outdoor playing time MVPA/ GPS + Accelerometer for 7-days	Neighbourhood	Sex/gender, age, SES of the neighbourhood (neighbourhood income) + neighbourhood perception (roads, personal safety, accessibility of facility)	Time spent outdoor was significantly associated with (MVPA) and associated with sex/gender, but age didn't play a role in this relation. Parental safety perception plays inverse relation with duration of outdoor play on weekdays. Parental concerns play a barrier role for children outdoor play. Association of outdoor play duration with the perception of safety (stranger danger and traffic safety), differ between weekdays and weekends.

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8	**(Janet E. Loebach & Jason A. Gilliland, 2016)/** During April and May of 2010 and 2011. n = 143 (49M/94F) two groups of 9- 11 and 12-13 years-old)/grade 5 - 8	Canada Cross-section from [STEAM] project	Parents & Children questionnaires	Neighbourhood Activity Space (NAS)/ GPS for 7-days	NAS found within 400, 800 m of home, the second set those found within 1,600 m.	Sex/gender, age, car availability + parents and children environmental perception +neighbourhood types + parental IM licenses to children	IM awarded to a child is associated with parents' perception of neighbourhood safety, and strong predictor for distance travelled. Distance to school predicts active travel. Gender or age was not associated to NAS size. No association between parents' perception of neighbourhood and child spend their time closer to home. Children perception correlate with time spend closer to home but were not as strongly predictive as parental mobility restriction. Neighbourhood types (Residential) was predictor time spend out and (commercial) to distance travelled (over 800 m). smaller home to school distance facilitate active travel and more frequent and distant neighbourhood travel
9	(Helbich et al., 2016)/between December 2008 – April 2009/ n = 97 (60%F) /Aged 6 – 11 years	The Netherlands, Part of [SPACE] project	Road traffic Safety exposures within 100 m buffer width around using GIS	Active trips to school and transport mode of choice GPS for 8-days	Home to school trip	Sex/gender, Age + active trips to school + built environmental variables (land use mix, density, closeness and in between indices) + traffic safety control variables (major roads availability, distance, the proportion of cycling path and road accidents) +street density) weather	Gender is significantly associated with AST. Age is significantly associated with AST. Negative association with the distance. Negative weak association of AST with major road. Traffic safety (exposure to major roads/highways) is negatively correlate with AST. Cycling path availability positively corollate with AST.
10	Lin et al. (2017)/between 2011 and 2012 n = 254 (100M, 133F) and 239 parents/ 8–13 years (mean age of 10.5)	New Zealand from [KITC]	CATI-Parents Survey (perception of neighbourhood cohesion using social cohesion scale, + neighbourhood social connection using intergenerational closure scale and parents' concerns of places that will not let their children go alone)	*Independent mobility /Travel diary for 7-days	Neighbourhood	Sex/gender, age, race/ethnicity, household profile, built environment (street connectivity, destination accessibility)	Distance to school accounted for the effect of neighbourhood cohesion and neighbourhood connect on children independent mobility. IM associated with having an older sibling and limited access to cars. Parents perception is not associated with children independent mobility though concerns of safety which differed by ethnicity (pacific/Asian).
11	Mehdizadeh, Mamdoohi, and Nordfjaern (2017)/-/ 735 parents of (364M, 371F)/ 7-9-year-old in 9 schools of 735	Iran Cross-section	Parents questionnaire	Perceived Walking Time to school (PWTS)/ questionnaires	Environment around school	Sex/gender, demographics, household characteristics (father/mother driving licence, car owner) SES	Perceived walking time to school (10 min) is the maximum threshold where the proportion of active mode started to decrease. Certain demographics: parental age, household income, accessibility to public transport, type of school (public or private) school service status and psychological factors (parents attitude towards walking in dirty, vandalised and unsafe streets), SES of children and household variables were significant predictors of a PWST to eligible school. Sex didn't have a consistent and definite role in active travel.
12	(Nguyen, Borghese, & Janssen, 2018)/ Jan. 2015 and Dec. 2016/ n= 458 (230M/228F)/ aged 10-12	Canada Cross Section	Parents questionnaire (perceived pedestrian safety) + Objective measures of Pedestrian safety	Average of minutes per day of active outdoor play/ Accelerometer + GPS in smartwatch for 7-days	1 km buffer zone around participants home	Sex/gender Race (white, non-white) Family characteristics (single or dual parents' household, number of siblings, household income, parental education, parents' value of outdoor and income) + Pedestrian safety (traffic volume, traffic speed, traffic calming and pedestrian infrastructure + season	No association between pedestrian safety measures and outdoor active play. No association between objective and perceived pedestrian measures. Perceived measures of traffic volume, traffic calming and pedestrian infrastructure were not associated with outdoor active play. Parents perceived moderate to high traffic speeds had higher outdoor active play value than children whose parents perceived low traffic speed. Objective measure of traffic volume, traffic calming and pedestrian infrastructure but not traffic speed was associated with outdoor active play.
13	Noonan, Boddy, Knowles, and Fairclough (2016), n = 194 children/ 9-10 years old Gender not reported	United Kingdom Cross-section	Parents questionnaire on environment perception using NEWS_Y + Children self-reported PA using PAQ-C	IM/ Self-Reported PA	High and low deprived areas	Home environment, Area deprivation, parent's perception walkability Index	Home environment for HD provides more opportunities for sedentary behaviour and less opportunity for PA, less access to bedroom media equipment and greater independent mobility were *strongly associated* with higher PA in HD and MD children, respectively.

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14	Oliver et al. (2015)/ Between 2011 and 2012/ n = 236 (104 M/132) for weekday analyses, n = 210 (91M/119F) children for weekend days analyses. /age 9-13 Age mean 9.8 from 9 schools/grade 5-8	New Zealand Cross-section from [KITC] project	CATI-Parents questionnaire (Perceived parents safety) + measured roads connectivity	*% MVPA/ accelerometer + GPS for 7-days	Buffer of 800-1000 m around school	Sex/gender, age, race/ethnicity (New Zealand European, Maori, Pacific Island, Indian/Asian/Other Ethnicity), SES, Neighbourhood exposures.	On weekdays: Females & access to car accumulated less %MVPA than males. On weekends: Female and ethnicity (Indian, Asian or "other" ethic group) mades less %MVPA. Street connectivity and distance to school were related to the proportion of active trips on both weekends and weekdays. The ratio of high-speed roads associated with %MVPA (weekdays after school). Improved streetscape for active travel was related to %MVP on weekdays. Ethnicity and %MVPA differ by day type. Age, access to cars was negatively associated with %MVPA. Inconclusive evident of socioeconomic association. Parents perceptions of neighbourhood safety Positive association with the proportion of active trips on weekdays.
15	(Oluyomi et al., 2014)/-/ n= 830 and their parents of 4th grade (412M/418F)	United States Cross-section from [T-COPPE] longitudinal project	Parents questionnaire adapted from several surveys including the National Centre for Safe Routes to School Parents Survey, SPAN, (UH-PEAK), NEWS, and EnVivo) Personal safety + Traffic Safety	Walking to school (WTS) /parents' questionnaire of National Safe Route to School. Captured safety of en-route to school, home neighbourhood, and school environment)	Within walking distance of 3.2 km to school from the residential address of students	Race/ethnicity (the majority were Hispanic) car ownership, public assistance) Examined two environments (home neighbourhood and en-route environment to school)	Parents Perception of road safety (higher sidewalk availability, well-maintained sidewalks and safe road crossing) is associated with students more likely to walk to school. On route to school: parents' perception of sidewalk, Speed and amount of traffic and intersection along school route, also associated with WTS Parents reported safe walking to school neighbourhood reported higher children WTS. On Personal safety, parents concern about general neighbourhoods' safety, stray or dangerous animals and availability of adults with whom child can walk en-route
16	(Page, Cooper, Griew, & Jago, 2010)/ between 2006 and 2008 n = 1307 (639 M, 661 F)/ 10-11 years old from 23 schools	United Kingdom Cross-section from [PEACH] longitudinal study	(computerised) children questionnaire perception of the environment (Aesthetics, Safety, Social Norms, Nuisance, constraints, accessibility, minutes of daylight from 3 pm till sunset)	Frequency of participation in outdoor play, exercise and active travel home to school/ questionnaire	School-home	Age, sex/gender, race/ethnicity (white, non-white, but not accounted in analysis), + perception of the environment + the level of deprivation (using Index of Multiple Deprivation (IMD) and derived from seven categories of deprivation + daylight + pubertal status	Boys had a more positive perception of the environment than girls (personal safety and traffic safety) for Local-IM and Area-IM. No gender differences in school travel Boys scores significantly lower in the Constrained scale than girls Physical activity, structured sport and active travel to school weakly significantly correlate to each other. Boys have greater independent mobility scores more time of playing out than boys who played less. Girls with a higher positive perception of neighbourhood correlate to play outdoor often. For boys and girls increased the likelihood of walking or cycling to school was associated with higher levels of local-IM. Distance from home to school is a predictor for both boys and girls
17	Roberts, Knight, Ray, and Saelens (2016)/Sep.-Dec.2014/ n = 144(72M,72F)/7-12 years old (mean age of 9.7 children)/	United States Cross-section	Parents questionnaire (perception of the environment)	Active children (met the 60 min daily PA)/	Metropolitan area	Sex, demographics race/ethnicity (Hispanic/Latino, African American, American, Indian/Alaska Native, Asian American, White, Other) + perceived parents' safety	Parents' better perception of neighbourhood associated with active kids yet reported a high crime rate and being a victim of a crime of their neighbourhood. Parental perception of street barriers associates with children physical activity. Closer proximity to play areas was significantly associated with greater odd of children meeting the 60 min/day play. Gender (male more active) and race disparities associated with active behaviour.
18	(Santos, Pizarro, Mota, & Marques, 2013)/ 2010/2011 n= 354 (156M) of grade 6th, (mean age 11.63) and their parents	Portugal Cross-section from [SALTA] longitudinal study	Parents questionnaire (parental physical activity, and perception (adapted from NEWS and previous studies) + Children questionnaire to derive mobility style	IM/ questionnaire of previous week physical activity based on IPAQ	Local destinations	Age, sex/gender, family demography (parents age, education, parental PA, parents' perception of neighbourhood safety (sidewalk, street safety, fear from strangers, crime and traffic safety).	Parental perception of sidewalks and street safety is associated with IM Parents physical activity was significantly associated with more active children Perception of fear from a stranger, crime and traffic safety was not significantly associated with child IM.
19	(Shokoohi, Hanif, & Dali, 2012)/ Jan.-Feb. 2009 /-/ Grades 3- 5 (48.8%M,51%F)	Iran Cross-section	Parents + children questionnaire on the perception of environmental factors that prevent children from walking to school	Walking to school from parent's survey. Differed the trips from home to school and from school to home.	Home-school	Socioeconomic status (three income groups of parents)	Parents and children with a negative perception of neighbourhood safety tended to use motor vehicles or to escort their children while walking to and from school.

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20	Stark, Frühwirth, and Aschauer (2018)/-/ n = 190 (49%F) from two public schools/ 6-9(10) years old	Austria Cross-section	Parents questionnaire	AIM/ One weekday and one weekend Travel Diary using (KONTIV-format)	School active travel	Parental attitude, Parents and household characteristics	Most influential variables on IM are: Shorter trip distance to school, higher age and parents perceived social safety and traffics. Working status of parents. The type of school (all-day/half-day primary school) is relevant. Parental attitude (Promoters, Pragmatists and Protectors) strongly influence the degree of AIM.
21	Stephanie H. Kneeshaw-Price et al. (2015)/between Sep. 2007 and Jan 2009 n = 145 (71M, 74F) / 6–11-years-old	United States Cohort study [NIK]	-Police report crime + parents' prior crime victimisation survey + parents' perception (stranger danger, general crime and disorder)	MVPA/ accelerometer for 7-days	Census block	Age, race/ethnicity (non-Hispanic white, Hispanic, and non-Hispanic non-white) Household income, NEWS + collective efficacy	Lower children's physical activity was associated with more neighbourhood crime but not with Parents perception of stranger danger, disorder and neighbourhood safety. Lower MVPA didn't account for race/ethnicity as the majority was non-Hispanic (white).
22	Suminski, Robson, May, Blair, and Orsega-Smith (2018)/ T1 n = 2108/50.5%F/ 5-11 years-old	United States longitudinal study	Measured crime using CRI Index (a higher number than 100 indicate a greater chance of crime) for each zip code	Body Mass Index/(BMI) score/ measured at baseline and three years later	Zipcode	(BMIz scores, Sex/gender, family characteristics (income, education Race/Ethnicity White, African American, Hispanic, Asian)	Actual, neighbourhood-level crime predicts changes in BMIz scores in white children This relationship varies as a function of race/ethnicity (i.e. in White children but not African American, Hispanic or Asian American)
23	Timperio, Crawford, Telford, and Salmon (2004)/ n = 291 (150M/141F)/ Aged 5–6 years n = 919 (424M/495F)/ aged 10–12 years from 19 primary schools	Australia Cross-section	Parents & Children Survey Compared parents to own children (10-12) view	Frequency of walking and cycling/ parents survey	Local destinations	Sex, Family background: the first language is spoken, marital status, parents' education (SES), cars' ownerships, siblings	Parents' perception of the neighbourhood is *associated* with a lower likelihood of walking or cycling (girls in particular). Children's perceive parents' perception of their local neighbourhood safety more negative than their own. Children are reporting less concern about heavy traffic, stranger dangers and road safety and lack of parks or sports ground than parents. Age & SES is associated positively with the frequency of trips made to destinations. Sex is not associated with the frequency of active trips for the young group, but Boys are more still active than girls in the 10 – 12 years old.
24	(Tung, Ng, Chin, & Mohd Taib, 2016) n = 256 (42%M/58%F and their parents aged 9-12 years	Malaysia Cross-section	Parents perception using NEWS Constrained behaviour of parents based on (Carver et al 2010)	Children PA level 7-days questionnaire	-	Sociodemographic (age, parents' gender and ethnicity (54%Malay, 35.2 Chinese, and 10 were Indian), highest education level, parent's occupation and monthly income) + parents' perception+ children PA	Sex/gender association with the level of PA (boys accumulate more than girls) Correlates found between parental perception of neighbourhood safety and constrained behaviour of children active play Land use mix (access) was correlated positively with higher PA level
25	(van den Berg, Waygood, van de Craats, & Kemperman, 2020)/ Fall of 2018/(315M/341F)/ n= 660 children and parents/ aged 7 – 12/ grade 5-8 (mean age 9.5)	The Netherland Cross-section	Parents survey for perceived pedestrian safety Objective measures of Pedestrian safety	Travel mode to school/ children survey (at school)	Participants were of Home-school Within 1 km distance	Age, Sex/gender Household (income, car ownership), weather, street connectivity	Parental perceptions are related to the child's age, income, perception of neighbourhood infrastructure, travel distance and social cohesion. Parents that are less concerned let their children travel actively
26	Vonderwalde, Cox, Williams, Borghese, and Ian Janssena (2019)/ between Jan. 2015 and Dec.2016/ n = 387(185M, 182F)/ 10–13 years old (mean age 11.5)	Canada Longitudinal from [Active Play Study]	Measured Crime (person, Property) 24 months before measures of active behaviour	Active transportation/ GIS loggers	Crime in 1 km distance to participants home	Sex/gender, age, and race (85% white) family profile (single or dual-parent, parent income education), walkability index. Season change	Children living in neighbourhoods in the highest neighbourhood crime rate quartile engaged in significantly more active transportation than children living in neighbourhoods in the lowest neighbourhood crime rate quartile and persisted after adjustment for several individual, family, and environmental covariates. Association of AS and crime against a person but not against a property.
27	**Villanueva et al. (2012)/** July – December 2007/ n = 926 (463M,463F)/ 10-12 years old included in the analysis.	Australia Cross-section from [TREK]	Parent Child questionnaires	Activity Space/ steps count Pedometer + mapping activity for 7-days	within 800 and 1600 meter of child's home high and low walkable areas	Age, sex/gender + maternal education + School-specific walkability level	Children in high walkable neighbourhood's schools had larger AS. Girls had smaller AS for if parents perceived living on a busy road. Utilitarian destinations were associated with a smaller AS. Sex is associated with AS areas and daily pedometer. Boys' and girls' own confidence in travelling independently was positively associated with covering larger AS areas. Despite some environmental barriers, Individual and social-cultural factors likely to encourage children to roam in their local environments.

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28	(Waygood & Susilo, 2015)/ 2006 survey/ n = Roughly 31,000 households' participants (52% boys) aged 10-11 years	Scotland Cross-section	Parents survey	Walking to school	Home to school	Sex/gender Family background +built environment, deprivation index	Cars availability is negatively associate with walking to school Distance to school was strong determinate for children walking to school Parental perception Good local shops is positively associated with children active travel to school and slow traffic or safe was negatively associated variable,
29	Zhu and Lee (2008)/-/ Children from 73 elementary school/-/	United States Cross-section	Measured Crime Geocoded 8 major crimes rate	Potential walkers/ neighbourhood-level walkability index	School attendance areas	Race/Ethnicity (Hispanic, non-Hispanic, and White), Poverty + Neighbourhood-level walkability + Neighbourhood-level safety SWI	Poverty associated with negative conditions and low perceived safety. Ethnicity associated with increased crime, traffic danger and poor safety. Unsafe neighbourhoods and poor street conditions may influence not only children's school travels but also their play activities and the overall physical activities of all residents.

Notes: /-/ = data was not reported, M = males, F = Female, BMI = Body Mass Index, IM = independent mobility, AST = active school transport, PA = physical activity, AS = activity space, SES = socioeconomic status, HD = high deprivation, MD = medium deprivation, WTS = walking to school, NAS = Neighbourhood Activity Space, CLAN = Children Living in Active Neighbourhoods, SES = Socioeconomic status, MVPA = Medium-to vigorous Physical Activity, SPEEDY = Sport, Physical activity and Easting Behaviour Environmental Determinants in Young People, BEAT = Built Environment and Active Transport, GPS = Global Positioning System, STEAM = Spatio-Temporal Exposure and Activity Monitoring, KITC = kids in the City, PAQ-C = Physical Activity Questionnaire for Older Children, NIK = Neighbourhood Impact on Kids, CRI = Crime Risk Index (measured crime using actual crime statistics) TREK = Travel Environment and Kids Project, CATI = Computer-aided Telephone Interview, KIC = Kinds in the City, T-COPPE survey = Texas Childhood Obesity Prevention Policy Education project, IMD = Index of Multiple Deprivation is a composite score based on seven categories of deprivation (income, employment, health and disability, education skills and training, housing and geographical access to service), PEACH = Personal and Environmental Association With Child's Health, SPAN = School Physical Activity and Nutrition, UH-PEAK = Urban Hispanic Perceptions of Environment and Activity Among Kids, En Vivo = TV reduction intervention study, NEWS = Neighbourhood Environment Walkability Scale, NEWS-Y = Neighbourhood Environment Walkability Scale for Youth used to assess parental perceptions of neighbourhood design, SALTA = Environmental Support for Leisure and Active Transport, KONTIV = format of travel diary survey for non-home activity patterns, GIS = geographic information systems, IPAQ questionnaire = International Physical Activity Questionnaire, Local-IM = destinations of best friend's house, school, local shops and park or playground, Area-IM = destinations of swimming pool, library, cinema, arcade, bus stop, sports and shopping centre, SWI = School Walkability Index derived from network connectivity and traffic volume, Neighbourhood-level walkability index derived from an (estimate of potential walkers, pedestrian facilities, residential density, street connectivity, land use mix), Neighbourhood-level safety = derived from (traffic danger and the crime rate in a year), TREK = Travel Environment and Kids

Studies denoted with * = Study measures and analysis accounted for temporal changes (weekend/weekdays or outside school, i.e. before and after school hours).

[1] = Objective road measures using GIS were: street network total length, local road index, No of intersections, the total length of walking track, no of speed humps, no of traffic /pedestrian lights, no of barriers

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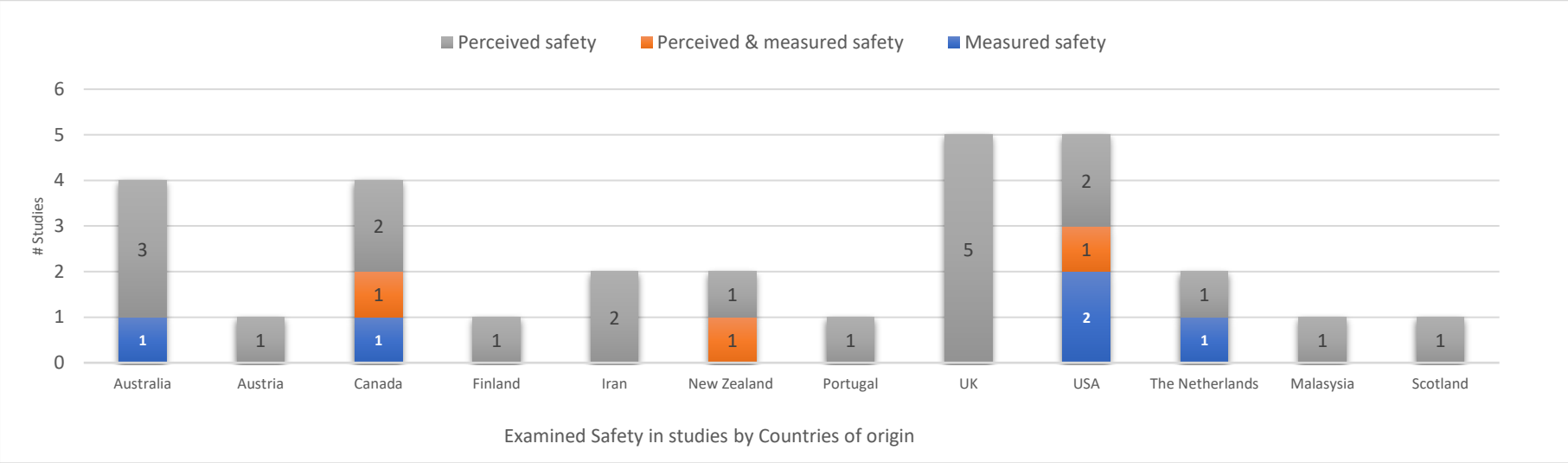


Diagram 1. Country of origin of the studies included in this review (with examined neighbourhood safety types)

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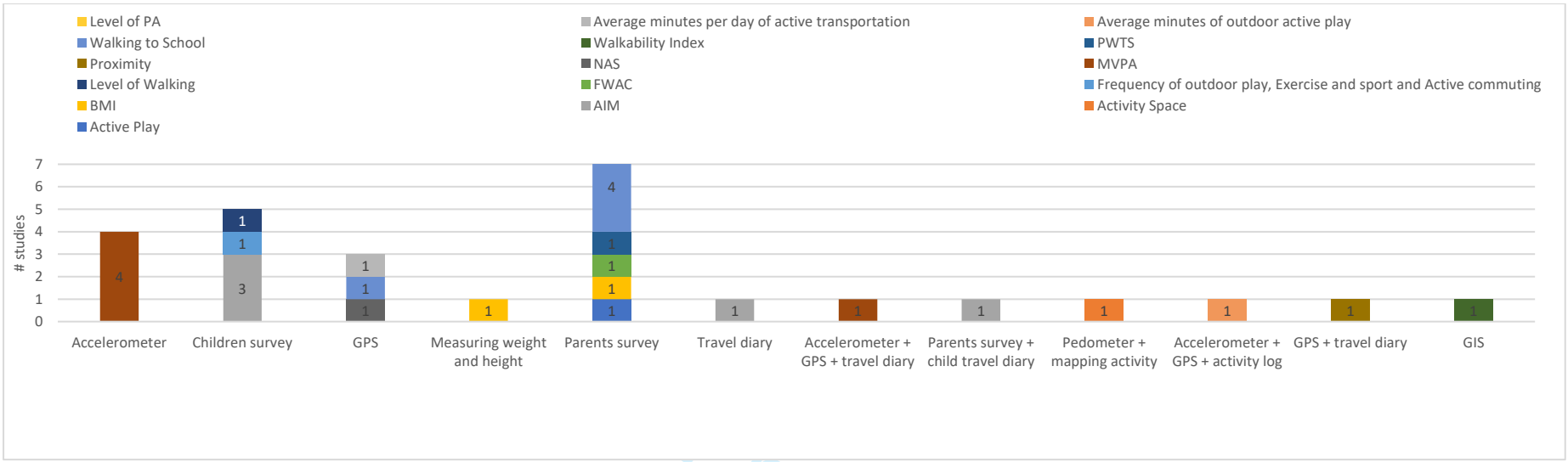


Diagram 2. Variances in measurement methods of children active mobility behaviour and the outcome of measures in 29 studies. Abbreviations: PA = physical activity, AIM = active independent mobility, BMI = body mass index, FWAC = frequent walking and cycling, MVPA = moderate-to-vigorous physical activity, NAS = neighbourhood activity space, PWTS = perceived walking to school, GPS = global positioning system, GIS = geographic information system

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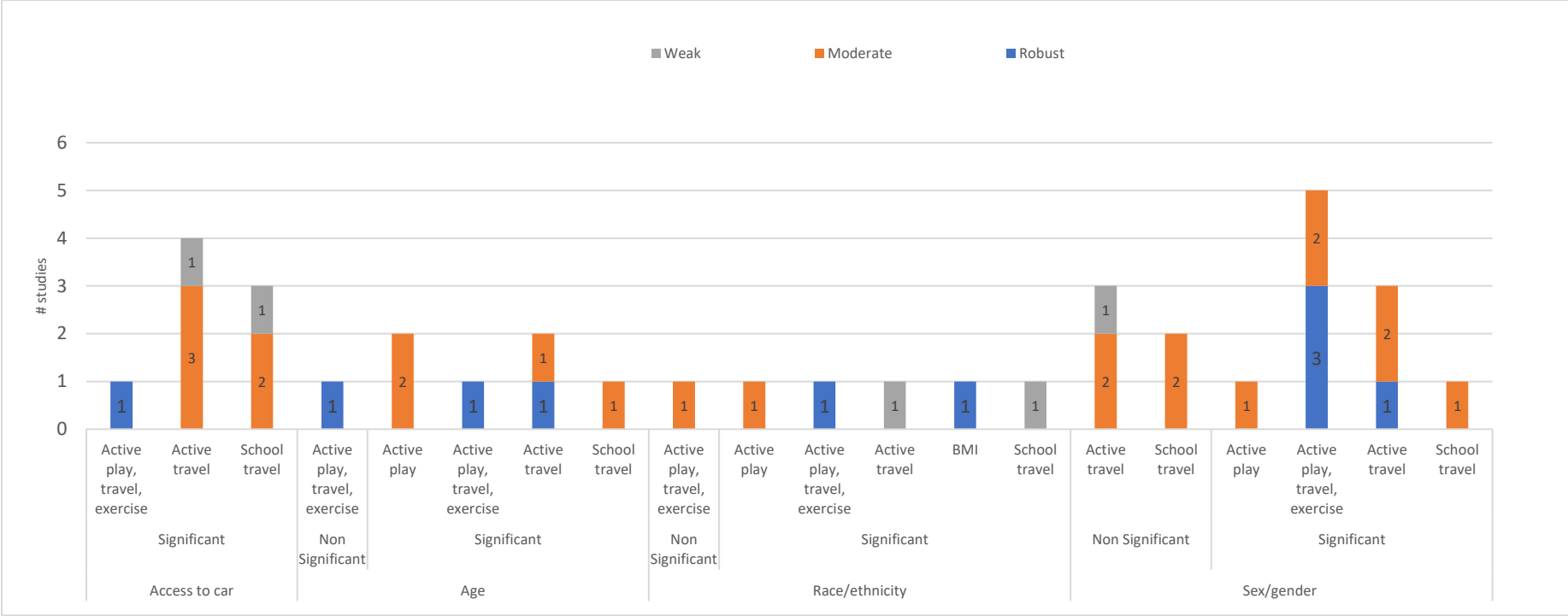


Diagram 3. Findings of Individual and family layers of association to children various type of active behaviour in studies examined neighbourhood safety. Studies were re-grouped each by the examined active behaviour into four categories of (Active travel, active play, school travel and active play, travel and exercise) and as appraised by the methodological quality assessment. x-Axis depicted variable addressed at the individual and family level and the examined type of child active behaviour. y-Axis represents the number of studies accumulated.

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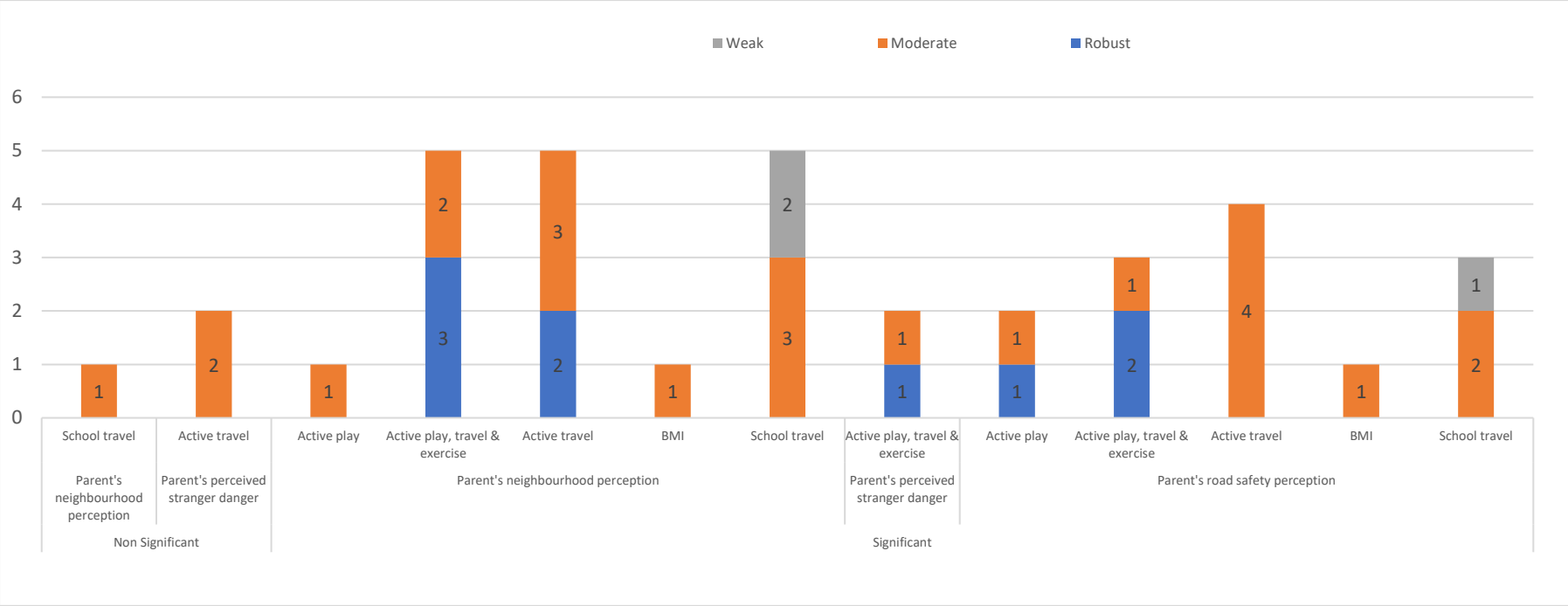


Diagram 4. Findings of perceived safety layers of association to children various type of active behaviour. Studies were re-grouped by active behaviour into four categories of (Active travel, active play, school travel and active play, travel and exercise) and as appraised by the methodological quality assessment. x-Axis depicted variable addressed at the neighbourhood perceived level and the examined type of child active behaviour. y-Axis represents the number of studies

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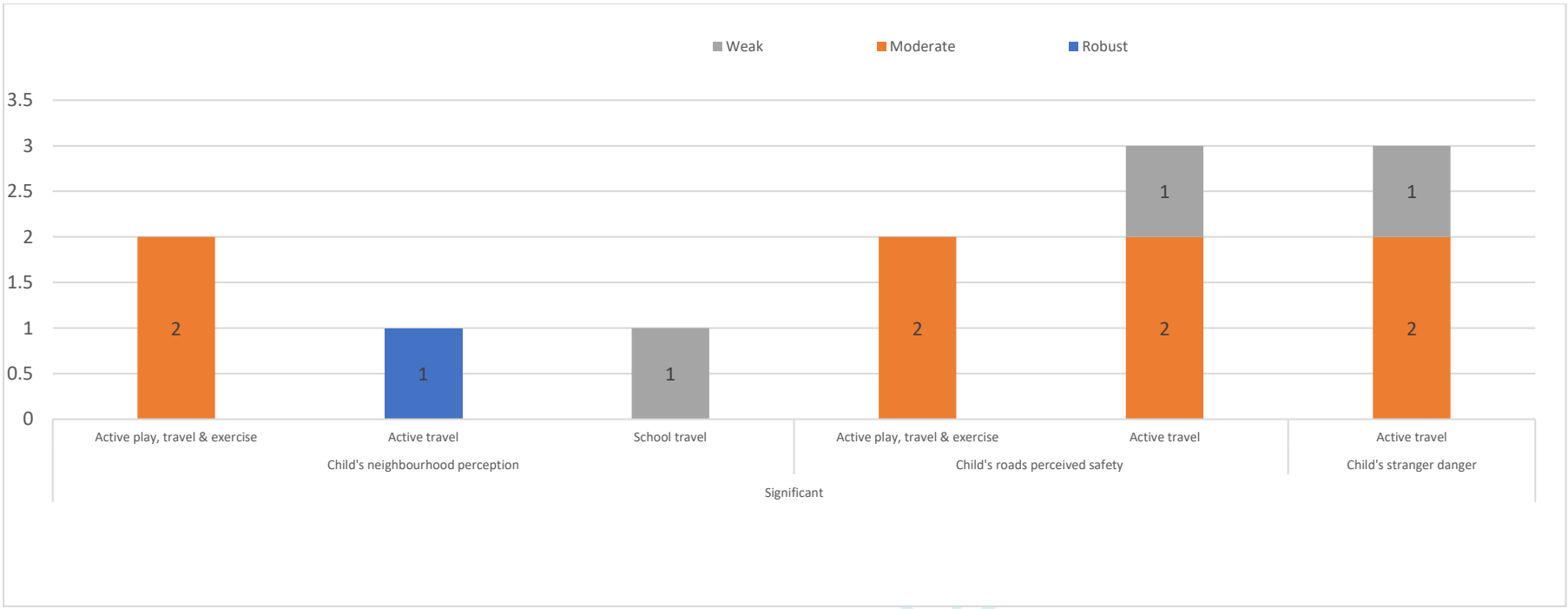


Diagram 5. Findings of perceived safety layers of association to children various type of active behaviour. Studies were re-grouped by active behaviour into four categories of (Active travel, active play, school travel and active play, travel and exercise) and as appraised by the methodological quality assessment. x-Axis depicted variable addressed at the neighbourhood perceived level and the examined type of child active behaviour. y-Axis represents the number of studies

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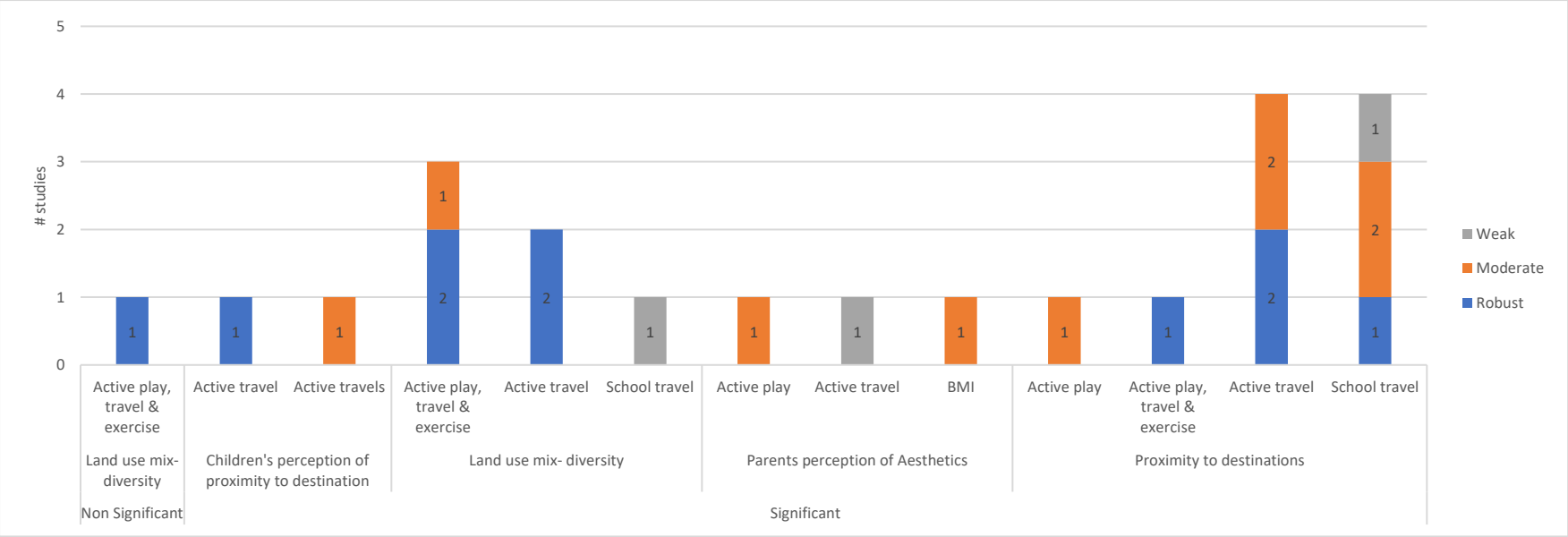


Diagram 6. Elements in the physical environment associated with types of child active behaviour. Studies were re-grouped by active behaviour into four categories of (Active travel, active play, school travel and active play, travel and exercise) and as appraised by the methodological quality assessment. x-Axis depicted variable addressed at the neighbourhood level and the examined type of child active behaviour. y-Axis represents the number of studies

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Table 2. Findings of influences across the socioecological levels and children shapes of outdoor active mobility behaviour.

Effective domains		**+ - 0**	**Correlation with children active mobility**
Individual (child)			
P_Survey Sex: Female	Carver et al. (2010)	+	avoidance behaviour[1]
	(Oliver et al., 2015)	–	%MVPA (F) weekdays
	(Timperio et al., 2004)	–	walking/cycling, after dark (F) of (10-12 YOLD)
		0	5- 6-year-old age group
	(Faulkner et al., 2015)	–	time outdoor (weekends and weekdays)
	(Villanueva et al., 2012)	–	activity Space (F)
	(Carver et al., 2014) greater land use mix (residence/retail outlets, sports facility) at T2	+	walking and cycling to school (F)
Sex: Male	Carver et al. (2010)	+	made more active trips (M)
	(Roberts et al., 2016)	+	active behaviour
	(Fagerholm & Broberg, 2011)	+	Distance travelled, and speed children travel
	(Villanueva et al., 2012)	+	walking/cycling, after dark (M) of (10-12 YOLD) Steps count
	(Mehdizadeh et al., 2017)	0	Perceived walking to school
	(Helbich et al., 2016)	0	Active school travel
	(Carver et al., 2014) being allowed by parents to play outside	+	odd of walking/cycling independently to school
	(Tung et al., 2016)	+	PA (reported)
C_Survey Gender	(Alton et al., 2007)	0	walking trips
	(Page et al., 2010) playing and taking part in structured sport	+	local-IM: (M) more than F
	(Page et al., 2010)	0	Active commuting to school
Age Adult Survey	(J. Loebach & J. Gilliland, 2016)	+	time spend outdoor, distance & IM (older).
	(Stark et al., 2018)	+	IM with older children
	(Timperio et al., 2004) of age 5-6 comparing to 10-12	+	walking or cycling with age in particular for (M).
	(Oliver et al., 2016)	+	%MVPA weekdays
	(Faulkner et al., 2015)	0	MVPA
	(Helbich et al., 2016)	+	ATS
	(van den Berg et al., 2020)	+	active school travel
Parents/Family background characteristics			
SES	(Timperio et al., 2004) (boys) 10 -12	+	Frequency of walking or cycling to public transport (active trips) in high SES more than in medium SES.
	(Timperio et al., 2004) 10- 12 F	+	walked and cycled more to school in high SES.
	(Timperio et al., 2004) age 5-6 years old) M	+	in high SES
	(Timperio et al., 2004) age 5-6 years old) F	0	association to SES
	(Tung et al., 2016) high SES	+	PA
	(Timperio et al., 2004)	0	concerns of stranger danger (F)
	(Mehdizadeh et al., 2017) mothers with driving liscence		Reduced active travel to school
	(Oliver et al., 2016)	–	(Inconclusive)
	(Noonan et al., 2016) in high deprivation	+	BMI level
		+	self-reported PA
C_Survey And SES	(Timperio et al., 2004) concerns of traffic in (M)	x	found in children in low SES more than children in high SES
Poverty	(Zhu & Lee, 2008)	–	safety & walking
Household income P_survey	Mehdizadeh et al., 2017	+	perceived walking to school
	(van den Berg et al., 2020)	x	increased income & increased perception of safety
Mother driving licence	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	–	active travel
Parental age	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	+	perceived walking to school
Employments Status (mother)	(Stark et al., 2018)	+	IM licence
Home environment (access to Media)	(Noonan et al., 2016)	x	in High Deprivation
Having older siblings	(Lin et al., 2017)	+	IM
	(Carver et al., 2014)	+	IM
P_Survey Cars	(Alton et al., 2007)	–	level of walking
	(Davis & Jones, 1996)	–	IM

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ownership/ access	(Lin et al., 2017)	–	IM
	(Mehdizadeh et al., 2017)	–	active travel
	(Oliver et al., 2015)	–	# of active trips (weekdays)
	(Timperio et al., 2004) Parent of 5-6 years old (F) who owned more than one car associated with	–	less likely of walking or cycling to destinations
	(Waygood & Susilo, 2015)	–	Walking to school
	(Carver et al., 2014)	–	odd of boys walking and cycling to school at T2
C_Survey rooms in the house	(Alton et al., 2007)	–	(significant only for 2.1%)
Parental attitude towards walking	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	+	short perceived walking time to school
Parental Physical Activity	(Santos et al., 2013) Parental physical activity	+	independent mobility
Ethnicity	(Alton et al., 2007)	–	level of walking in minority
	*(Oliver et al., 2016)	+	%MVA Pacific, European and Maori accumulates (weekdays only). %MVA Indian/Asian/others (weekend days) than their counterparts/Pacific on weekdays
	(Suminski et al., 2018) crime risk index	–	BMI (white children)
	(Roberts et al., 2016)	+	in minority
	(Zhu & Lee, 2008) (Hispanic)	+	in crime & traffics
Access to public transport	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	–	active travel
Attending Public School	(Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	–	perceived walking to school
Neighbourhood, Perceived (Personal and road Safety)			
P_Survey low personal safety perception	(Fagerholm & Broberg, 2011)	–	Independent mobility after dark and by distance (only F)
	(Stark et al., 2018)	–	IM
	(Timperio et al., 2004)	–	frequency of walking
	(Mehdizadeh et al., 2017)	–	activity space after 10 min walking
	(J. Loebach & J. Gilliland, 2016)	–	time outdoor and travelled distance
	(Faulkner et al., 2015)	–	outdoor playtime (outdoor plat and MVPA)
	(Roberts et al., 2016)	+	active children
	(Villanueva et al., 2012)	–	Activity space F
	(Oliver et al., 2016)	–	active trips (weekdays)
	(Noonan et al., 2016)	–	outdoor play in MD
	(Carver et al., 2010)	–	active transport (M/F) in 10 -11 years old
	Constrained behaviour (avoidance and Defensive behaviours)	–	MVPA (M) weekends
	(Lin et al., 2017)	0	IM (though parents had concern of safety)
	(Oluyomi et al., 2014)	–	walking to school general neighbourhoods' safety, stray or dangerous animals and availability of adults
	(Carver et al., 2014)	–	walking and cycling to school at T1
	(Shokoohi et al., 2012)	–	walking to school
	(van den Berg et al., 2020)	–	children travel actively
	(Waygood & Susilo, 2015)	–	School active travel
	(Tung et al., 2016)	–	PA
C_Survey Low perceived safety	(Villanueva et al., 2012)	-	activity space
	(Janet E. Loebach & Jason A. Gilliland, 2016)	-	time spend close to home
	(Shokoohi et al., 2012)	–	walking to school
	(Page et al., 2010) positive perception	-	playing out for girls
	(Page et al., 2010) positive perception of the environment	-	greater IM
	(Page et al., 2010) exercising or doing sport every day associated	-	local-IM (boys)
P_Survey Perceived Stranger danger	(Faulkner et al., 2015)	+	on weekdays
	(Lin et al., 2017)	0	though parents have safety concerns
	(Stephanie H. Kneeshaw-Price et al., 2015)	–	MVPA
	(Santos et al., 2013)	0	IM
C-Survey Perceived Stranger danger	(Alton et al., 2007)	–	less walking
	(Davis & Jones, 1996)	–	in particular (F)
	(Timperio et al., 2004)	–	low SES (F) than high SES

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P_Survey Increased social Cohesion/Social Norms	(Lin et al., 2017)	+	IM
	(van den Berg et al., 2020) increase perception of social cohesion	+	parents perceived safety
C-Survey Neighbourhood high aesthetics / Nuisance Social Norms	(Noonan et al., 2016) Neighbourhood aesthetics	0	self-reported PA
	(Noonan et al., 2016) Neighbourhood aesthetics	–	BMI
	(Page et al., 2010) higher score of social norms	+	frequency of outdoor play
P_Survey Perception of Road Safety Low perceived road safety	(Noonan et al., 2016)	–	independent mobility
	(Stark et al., 2018)	–	independent mobility (general traffic safety)
	(Timperio et al., 2004) Parents believe of (heavy traffic) in (5-6 years old) boys	+	Frequency of walking (sidewalks or bike lanes, safe crossing
	(Timperio et al., 2004) Parents believe (for M) of 10 – 12 years old of no lights or crossing	–	walking and cycling
	(Timperio et al., 2004) The parental belief of F need to cross many roads to reach play area 10 - 12	–	likelihood of walking or cycling
	(Villanueva et al., 2012) safe neighbourhood crossing	–	activity space in particular girls,
	(Faulkner et al., 2015) Fast drivers (weekdays),	–	MVPA (active play)
	(Roberts et al., 2016) if parents perceived lack of sidewalk and signals on a busy street	–	active children
	(Santos et al., 2013) perceived sidewalk and street safety	–	independent mobility
	(Oluyomi et al., 2014) road safety in the home environment of higher sidewalk availability, Speed, amount of traffic, intersection safety, road crossing problem and availability of crossing guard	–	Likelihood of walking to school
	(Carver et al., 2014) concern about traffic	-	likely for boys and girls to cycle independently
	(Tung et al., 2016) traffic hazards (perceived_)	–	PA
	(Waygood & Susilo, 2015) slow or safe traffic	+	Walking to school
	(Davis & Jones, 1996) traffic	–	Independent mobility
	(Nguyen et al., 2018) parents' perception of high or moderate traffic speed higher	+	outdoor active play
Perceived increased street connectivity	(van den Berg et al., 2020) increased connectivity	+	parental perception of safety
Destination accessibility	(Oliver et al., 2016)	+ –	the proportion of trips in active mode (weekdays) %MVPA (weekend)
Improved streetscape (measured)	(Villanueva et al., 2012)	+	Activity space
C_Survey Road safety	(Alton et al., 2007)	–	road safety and heavy traffic and level of walking
	(Villanueva et al., 2012)	–	activity space
	(Davis & Jones, 1996)	–	IM in particular F (traffic danger)
	(Timperio et al., 2004)	–	low SES (M) than in high SES
	(Page et al., 2010)	+	local and Area-IM with traffic safety (for girls) positive perception of road safety (+ IM) for girls
Neighbourhood's Safety –Perception of other elements related to safety in the Physical environment			
P_Survey Perceived **lack of leisure facility**, Parks and sport ground	(Timperio et al., 2004)	–	likely to walking &cycling three times a week (10-12) F
	(Alton et al., 2007)	–	with a crossing of no light or crossings
	(Faulkner et al., 2015)	0	duration of playout
Land use diversity: Residential/ Commercial/ Industrial/	(Fagerholm & Broberg, 2011)	+	mobility mainly in urban structure (residential, commercial and traffic areas)
	(Villanueva et al., 2012)	–	AS in utilitarian destination (within 800 m)
	(Janet E. Loebach & Jason A. Gilliland, 2016)	+	Time spend close to home (residential) and (+) distance travelled and time spends in commercial (beyond 800m).
		–	mobility in agricultural and industrial (on 400-800m buffer)
	(Faulkner et al., 2015)	+	time playing outside (Residential &Commercial)
	*(Oliver et al., 2016)	0	%MVPA

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	(Helbich et al., 2016)	0	Active school travel
	(Carver et al., 2014) (land use mix of residential, retail shops, sports centre)	+	walking/cycling independently to school (F)
Land use mix (access)	(Tung et al., 2016)	+	PA
P_Survey proximity to destinations	(Stark et al., 2018)	–	AIM
	*(Fagerholm & Broberg, 2011)	–	trajectory (with increased distance),
	*(Oliver et al., 2016) distance to school	–	the proportion of trips made in active mode, + distance to school
	(Lin et al., 2017)	–	AIM
	(Mehdizadeh et al., 2017) perceptiom of more than 10 min walking	–	perceived walking to school distance and active travel
	(Roberts et al., 2016)	+	to play area, active behaviour
	(Janet E. Loebach & Jason A. Gilliland, 2016)	–	majority of time spent closer to home (400 m buffer around the home)
		–	school travel and neighbourhood travel
	(Helbich et al., 2016)	–	AST
	(Waygood & Susilo, 2015)	–	Walking to school
	(van den Berg et al., 2020)	–	school active travel
C_Survey proximity	(Janet E. Loebach & Jason A. Gilliland, 2016)	+	distance travelled and activity space near home
	(Page et al., 2010)	–	longer route and active commuting to school (for both boys and girls Active commuting
P_Survey perception of limited access to public transportation	(Timperio et al., 2004)	– –	likely of walking or cycling F of 5-6-year-old likely of walking or cycling F of 10-12-year-old
P_Survey Availability of Neighbourhood Amenities	(J. Loebach & J. Gilliland, 2016)	+	neighbourhood activity space
P_Survey High walkable neighbourhood	(Villanueva et al., 2012)	+	activity space
P_Survey High Neighbourhood aesthetics	(Roberts et al., 2016)	+	active children
	(Noonan et al., 2016)	–	BMI z-score and waist circumference
	Mehdizadeh et al., 2017	+	Perceived walking to school
P_Survey Owning a dog	(Timperio et al., 2004)	+	frequency of walking
C-Survey Accessibility to destination	(Timperio et al., 2004) child believe of no parks access to parks (F/M)	–	walking and cycling in 10-12 years old (M)
	(Page et al., 2010) easy access to range pf destination (Page et al., 2010) greater perceived accessibility	+ +	taking part in structured exercise/sport every day Active commuting to school
Neighbourhood– Measured safety			
Measured safety from crime (high level of actual crime)	(Stephanie H. Kneeshaw-Price et al., 2015)	–	MVPA
	(Vonderwalde, Cox, Williams, Borghese, & Janssen, 2019)	+	active transportation in High crime areas
	(Zhu & Lee, 2008)	–	walkability
	(Suminski et al., 2018)	+	BMI z-score
Road safety	(Nguyen et al., 2018) children from the highest traffic volume	+	outdoor active play
	(Carver et al., 2014) the proportion of main roads	–	odd of walking and cycling (F) walking and cycling to school independently
	*(Oliver et al., 2015) ratio of high-speed roads around schools (weekdays)	–	%MVPA
	street connectivity	+	the proportion of trips made in active mode (on weekend and weekdays)
	(Helbich et al., 2016) exposures to major roads/highway Availability of cyclist path	– +	AST AST
	(Carver et al., 2008)* measured road environment	0	Likelihood of making at least 7 walking and cycling active trips per week
Perceived and measured road safety	(Nguyen et al., 2018) perceived road safety	x	measured road safety

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Perceived and measured personal safety	(Stephanie H. Kneeshaw-Price et al., 2015) Measured crime	0	with parents' perception
Parents to children perceived safety	(Timperio et al., 2004)		parent of 10 – 12 perception was more negative than their own children

Note: Association identified between safety and active mobility behaviour: (+) positive association, (-) negative association, (0) Non-Associated/Not Significant difference, x = association not to active behaviour. Abbreviation: P_Survey = studies examined the perception of safety among parents; C_Survey = studies examined the perception of safety among children. M = male (boys), F = female (girls), IM = independent mobility, MBI – body mass index, SES = socioeconomic status, HD = high depreciation, MD = Medium Deprived, CRI = Crime Risk Index, BMI Body Mass Index, SES = Socioeconomic level, * = behaviour show significant correlate on Weekdays, ** = Behaviour show on weekend, MVPA = Moderate to vigorous physical activity. [1] "Avoidance behaviour is where no further engagement in habits and activities due to perceived risk, e.g. parents driving children to school instead of walking or cycling., [2] Defensive behaviour is where habits are altered in an attempt to reduce perceived risk, e.g. parental accompaniment to children while walking to school", Ferraro, 1995

Search Terms used

(1) **target population** using terms such as "Children" "youth" or "kids", (2) **exposures of social neighbourhood** "Perceived safety from crime", "Social neighbourhood", "Neighbourhood safety", "Measured crime", "Fear of crime", or "Perception of crime", "road safety", "traffic safety". (3) **moderators and mediators** of "Parents perception of safety from crime" "parents and children perception of safety from crime" or "children perception" "age 5-11" "gender", "ethnicity". (4) The **spatiotemporal** aspects: "spatiotemporal measures, Geographic Information Systems GIS, time geography", "space and time geography" "Objective measures" or "GPS". (5) the **outcome** such as "Active Mobility", "Free play", Medium to Vigorous Physical Activity". "Outside school active behaviour". Combination of at least three keywords was alternated to find the possible variety of papers related that could be scanned and retrieved.

Search strategy in Scopus

Search by document type (ALL), Search in Article title, Abstract, Keywords), Access Type (ALL), Limit is (Published all years). The source type is articles. Language is in English. Keywords (Children OR kids) AND (active behaviour OR "physical activity " OR "spatial measures" OR "GPS " OR "active mobility" OR "spatiotemporal measures ") AND (Safety "crime") OR ("road safety") AND ("neighbourhood").

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Table 3. Studies re-grouped output measure of active behaviour to enable synthesis of evidence

#	citation	Neighbourhood Safety	Active Type	Re-Grouped Active Mobility Behaviour
1	(Alton et al., 2007)	Perceived	Level of Walking	Active travel
2	(Carver et al., 2010)	Perceived	MVPA	Active play, travel & exercise
3	(Carver et al., 2014)	Perceived	AIM	School travel
4	(Carver et al., 2008)			
5	(Davis & Jones, 1996)	Perceived	AIM	Active travel
6	(Fagerholm & Broberg, 2011)	Perceived	Proximity	Active travel
7	(Faulkner et al., 2015)	Perceived	MVPA	Active play, travel & exercise
8	(Helbich et al., 2016; Oliver et al., 2015)	Perceived	MVPA	Active play, travel & exercise
9	(Janet E Loebach & Jason A Gilliland, 2016)	Perceived	NAS	Active travel
10	(Lin et al., 2017)	Perceived	AIM	Active travel
11	(Mehdizadeh et al., 2017)	Perceived	PWTS	School travel
12	(Noonan et al., 2016)	Perceived	BMI	BMI
13	(Nguyen et al., 2018)	Perceived & measured	Average minutes of active outdoor play	Active play
14	(Oliver et al., 2015)	Perceived & measured	MVPA	Active play, travel & exercise
15	(Oluyomi et al., 2014)	Perceived	Walking to School	School travel
16	(Page, Cooper, Griew, Davis, & Hillsdon, 2009)	Perceived	Frequency of outdoor play, Exercise and sport and Active commuting	Active play, travel & exercise
17	(Stephanie H. Kneeshaw-Price et al., 2015)	Perceived & measured	MVPA	Active play, travel & exercise
18	(Roberts et al., 2016)	Perceived	Active Play	Active play
19	(Stark et al., 2018)	Perceived	AIM	Active travel
20	(Suminski et al., 2018)	Measured	BMI	BMI
21	(Shokoohi et al., 2012)	Perceived	Walking to school	School travel
22	(Santos et al., 2013)	Perceived	AIM	Active travel
23	(Timperio et al., 2004)	Perceived	FWAC	Active travel
24	(Tung et al., 2016)	Perceived	MVPA	Active play, travel & exercise
25	(van den Berg et al., 2020)	Perceived	Walking to School	School travel
26	(Villanueva et al., 2012)	Perceived	Activity Space	Active play, travel & exercise
27	(Vonderwalde, Cox, Williams, Borghese, & Ian Janssena, 2019)	Measured	Average minutes per day of active transportation	Active travel
28	(Waygood & Susilo, 2015)	Perceived	Walking to school	School travel
29	(Zhu & Lee, 2008)	Measured	Walkability Index	Active travel

Notes: we categorised each study as per the active behaviour addressed: A study addressed total MVPA = total active travel, play and exercise. If walking to school = Active school travel. If addressed frequency of walking or cycling or independent mobility travel (IM) = active travel, and when addressed active playing behaviour = active play, if addressed BMI = BMI.

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Table 4. Quality and methodological measure assessments and output of strength per study.

Study citation	Study Objectives, design, target population, random sampling (0.25 each)	study participants, inclusion/exclusion, study population, participants recruitments (0.25 each, total 1)	Response rate, data collection, data sources, missing data. (0.25 each, total 1)	Did the active behaviour data collection was objectively measured? (0.25 for each method, total 1 point,)	Did the measure of active behaviour account for temporal characteristics? (0.5 point)	Has the perceived safety measured the temporal characteristics? Or has measured safety used geocoded data in actual crime for personal safety or in road safety? (0.5 point)	Did the study delineate the exposure area "neighbourhood" objectively (0.5 point)	Did the study evidence Accounted For spatiotemporal behaviour in output (1 point)	Was it clearly described the statistical method and to assess significance association or did the study describe the method of spatial analysis? (0.5)	Did the study account for the four cofounders (age, sex, ethnicity and family characteristics) (0.25 each, total 1 point)	Quality score total	%	overall rating of evidence Quality
(Alton et al., 2007)	1	0.75	0.5	0.25	0	0	0	0	0.5	1	4	50.0	Moderate
(Carver et al., 2010)	1	1	0.75	0.5	0.5	0.5	0	1	0.5	0.5	6.25	78.1	Robust
(Carver et al., 2014)	1	1	0.75	0.25	0	0	0.5	0	0.5	0.75	4.75	59.4	Moderate
(Carver et al., 2008)	1	1	0.75	0.25	0	0	0.5	0	0.5	0.75	4.75	59.4	Moderate
(Davis & Jones, 1996)	1	0.5	0.25	0.25	0	0	0	0	N/A	0.5	2.5	35.7	Weak
(Fagerholm & Broberg, 2011)	1	0.75	1	1	0.5	0	0.5	0.5	0.5	0.5	6.25	78.1	Robust
(Faulkner et al., 2015)	1	1	1	0.5	0.5	0	0	0.5	0.5	0.75	5.75	71.9	Robust
(Lin et al., 2017)	1	0.5	0.75	0.25	0	0	0	0	0.5	1	4	50.0	Moderate
(Janet E Loebach & Jason A Gilliland, 2016)	1	1	1	0.5	0.5	0	0.5	0	0.5	0.75	5.75	71.9	Robust
(Helbich et al., 2016)	1	0.75	0.5	0.25	0	0.5	0.5	0.5	0.5	0.5	5	62.5	Moderate
(Mehdizadeh et al., 2017)	1	1	1	0.25	0	0	0	0	0.5	0.75	4.5	56.3	Moderate
(Noonan et al., 2016)	1	1	0.75	0.25	0	0	0	0	0.5	0.25	3.75	46.9	Weak
(Nguyen et al., 2018)	1	1	0.75	1	0	0.5	0.5	0	0.5	1	6.25	78.1	Robust
(Oliver et al., 2015)	1	1	1	1	0.5	0.5	0.5	1	0.5	1	8	100.0	Robust
(Oluyomi et al., 2014)	1	1	1	0.25	0	0	0.25	0	0.5	0.5	4.5	56.3	Moderate
(Page et al., 2010)	1	1	0.75	0.25	0	0	0	0	0.5	0.75	4.25	53.1	Moderate
(Roberts et al., 2016)	1	1	0.75	0.25	0	0	0.25	0	0.5	0.75	4.5	56.3	Moderate
(Stephanie H. Kneeshaw-Price et al., 2015)	1	1	0.5	0.5	0	0.5	0	0	0.5	0.25	4.25	53.1	Moderate
(Santos et al., 2013)	1	1	0.75	0.25	0	0	0	0	0.5	0.5	4	50.0	Moderate
(Shokoohi et al., 2012)	1	0.5	0.5	0.25	0	0	0.25	0	0.25	0	2.75	34.4	Weak
(Stark et al., 2018)	1	1	0.75	0.25	0	0	0	0	0.5	0.5	4	50.0	Moderate
(Stephanie H. Kneeshaw-Price et al., 2015)	1	1	0.5	0.5	0	0.5	0	0	0.5	0.25	4.25	53.1	Moderate

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(Suminski et al., 2018)	1	1	1	N/A	N/A	0.5	0.5	1	0.5	1	6.5	108.3	Robust
(Timperio et al., 2004)	1	1	0.75	0.25	0	0	0	0	0.5	0.75	4.25	53.1	Moderate
(Tung et al., 2016)	1	1	0.5	0	0	0	0	0	0.5	1	4	50.0	Moderate
(van den Berg et al., 2020)	1	0.75	1	0.25	0	0	0.25	0	0.5	0.5	4.25	53.1	Moderate
(Villanueva et al., 2012)	1	1	0.75	1	0	0	0.5	0	0.5	0.75	5.5	68.8	Moderate
(Vonderwalde, Cox, Williams, Borghese, & Ian Janssena, 2019)	1	1	1	1	0	0.5	0.5	0.5	0.5	1	7	87.5	Robust
(Waygood & Susilo, 2015)	1	0.75	0.5	0	0	0	0	0	0.5	0.5	3.25	40.6	Weak
(Zhu & Lee, 2008)	1	N/A	N/A	0	0	0.5	0.5	0	0.5	0.5	3	50.0	Moderate

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Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist

SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
TITLE			
Title	1	Identify the report as a scoping review.	1
ABSTRACT			
Structured summary	2	Provide a structured summary that includes (as applicable): background, objectives, eligibility criteria, sources of evidence, charting methods, results, and conclusions that relate to the review questions and objectives.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known. Explain why the review questions/objectives lend themselves to a scoping review approach.	3
Objectives	4	Provide an explicit statement of the questions and objectives being addressed with reference to their key elements (e.g., population or participants, concepts, and context) or other relevant key elements used to conceptualize the review questions and/or objectives.	3
METHODS			
Protocol and registration	5	Indicate whether a review protocol exists; state if and where it can be accessed (e.g., a Web address); and if available, provide registration information, including the registration number.	N/A
Eligibility criteria	6	Specify characteristics of the sources of evidence used as eligibility criteria (e.g., years considered, language, and publication status), and provide a rationale.	4
Information sources*	7	Describe all information sources in the search (e.g., databases with dates of coverage and contact with authors to identify additional sources), as well as the date the most recent search was executed.	3-4
Search	8	Present the full electronic search strategy for at least 1 database, including any limits used, such that it could be repeated.	SF 2
Selection of sources of evidence†	9	State the process for selecting sources of evidence (i.e., screening and eligibility) included in the scoping review.	4
Data charting process‡	10	Describe the methods of charting data from the included sources of evidence (e.g., calibrated forms or forms that have been tested by the team before their use, and whether data charting was done independently or in duplicate) and any processes for obtaining and confirming data from investigators.	4
Data items	11	List and define all variables for which data were sought and any assumptions and simplifications made.	NA
Critical appraisal of individual sources of evidence§	12	If done, provide a rationale for conducting a critical appraisal of included sources of evidence; describe the methods used and how this information was used in any data synthesis (if appropriate).	4-5

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SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
Synthesis of results	13	Describe the methods of handling and summarizing the data that were charted.	4
RESULTS			
Selection of sources of evidence	14	Give numbers of sources of evidence screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally using a flow diagram.	6
Characteristics of sources of evidence	15	For each source of evidence, present characteristics for which data were charted and provide the citations.	SF2_table1
Critical appraisal within sources of evidence	16	If done, present data on critical appraisal of included sources of evidence (see item 12).	5 and SF2
Results of individual sources of evidence	17	For each included source of evidence, present the relevant data that were charted that relate to the review questions and objectives.	SF2Table2
Synthesis of results	18	Summarize and/or present the charting results as they relate to the review questions and objectives.	6-8
DISCUSSION			
Summary of evidence	19	Summarize the main results (including an overview of concepts, themes, and types of evidence available), link to the review questions and objectives, and consider the relevance to key groups.	8-10
Limitations	20	Discuss the limitations of the scoping review process.	10-11
Conclusions	21	Provide a general interpretation of the results with respect to the review questions and objectives, as well as potential implications and/or next steps.	11
FUNDING			
Funding	22	Describe sources of funding for the included sources of evidence, as well as sources of funding for the scoping review. Describe the role of the funders of the scoping review.	12

JBI = Joanna Briggs Institute; PRISMA-ScR = Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews.

* Where *sources of evidence* (see second footnote) are compiled from, such as bibliographic databases, social media platforms, and Web sites.

† A more inclusive/heterogeneous term used to account for the different types of evidence or data sources (e.g., quantitative and/or qualitative research, expert opinion, and policy documents) that may be eligible in a scoping review as opposed to only studies. This is not to be confused with *information sources* (see first footnote).

‡ The frameworks by Arksey and O'Malley (6) and Levac and colleagues (7) and the JBI guidance (4, 5) refer to the process of data extraction in a scoping review as data charting.

§ The process of systematically examining research evidence to assess its validity, results, and relevance before using it to inform a decision. This term is used for items 12 and 19 instead of "risk of bias" (which is more applicable to systematic reviews of interventions) to include and acknowledge the various sources of evidence that may be used in a scoping review (e.g., quantitative and/or qualitative research, expert opinion, and policy document).

From: Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMAScR): Checklist and Explanation. Ann Intern Med. 2018;169:467–473. doi: 10.7326/M18-0850.

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Is there a correlation between children's outdoor active mobility behaviour and neighbourhood safety? A systematic review of the evidence

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Is there a correlation between children's outdoor active mobility behaviour and neighbourhood safety? A systematic review of the evidence

Aim: BMJ Open Journal

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ABSTRACT

Objective: To identify, summarise and evaluate evidence on the correlation between perceived and actual neighbourhood safety (personal and road danger) and diverse forms of outdoor active mobility behaviour (i.e., active play, exercise, and travel) among primary-school-aged children.

Design: A systematic review of evidence from observational studies exploring children's active mobility behaviour and safety.

Data sources: Six electronic databases were searched: Google Scholar, PubMed, Scopus, Science Direct, ProQuest and Web of Science from study inception until July 2020.

Data extraction and synthesis: Study selection and quality assessment were conducted independently by two reviewers. We expanded on a quality assessment tool and adopted a vote-counting technique to determine strength of evidence. The outcomes were categorised by individual, family and neighbourhood levels.

Results: A total of 29 studies were included, with a majority of cross-sectional design. Higher parental perceived personal safety correlated with increased children's active mobility behaviour, but most commonly in active travel (e.g. independent walking or cycling to a local destination). Increased concerns regarding road danger correlated with a decrease in each type of children's active behaviour; active travel, play and exercise. However, these correlations were influenced by child's sex/gender, age, car ownership, neighbourhood types, across time, and proximity to destination. Limited or inconclusive evidence was found on correlate of COAMB to 'stranger danger', children's perceived personal safety, race/ethnicity, socio-economic status or measured safety.

Conclusion: Children are restricted by perception of safety. Encouraging children's active travel may require future strategies to address characteristics relevant to types of the neighbourhood that promote a high sense of personal safety. Children and parents may embrace other types of active mobility behaviour if road danger is mitigated. Sex/gender and age-specific interventions and redesign of public places could lead to child-friendly cities. Future studies may benefit from adopting validated measurement methods and fill existing research gaps.

Keywords: Children's outdoor active mobility behaviour, Parents' perceived safety, Children's perceived safety, Measured safety, Neighbourhood safety, Socio-ecological framework.

Article Summary: Strengths and limitations of this study

- This systematic review adapted a methodological quality appraisal that integrated studies' reporting and measurement methods along with a vote-counting technique to understand the strength of the evidence.
- To our knowledge, this is the first systematic review that comprehensively synthesises evidence on neighbourhood safety correlates to forms of primary school-aged children's outdoor active mobility behaviours.
- Peer-reviewed studies published in languages other than English were excluded, and meta-analysis review was not possible due to the heterogeneity in studies measurement methods and outputs.

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INTRODUCTION

Children's daily active mobility behaviour is linked with profound long-term positive outcomes.[1] Recommended daily level of children's physical activity is primarily achieved through various forms of structured (i.e. organised sport) or unstructured active behaviour (i.e. active play in open spaces, walking, cycling or scootering to school and other local destinations).[2 3] Neighbourhood, the place where a child's majority of daily routines occur, has been identified as a primary venue for children to meet the daily physical activity recommendations.[2] However, variability in the neighbourhood (i.e. residential, commercial, industrial and agricultural) have been found to affect types and extent of children's various shapes of active behaviour.[4 5] For example, different types of neighbourhood were linked to changed levels of active transportation in children aged 10–11 years living in Finnish urban areas.[6] Canadian children travel longer distance actively in residential or commercial areas than other types of neighbourhood.[5] Urban and road structures in The Netherlands reversed the negative effect of reduced active travel with longer distances.[7] Nevertheless, neighbourhood impact on other aspects of children's behaviour such as play and peer interaction is mixed.[8]

Safety, including personal and road danger,[9 10] is a complex concept.[3] Perceived (by parents or children) or measured safety (actual crime and traffic danger) may create obstacles resulting in children's lower confidence in exploring their surroundings,[2] or contributing to their community.[11] Lack of safety may have contributed to urban streets and public spaces becoming increasingly inhospitable environments. With more people living in urban settings than ever before,[12] and increasing safety concerns, children's territory for active outdoor mobility behaviour, such as walking, playing, cycling and scootering, has shrunk.[13-16] The damage caused by a restriction of children's active mobility behaviour may extend beyond their immediate physical health,[17] to overall social skills[18] and healthy brain development.[19]

Nevertheless, reviews that addressed neighbourhood safety and children's active behaviour have reported conflicting results. The disagreements could be due to: narratively investigating safety,[3] examining one shape of active behaviour such as active school travel [9 20-23] or independent mobility [26] or combinedly addresses safety with other environmental correlates and from a wide age range of children age (three or five to 18 years).[16 24 25] Over and above that, variability in measurement methods and output adds complexity in evaluating available evidence.[20 22] The absence of suitable checklists that account for measurement methods when assessing the quality of observational studies,[26] called research for alternatives approach to increase confidence in the synthesis of studies outcome[20 22] to inform evidence-based policy decisions. On the other hand, active school travel is a significant contributor to overall child active behaviour,[27] but other forms of activity incorporated into children's daily routines[26] are vital contributors to children's daily movement targets.[28] No review up to now has exclusively focussed on the varying impact of measured and perceived safety in urban settings across primary-school-aged children's various forms of daily active mobility behaviours. Thus, this review aims to (1) systematically synthesise evidence of correlations between primary-school-aged children's outdoor active mobility behaviour (COAMB) and neighbourhood measured and perceived safety, and (2) derive the strength of the evidence by evaluating quality and methodological measurement method for each study and vote counting technique as justified from previous reviews.

METHOD

This systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA). The corresponding PRISMA checklist was completed during the review, as illustrated in online supplemental file 1. Though the search was planned, no protocol was registered.

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Search strategy

A keyword search was designed and conducted to identify all relevant studies in six electronic databases: *Google Scholar*, *PubMed*, *Scopus*, *Science Direct*, *ProQuest* and *Web of Science*. All databases were searched between March 2019 and July 2020. The search algorithm included all possible combinations of keywords. Terms pertaining to this review were grouped by (1) target population, (2) AND active behaviour, AND (3) neighbourhood safety, OR (4) moderators and mediators, OR (5) spatiotemporal aspects (Global positioning system, GPS, geographic information system, GIS, out of school hours). The entire search thesauruses and strings are available in online supplemental file 2. Additional papers were identified from the citation lists of individual and review papers.

Eligibility criteria

Studies were included for review if they were: published in peer-reviewed journals, in English, were observational studies that collected data directly on participants aged between five and 12 years (primary-school-aged children), assessed neighbourhood safety (personal and road safety) either as perceived by parents and/or children or measured (crime or road conditions), and reported or measured children's active mobility outside school hours. Included publications were not restricted to a single geographic location or a specific time. Studies were excluded for failing to meet the inclusion criteria. Systematic, scoping, or narrative reviews, opinion and editorial pieces, and other non-original research publications were also excluded.

Study selection process

Using search terms developed for this review, one reviewer (RZ) screened titles and abstracts to identify those works that met the inclusion criteria. Abstracts of the downloaded papers were then reviewed by two independent reviewers (RZ, CX) against the selection criteria. The final list of included studies was agreed by consensus involving a third reviewer (RN). Studies that met inclusion criteria were reviewed in full by RZ.

Data extraction process

The first author (RZ) extracted, into an excel datasheet relevant information on: author(s) year of publication, year of data collection, location of study, sample size (separated by gender), age/school grade, characteristics regarding methods of measuring or assessing neighbourhood safety (perceived or measured), children's active behaviour and outcome measures (e.g., body mass index [BMI] or medium-to-vigorous physical activity [MVPA]), child's neighbourhood, other variables accounted in studies analysis including cofounders (e.g., sex/gender/family characteristics) and summary of findings. The data was collated into a systematic narrative summary table of the relevant papers.

Patient and public involvement

No patients involved.

Data synthesis

Correlates between COAMB and safety were organised according to aspects of the socio-ecological framework (i.e. individual, family, and neighbourhood). This framework is frequently used in active health behaviour research[29] as a conceptualisation tool. This tool provides a structure for organising and understanding multiple factors that cause changes in health behaviour. Additionally, to enable the review investigating influences specific to forms for active behaviour, the outcome of correlates were compared by grouping findings into one of four categories: active travel; school travel; active play; active play, travel, and exercise. The subgrouping corresponds to children's main types of structured and unstructured active behaviour outside of school hours.[30]

Quality assessment

Two independent reviewers (RZ, BJ) appraised the selected studies using priori defined quality criteria. Previously practised quality assessment by Schoeppe et al.[26] for active travel, Marzi et al.[31] on independent mobility, and by Lubans et al.[27] on active school travel, were reviewed. We adopted thirteen existing criteria used earlier that covers all pertinent studies quality reporting characteristics. We expanded on six items to the checklist based on findings from a recent systematic review of which

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methodological measures in observational studies were assessed and found to be fundamental to increase reliability in children's active behaviour research.[32] Those measures were covering: COAMB (objective or subjective), temporality elements such as change in time of day, weekdays or weekends (addressed in both perceptions of safety as well as COAMB), the delineation of the neighbourhood area (arbitrary or measured) and accounting for cofounders. We adopted the scoring system applied by Marzi et al.[31] and Schoeppe et al.[26] to quantify study robustness. Each criterion in the quality assessment checklist was given the assigned score if coded "yes", 0 if coded "no" or unclear, and half the score if coded 'partial'. A total overall score out of eight points was assigned, showing in Table 1. As included studies have employed a variety of tools and sometimes composite techniques to measure active behaviour, we adopted Marzi et al. method for criterion number fourteen that was scored on a scale ranging from 0 to 1. For example, if a study employed an accelerometer and GPS, the study accumulates 0.5 points. However, when the study only used one measurement tool, it was assigned 0.25 points. A study variable that was measured appropriately but was not applicable to a specific criterion used for the quality assessment was discounted in the sum of total points used to derive quality score.[26] For example, the quality assessment criterion related to temporal and spatial measurement was not applicable for a paper where the primary outcome measure was BMI. In that instance, the criterion was removed from the total quality score of that paper.[26] The quality score for each study was calculated by converting the total accumulated score into a percentage (total accumulated score/8 (maximum possible score) x 100).[26] Adopting Schoeppe et al. cut-off for levels of studies' quality, a percentage score of ≥ 66.7% was deemed robust, a score between ≥ 50 and < 66.6% was considered moderate, and < 50% was rated poor.

Table 1. Criteria for quality assessment and scoring for each criterion adapted from previous published systematic reviews.

Criteria			Description	Quality score per criterion
Adopted quality assessment characteristics	1.	Study objectives	Are objectives clear?	(Yes=0.25-point No=0 of each criterion)
	2.	Design	Was the study design appropriate for research undertaken?	
	3.	Target population	has the target population defined?	
	4.	Random sampling	Was a random sampling of the target population taken or was sampling appropriate for the study design?	
	5.	Study participant number	Was the number of participants at each stage reported?	(Yes=0.25-point No=0 of each criterion)
	6.	Participants' inclusion/exclusion	Were criteria for inclusion and exclusion of the participants used?	
	7.	Study population	Was the study population sufficiently described, i.e., sample size, gender, age, indicators of socio-economic status?	
	8.	Participant recruitment	Was participant recruitment described or referred to?	
	9.	Response rate	Was the response rate 60% or more?	(Yes=0.25-point No=0 of each criterion)
	10.	Data collection	Did the study describe data collection, i.e., by mail, by interview, objective measure?	
	11.	Data sources	Did the study describe source of data, i.e., questionnaire, survey, focus group, accelerometer, GPS?	
	12.	Missing data	Were numbers/percentages of the participants with missing active behaviour data reported, and did at least 80% of enrolled participants provide complete data to include?	
	13.	Statistical method	Was it clear what was done to determine statistical or spatial statistical significance, e.g., p-value, confidence interval?	Yes = 0.5 No = 0
Methodological measures	14.	Active behaviour measures	Did the study objectively measure active mobility, i.e., activity tracking. Spatial technology or web application? Were measures reliable and valid for subjective measures?	GPS = 0.25 (accelerometer) = 0.25 questionnaire = 0.25 Travel diary = 0.25 N/A
	15.	Temporal active behaviour measures	Was measured active behaviour related to temporal characteristics, i.e., weekends vs weekdays or before and after school?	Yes = 0.5/No = 0 N/A
	16.	perceived safety measures characteristics	Did the study account for spatial or temporal features of the safety feelings? (Weekdays vs weekends or daytime (before and after school)) Or was measured safety conducted using geocoding?	Yes = 0.5 No = 0
	17.	Area of exposures	Did the study delineate the exposed area "neighbourhood" objectively or evaluate arbitrarily?	Objectively = 0.5 Arbitrary = 0 N/A
	18.	Evidence depicted temporal variation	Did the analysis account for the spatiotemporal behaviour of evidence?	Yes = 1/No = 0
	19.	Adjustments (Cofounders)	Did the study account for cofounders of age, sex, ethnicity, and family characteristics?	Each is 0.25/Total = 1

Evidence synthesis and strength

Evidence synthesis described by Rothman et al.[22] and Ikeda et al.[20] was utilised in this review. Each correlation and its direction (positive (+), negative (-), or inconclusive/no correlation(0)) was identified and coded in terms of statistically significant or non-significant findings organised by the socio-ecological levels. To facilitate synthesis of evidence, we performed a reverse coding of the original direction of associations (i.e. positive to negative or vice versa) as necessary, depending on the wording

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used in each work.[20] A minimum of five studies with significant or non-significant findings was adopted to generate a consensus.[20 22] Evidence with less than five studies available for synthesis was rated as limited,[26] and with three studies was rated insufficient. Deriving strength of evidence was also adapted from previous reviews[26 27] using the proportion and quality of studies reporting a significant association. The ratio of studies found with robust quality in this review to those found in an earlier review[26] was used to rescale the quality thresholds. In a minimum of five studies, if ≥ 27% of the total synthesised significant evidence that agrees in a direction (i.e. +, – or 0) was of robust studies, the evidence was rated strong, from ≥ 20%–< 27% was moderate strength, and at < 20% was rated weak.

RESULTS

Study selection

Following the title screening, a total of 231 papers across the six databases were initially identified that deemed potentially relevant. After removing 128 duplicates, the abstract review resulted in 64 articles that underwent a full-text review. Seven additional studies were identified from a manual search of individual reference lists. Following full-text review, a total of 29 studies met inclusion criteria, Figure 1.

Study characteristics

The characteristics of the 29 studies, including settings, measurement methods and evidence output, are summarised in Table 1 in supplementary file 2. There were three longitudinal studies,[10 33 34] and the remaining (25 studies: 86%) had a cross-sectional design. Twelve of the 29 studies (41%) featured analysis using data from other larger scale or government projects (e.g., the Children Living in Active Neighbourhoods [CLAN] study,[35 36] the Built Environment and Active Transport [BEAT] project[37] and the Spatio-Temporal Exposure and Activity Monitoring [STEAM] project[38]). Populations varied across studies, with both parents and children represented. The sample sizes ranged from 35 children[6] to 31,000 households.[39] Studies alternatively used age groups or school grade to refer to children's age, but the mean ages were within the specified range of five to 12 years. Included works examined associations between forms of COAMB and perceived safety (21 of 29 studies; 73%), measured safety (five of 29; 17%) or both measured and perceived safety (three of 29; 10%). Collectively, the publications spanned Asia, Europe, North America, and Oceania, Diagram 1 of online supplemental file 2. Aside from two Iranian studies,[40 41] and one from Malaysia,[42] the remaining publications (26 of 29; 90%) addressed populations in developed countries, with five each from the United States[34 43-46] and the United Kingdom;[10 47-49] four each from Canada[5 33 37 50] and Australia;[35 36 51 52] two each from New Zealand[53 54] and the Netherlands,[7 55] and one each from Portugal,[56] Finland[6] and Austria.[57] Except for one study,[58] remaining papers were published between 2007 and 2020. Two studies used BMI[34 48] as a measure of children's active health indices, yet one article has combined BMI with self-reported physical activity. Subjective assessment was the most common method of measurement of children's active behaviour. Ten of 29 studies (34%) employed objective measures such as accelerometers,[35 36 45] pedometers,[52] GPS[6] alone or in combination with an accelerometer,[37 50 53] or GIS loggers.[33] In view of the heterogeneity in the measurement methods, disparity in studies' units of outcome were apparent as exhibited in Diagram 2 of online supplemental file 2. Of 24 studies which examined personal safety, the majority assessed safety as perceived by parents, however seven studies featured safety perception among children.[5 41 47 49 51 52 58] Five studies[7 10 36 50 53] examined objective measures of road safety and 12 studies assessed road danger as perceived primarily by parents. The findings in 25 of the 29 studies (86%) were reported separately for boys and girls.

Methodological quality assessments and strengths in evidence

Nine of 29 studies (31%) were rated of robust quality, 16 studies (55%) were of moderate, and four (14%) were rated as poor quality, as presented in Table 2 in online supplemental file 2. Most studies

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focus on correlates of safety to active travel behaviour, such as independent walking or cycling to local destinations. Table 3 in online supplemental file 2 showed that active school travel was the focus in seven of 29 (24%), ten of 29 (34%) addressed active travel, and seven of 29 (24%) addressed active play, active school travel and exercise. Active play alone was investigated in 3 of 29 (10%) studies, and two (8%) used BMI, but one study combined BMI with self-reported physical activity, Figure 2.

Synthesis of the evidence

Derived from Table 4 in supplementary file 2, we summarised below statistically significant and non-significant correlations in 29 studies examining influences on COAMB by aspects of socio-ecological levels that were.

Individual level: Without considering safety and supported by strong evidence (40% in ten out of 15 studies reporting significance and agrees in the direction were of robust quality), we found a correlation between sex/gender and COAMB. A higher level of COAMB was accumulated by males than female children, Figure 3. Boys had more active behaviour[44], took more active trips,[35] accessed larger activity spaces,[52] and increased their active travel of walking or cycling to school,[10] or other destinations.[52] However, this correlates influenced by: temporal variation between weekdays and weekends,[37] time segments of the day (after dark)[51 52] or actual distance travelled.[6] Similarly, strong evidence (33.3% in six studies reporting a significant positive correlation were of robust quality) showed that children of older age groups or higher school grades accumulated more COAMB.

Family level: Eight studies found that correlates of COAMB and car ownership were consistently associated with lower levels of COAMB. Despite that, the level of evidence was weak, with only 12.5% of the total studies were robust quality. The inconsistent finding across five studies suggests no clear correlation between ethnicity/race or socio-economic status (SES) and COAMB. Children of minority groups in the United Kingdom achieved lower levels of active behaviour,[47], whereas children of minority groups in the United States accumulated higher levels of active behaviour.[44] In New Zealand, children's active behaviour differed by race/ethnicity and the day of the week: Indian/Asian background had increased their physical activity levels of medium-to-vigorous (MVPA%) on weekends while Pacific, European and Maori children accumulated higher (MVPA%) on weekdays.[53] Findings on (SES) and COAMB were either: non-significant for female children (five to six years); increased active behaviour in high (SES) groups;[51 42] inconclusive;[59] and increased BMI level in a high deprivation area yet reported higher self- physical activity.[48]

Neighbourhood level: We found strong evidence of the increased parental perception of safety correlated with lower COAMB (29.4% of the 17 out of 18 studies reporting significant and agreeing in the direction were of robust quality), Figure 4. We found limited and inconsistent evidence on the correlation of 'stranger danger' among parents to COAMB; no relationship,[54 56] reduced MVPA,[45] and has a temporal correlation (son weekdays).[37] We found moderate evidence (25% of the 11 studies were rated robust) but consistent in the direction of correlates of higher parental road safety concerns and decreasing levels of COAMB, including one study finding a gender difference (girls). Weak evidence yet consistent (i.e. agree in the direction of the significance) of lower levels of COAMB correlate with children's increased perception of danger from strangers, roads and personal safety. Limited (four studies) evidence related to children's perception of personal safety,[5 41 49 52] included a study that depicted variation by gender.[52] Perception of 'stranger danger'[51 55 62] was insufficiently (three studies) addressed. Road danger[47 49 51 52 58] depicted variation in two studies by sex/gender.[49 58] Despite robust evidence (six out of nine studies are of robust quality), a definitive conclusion cannot be drawn regarding measured safety (i.e. crime or road conditions) and COAMB. A synthesis of four studies showed inconsistency in the direction of the relationship with measured personal safety,[33 34 45 46] as one study depicted increased active transportation in high crime areas.[33] Road danger[3 7 10 50] showed increased outdoor active play in highest traffic areas[50], and one study lacked correlation to measured roads environment.[36] Other characteristics in the neighbourhood that are likely to be influenced by perceived safety were: strong evidence of distances travelled (perceived or objectively measured) with

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10 studies and 40% of robust quality and variability in neighbourhood type (6 studies with 66.6% of robust quality). Other addressed variables demonstrated associations with COAMB, but insufficient evidence to conclude are in table 4 of online supplemental file 2.

DISCUSSION

A review of 29 studies investigating multifaceted influences on COAMB and neighbourhood safety prompted the discussion over three primary topics.

Inequality in opportunity for children's active behaviour

Differences in individual, family, and neighbourhood safety, created an unequal opportunity for COAMB that was ubiquitous across geographical regions, Diagram 3 of online supplemental file 2. Our review depicted variability of COAMB by sex/gender that was predominant in active play and exercise forms of active behaviour than in active travel. This explains lacking of agreement with previous reviews on independent mobility[60] and active school travel,[22] yet agreed with reviews on physical activity.[15 61]

Our review suggest that with more access to cars, parents conveniences,[22] or other influences such as weather conditions, children at young age are more likely, regardless of their sex/gender, to be driven more often than actively travelling (e.g. school trips). However, variability in sex/gender becomes more apparent in physical activity as it is an elective form of active behaviour behind school or government jurisdiction. Increased COAMB with age undetected in previous reviews[15 61] could be owned to earlier reviews combining a broad range of children's age (three to 12 years[61] or three to 18 years[15]). A longitudinal study suggested that some forms of active travel (such as school trip) become more prevalent with age, starting from six years.[62]

Studies variability restrained clear conclusions on family race/ethnicity or neighbourhood (SES) correlates to COAMB also concord with earlier reviews findings.[20 22] Children from low SES areas in the UK[48] were more active. However, children from wealthy families in Australia and Malaysia[42 51] had increased active behaviour. Children from less affluent areas may be active by necessity, whereas more accessibility opportunities to open spaces and recreational centres are offered to children from high (SES) areas. A study on obesity rates showed that migrant children (in lower SES) were more active than resident children with higher (SES).[63] This socio-economic influence appeared to reverse its effect on COAMB by geographic location (between developed and developing countries). Nevertheless, we found variation in the correlates of COAMB to (SES) by gender (female children),[51 55 62] age group[55] and higher-income people feeling safer, which in turn encouraged active school travel.[55] Despite an agreement with earlier reviews,[20 22], the decline in COAMB concerning family ownership of car conflicts with some studies suggesting that wealthier families are more active, thus calling for further investigation.

Every child is entitled to feel safe at all times. Nevertheless, findings of this review suggest a distinct inequality as children living in low perceived safe areas are declined the opportunity to be active. Our finding conflicts with two previous reviews on active travel[16] and physical activity[15], respectively. The broad range of children age addressed in the two reviews may have contributed to the disagreements in findings. Parents have a larger influence on younger children's active health behaviour[9] and perhaps hold greater fears regarding road safety for younger children that may not contribute significantly in the older age group, as was explained by a longitudinal national study in the US.[64] However, concerns regarding personal safety are nuanced by gender (females),[6 35 52] temporal variations (on weekdays only[53] or weekends[35]), and intersect with gender (females) and temporality when after dark.[6] The prevalence effect of 'stranger danger' on COAMB among studies surveying children suggests that vulnerability to strangers' harm is more significant for children, particularly for girls.

Influences on COAM are specific to the type of active behaviour

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We acknowledged earlier that sex/gender correlate to COAMB was most apparent in children's active play and physical activity rather than active travel. However, this review also detected that perceived personal safety risk is most significant in restricting children's active travel, limiting their active space. This result concurs with earlier research revealing shrinking children's active space with the rise of personal safety fears.[2] Nevertheless, parents' concerns over road danger restricted all types of COAMB, Diagram 4 of online supplemental file 2. More active travel was regarded in the urban residential and commercial neighbourhood, was also concluded previously[31] suggesting that dense urban area may offer a sense of safety, encouraging more active travel amongst children. The correlate was also confirmed by the reversed influence of reduced active travel correlates to perceived or measured distance occurred with changes to roads and infrastructure.[55]

Methodological challenges

The inconsistencies seen in studies investigating neighbourhood safety correlates to COAMB, limit the ability to draw definite conclusions in some areas. Our methodological quality assessment revealed that most studies fulfilled elements relevant to studies reporting but essentially lacked addressing components in the design, method of measurement, and analysis for evidence-based research on children's active health.[32] This may have resulted in the majority of studies were of moderate and poor quality. Subjective measures using questionnaires rely on human recall distant events.[65] Yet, we found some inconsistencies among studies employing objective measures. Although a review of measurement methods is out of the scope of this paper, the discrepancies in some of the outcome necessitate highlighting some critical research challenges for future research attention yet were elaborated further elsewhere by Zougheibe et al.[32] Study design is likely to affect the outcome. Longitudinal studies limit selection bias associated with cross-sectional design. Inconsistency in the direction of correlates among studies employed objective measures showed most problematic measurement methods were in (a) inconsistent inclusion of crime categories for measured personal safety studies or elements addressed roads conditions, (b) measuring COAMB using spatial activity tracking (i.e. Global positioning system, GPS) was faced with an absence of standardised measurement protocols that caused variabilities in few areas such as the inclusion of surveyed days and threshold of counts per minutes of registered data, that could impact a true interpretation of results, (c) the spatial extent of the child's neighbourhood to derive safety exposures was primarily defined arbitrarily. Still, studies that adopted spatial measures had employed diverse methods, (d) inconsistent accounting for cofounders and (e) infrequent accounting for temporal safety and behaviour characteristics. We found scarcity in research on correlates of COAMB to variation of safety perceived by parents to children residing in the same neighbourhood and actual crime.

Implications of Findings

Successful implementation of any intervention strategy to encourage parents and children to embrace an active mobility culture require collaboration among all levels of children, families, schools, community-based organisations, city planners, policymakers and researchers.[28 66] The revealed multiple influences contributing to children's active behaviour guided our recommendations below.

Interventions for vulnerable groups of the population: Ignoring gendered primary-school-aged children's active mobility behaviour may result in only male adult's regarding physical activity as essential or accessible. Therefore, strategies to increase active behaviour must be sex/gender and age-specific. Additionally, the indication of influence on COAMB by diverse (SES) and race/ethnic backgrounds suggests that a subgroup of the population is constantly inactive. Parental involvement in educational programmes to promote children's free play and active travel or children's encouragement in school and community programmes to participate in active travel and leisure active play could be useful. Earlier evidence also supports the positive influences of active parents[56] or positive peer influence.[67]

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Promote child-friendly cities:

Revitalise cities design may offer play opportunity and safer travel for children. Perceived low safety linked to declined children's active travel suggests that current city and road designs are seen as unwelcome and unsafe for children to be outdoors actively playing and travelling safely to local destinations independently. Most importantly, interventions intended to encourage children to travel further actively call for redesign of urban structures to enable active travel and safety at the same time. This may include upgrading local features that promote walkability in desirable neighbourhood characteristics (e.g., schools, shops, libraries) and active playing in local parks or exercise in an 'activity-friendly neighbourhood'.[68] More voices are calling for children and families to be included in designing new urban spaces. Improvements in road safety may increase overall child outdoor active mobility behaviour and enable children and families to embrace an active mobility culture. Existing streets design is seen as unsafe and unappealing for children and their parents. Prompt policy interventions to address most concerned roads conditions(e.g. heavy traffic, fast drivers, high-speed roads, lack of road signals, availability of sidewalks and cycling infrastructure) may alleviate parents' and children's concerns over road danger.

Advancing Research: Methodological improvements to reduce inconsistency in the outcome and increase reliability in future research can better inform evidence-based policy. Namely: encouraging longitudinal study design and adopting a three-dimensional conceptual framework in research as suggested earlier[32] to account for critical elements of (a) what we measure: the determinants that affect the outcome and consistent accounting for cofounders (sex/gender, SES background, and ethnicity/race background), (b) how we measure: involves frequency and intensity for perceived safety[69-71] (personal and roads) and active behaviour whether reported or measured consistency in addressing crimes only relevant to personal safety, and (c) spatiotemporal characteristics (i.e. locality of active behaviour or safety incident, spatial extents of children neighbourhoods, and time[72]). Further research on COAMB correlates to variability in parents to their children safety perception or measured to perceived safety, and inequality in the neighbourhood (SES) would answer more profound research questions.

Strengths and limitations

This review was limited to English-language peer-reviewed observational studies. Despite every effort to identify all relevant studies through a variety of terms used, some relevant studies may have been omitted due to multiple synonyms used in works. The observed heterogeneity in study measures prevented a straightforward meta-analysis; thus, we conducted quantitative vote counting to overcome existing disparities in outcome and methods. The majority of studies were cross-sectional design and were of moderate or poor quality. Despite these limitations, this review has important strengths. This work synthesised findings and derived a quantitative understanding of evidence strength by combining a measurement of study reporting quality, involving critical characteristics that improve reliability in observation studies. This paper has comprehensively addressed the impact of safety in primary-school-aged children's active behaviour, eliminating inconsistencies resulting from a broad age range of study participants. To the best of our knowledge, this review is the first to consider the impact of various aspects of neighbourhood safety across multiple types of COAMB. Finally, this review was not limited to publication date or geographical region.

CONCLUSION

This systematic review of evidence revealed that there is inequality in COAMB by children's sex/gender and age and some indication regarding race/ethnicity and (SES) despite the evidence being inconclusive with regard to the latter two examined variables. Compared to safer neighbourhoods, children living in perceived unsafe areas correlated with lower outdoor active mobility behaviour and reduced active play. However, perceived personal safety risk has primarily restricted children's active

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travel to local destinations, while perceived danger from traffic reduced every type of COAMB. Nevertheless, the direction and strength of the correlates are affected by individual and family characteristics, distance travelled, and time (weekend/weekday/time of day). These findings were consistent across countries. There is a need to use validated measurement methods. Deeper understanding of safety (perceived or measured) correlates to race/ethnicity or (SES) variances, and COAMB (spatial extent of active behaviour or intensity of physical activity) may answer more profound behavioural research questions. Current inequalities in children's opportunities to engage in active mobility behaviour require sex/gender and age-based interventions. Most importantly, interventions aimed at improving personal safety and engaging children in urban design to promote child-friendly cities may prompt children to travel further actively. Improvements in road conditions may increase overall child outdoor active mobility behaviour and enable children and families to embrace an active mobility culture.

Figures Legend

Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses, PRISMA flow diagram for scoping review of neighbourhood safety correlates to COAMB.

Figure 2. Studies grouped by the type of active behaviour indicating studies' level of methodological quality assessment. The x-axis represents studies grouped by type of active behaviour examined in each study. y-axis denotes the number of studies giving the different levels of evidence 'strength' when total methodological review score was summed. Strength of evidence was robust when summing total score was ≥ 66.66% moderate sum was ≥ 50–< 66.6% and poor when an accumulated score was < 50%. Adapted from previous review by Schoeppe et al.[26]

Figure 3. Correlation between individual (child) and family characteristics with children's active mobility behaviour. The x-axis represents variables that accumulated the minimum number of studies to synthesise evidence of correlates between COAMB and sex/gender, age, access to a car, and ethnicity. y-axis denotes the number of studies that examined the correlation showing accumulated significance or non-significance. Note: * = in the sex/gender variable, we saw variability by temporal characteristics (i.e. changes between weekdays and weekends or time of the day (before and after school)) in three studies and reduced COAMB by distance travelled in one study. * = In Race/ethnicity, one study with a significant correlate depicted temporal change (i.e. weekend/weekdays).

Figure 4. Perceived safety (parents and children) of personal and road danger correlates to reported and measured children's active mobility behaviour. The x-axis represents examined variables that accumulated a minimum number of studies to synthesise evidence for perceived personal and road safety by parents or children. y-axis denotes the number of studies that accumulated significant or non-significance. Note: = Parents' perceived neighbourhood safety showing significant correlates included three studies depicting temporal characteristics (weekdays/weekends or before and after school), three studies varied by sex/gender and one study by age with one study in perceived road safety varied by gender. * = For children, perceived personal safety varied by gender in one study, but for perceived road safety, one study varied by SES and gender, and two studies depicted variation by gender*

Supplementary information

Supplemental File 1 PRISMA checklist

Supplemental File 2: Details of search terms used, Search Strategies across six electronic databases. Table 1: Summary of studies' characteristics and evidence, Table 2: Methodological measurement and quality assessment, Table 3: Studies grouping by active mobility behaviour type, Table 4: Findings of association by the socio-ecological framework of the included studies, Diagram 1: Studies distribution by country of origin, Diagram 2: Variances in measurement methods of children active mobility behavoiur and units of outcome, Diagrams 3 and 4:Findings of individual, family and neighbourhood safety correlates grouped by type of active behaviour.

Declarations

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References

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1. Steinbeck KS. The importance of physical activity in the prevention of overweight and obesity in childhood: a review and an opinion. *Obesity Reviews* 2001;2(2):117-30. doi: 10.1046/j.1467-789x.2001.00033.x
2. Jack G. Place matters: The significance of place attachments for children's well-being. *British Journal of Social Work* 2010;40(3):755-71. doi: https://doi.org/10.1093/bjsw/bcn142
3. Carver A, Timperio A, Crawford D. Playing it safe: The influence of neighbourhood safety on children's physical activity—A review. *Health & Place* 2008;14(2):217-27. doi: https://doi.org/10.1016/j.healthplace.2007.06.004
4. Stone MR, Faulkner GE, Mitra R, et al. Physical Activity Patterns of Children in Toronto: The Relative Role of Neighbourhood Type and Socio-economic Status. *Canadian Journal of Public Health* 2012;103(3):S9-S14. doi: 10.1007/BF03403829
5. Loebach JE, Gilliland JA. Free Range Kids? Using GPS-derived activity spaces to examine children's neighborhood activity and mobility. *Environment and Behavior* 2016;48(3):421-53.
6. Fagerholm N, Broberg A. Mapping and characterising children's daily mobility in urban residential areas in Turku, Finland. *Fennia* 2011;189(2):31-46.
7. Helbich M, Emmichoven MJ, Dijst MJ, et al. Natural and built environmental exposures on children's active school travel: A Dutch global positioning system-based cross-sectional study. *Health Place* 2016;39:101-9. doi: 10.1016/j.healthplace.2016.03.003 [published Online First: 2016/03/25]
8. Lia K, Willem van V, x. Children in the City: Reclaiming the Street. *Children, Youth and Environments* 2006;16(1):151-67.
9. McMillan TE. Urban Form and a Child's Trip to School: The Current Literature and a Framework for Future Research. *Journal of Planning Literature* 2005;19(4):440-56. doi: 10.1177/0885412204274173
10. Carver A, Panter JR, Jones AP, et al. Independent mobility on the journey to school: A joint cross-sectional and prospective exploration of social and physical environmental influences. *Journal of Transport & Health* 2014;1(1):25-32. doi: https://doi.org/10.1016/j.jth.2013.12.003
11. Donovan J. Designing the compassionate city to overcome built-in biases and help us live better La Trobe University: Routledge; 2018 [Available from: https://theconversation.com/designing-the-compassionate-city-to-overcome-built-in-biases-and-help-us-live-better-92726 accessed 1/11/2020 2020.
12. Schwab K. Play access in cities is critical to children's 2020 [cited 2020 3/11/2020]. accessed 9/10/2020 2020.
13. Garofalo J. The Fear of Crime: Causes and Consequences. *THE JOURNAL OF CRIMINAL LAW & CRIMINOLOGY* 1981;72(2):839.
14. Lorenc T, Petticrew M, Whitehead M, et al. Crime, fear of crime and mental health: synthesis of theory and systematic reviews of interventions and qualitative evidence. : Public Health Research 2014.
15. Davison K, Lawson C. Do attributes in the physical environment influence children's physical activity? A review of the literature. *International Journal of Behavioral Nutrition and Physical Activity* 2006;3(1):19. doi: 10.1186/1479-5868-3-19
16. Panter JR, Jones AP, van Sluijs EMF. Environmental determinants of active travel in youth: A review and framework for future research. *International Journal of Behavioral Nutrition and Physical Activity* 2008;5(1):34. doi: 10.1186/1479-5868-5-34
17. WHO. Global Status Report on Non Communicable Diseases 2010. 2010:176.
18. Pellis SM, Pellis VC, Bell HC. The Function of Play in the Development of the Social Brain. *American Journal of Play* 2010;v2 (n3):p278-96
19. Council IoMNR. From Neurons to Neighborhoods: The Science of Early Childhood Development. Washington, DC: The National Academies Press 2000.
20. Ikeda E, Hinckson E, Witten K, et al. Associations of children's active school travel with perceptions of the physical environment and characteristics of the social environment: A systematic review. *Health & Place* 2018;54:118-31. doi: https://doi.org/10.1016/j.healthplace.2018.09.009
21. Wong BY-M, Faulkner G, Buliung R. GIS measured environmental correlates of active school transport: A systematic review of 14 studies. *International Journal of Behavioral Nutrition and Physical Activity* 2011;8(1):39. doi: 10.1186/1479-5868-8-39

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

22. Rothman L, Macpherson AK, Ross T, et al. The decline in active school transportation (AST): A systematic review of the factors related to AST and changes in school transport over time in North America. *Preventive Medicine* 2018;111:314-22. doi: https://doi.org/10.1016/j.ypmed.2017.11.018
23. Pang B, Kubacki K, Rundle-Thiele S. Promoting active travel to school: a systematic review (2010–2016). *BMC Public Health* 2017;17(1):638. doi: 10.1186/s12889-017-4648-2
24. Pont K, Ziviani J, Wadley D, et al. Environmental correlates of children's active transportation: A systematic literature review. *Health & Place* 2009;15(3):849-62. doi: https://doi.org/10.1016/j.healthplace.2009.02.002
25. Davison KK, Werder JL, Lawson CT. Children's active commuting to school: current knowledge and future directions. *Preventing chronic disease* 2008;5(3):A100-A00.
26. Schoeppe S, Duncan MJ, Badland H, et al. Associations of children's independent mobility and active travel with physical activity, sedentary behaviour and weight status: A systematic review. *Journal of Science and Medicine in Sport* 2013;16(4):312-19. doi: https://doi.org/10.1016/j.jsams.2012.11.001
27. Lubans DR, Boreham CA, Kelly P, et al. The relationship between active travel to school and health-related fitness in children and adolescents: A systematic review. *The International Journal of Behavioral Nutrition and Physical Activity* 2011; 8(5) doi: http://dx.doi.org.dbgw.lis.curtin.edu.au/10.1186/1479-5868-8-5
28. ABS. Australian Health Survey: Physical Activity, 2011-12 In: Australian Bureau of Statistics, ed. Canberra: Australian Bureau of Statistics, 2013.
29. Mehtälä MAK, Sääkslahti AK, Inkinen ME, et al. A socio-ecological approach to physical activity interventions in childcare: a systematic review. *International Journal of Behavioral Nutrition and Physical Activity* 2014;11(1):22. doi: 10.1186/1479-5868-11-22
30. Borghese MM, Janssen I. Development of a measurement approach to assess time children participate in organised sport, active travel, outdoor active play, and curriculum-based physical activity. *BMC Public Health* 2018;18(1):396. doi: 10.1186/s12889-018-5268-1
31. Marzi I, Reimers A. Children's Independent Mobility: Current Knowledge, Future Directions, and Public Health Implications. 2018;2018:2441. doi: 10.3390/ijerph15112441
32. Zougheibe R, Xia J, Dewan A, et al. Children's outdoor active mobility behaviour and neighbourhood safety: a systematic review in measurement methods and future research directions. *International Journal of Health Geographics* 2021;20(1):2. doi: 10.1186/s12942-020-00254-w
33. Vonderwalde M, Cox J, Williams GC, et al. Objectively measured crime and active transportation among 10–13year olds *Preventive Medicine Reports (PMR)* 2019;13 (2019) (48-51)
34. Suminski RR, Robson SM, May LL, et al. Actual neighborhood-level crime predicts body mass index z-score changes in a multi-racial/ethnic sample of children. *Preventive Medicine Reports* 2018;12:164-69. doi: https://doi.org/10.1016/j.pmedr.2018.09.012
35. Carver A, Timperio A, Hesketh K, et al. Are children and adolescents less active if parents restrict their physical activity and active transport due to perceived risk? *Social Science & Medicine* 2010;70(11):1799-805. doi: https://doi.org/10.1016/j.socscimed.2010.02.010
36. Carver A, Timperio AF, Crawford DA. Neighborhood road environments and physical activity among youth: the CLAN study. *J Urban Health* 2008;85(4):532-44. doi: 10.1007/s11524-008-9284-9 [published Online First: 2008/04/25]
37. Faulkner G, Mitra R, Buliung R, et al. Children's outdoor playtime, physical activity, and parental perceptions of the neighbourhood environment. *International Journal of Play* 2015;4(1):84-97.
38. Loebach J, Gilliland J. Neighbourhood play on the endangered list: examining patterns in children's local activity and mobility using GPS monitoring and qualitative GIS. *Children's Geographies* 2016;14(5):573-89. doi: 10.1080/14733285.2016.1140126
39. Waygood EOD, Susilo YO. Walking to school in Scotland: Do perceptions of neighbourhood quality matter? *IATSS Research* 2015;38(2):125-29. doi: https://doi.org/10.1016/j.iatssr.2014.12.002
40. Mehdizadeh M, Mamdoohi A, Nordfjaern T. Walking time to school, children's active school travel and their related factors. *Journal of Transport & Health* 2017;6:313-26. doi: https://doi.org/10.1016/j.jth.2017.01.012

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

41. Shokoohi R, Hanif NR, Dali MM. Children Walking to and from School in Tehran: Associations with Neighbourhood Safety, Parental Concerns and Children's Perceptions. *Procedia - Social and Behavioral Sciences* 2012;38:315-23. doi: https://doi.org/10.1016/j.sbspro.2012.03.353
42. Tung SEH, Ng XH, Chin YS, et al. Associations between parents' perception of neighbourhood environments and safety with physical activity of primary school children in Klang, Selangor, Malaysia. *Child: Care, Health and Development* 2016;42(4):478-85. doi: 10.1111/cch.12355
43. Oluyomi AO, Lee C, Nehme E, et al. Parental safety concerns and active school commute: correlate across multiple domains in the home-to-school journey. *International Journal of Behavioral Nutrition and Physical Activity* 2014;11(1):32. doi: 10.1186/1479-5868-11-32
44. Roberts JD, Knight B, Ray R, et al. Parental perceived built environment measures and active play in Washington DC metropolitan children. *Preventive Medicine Reports* 2016;3:373-78. doi: https://doi.org/10.1016/j.pmedr.2016.04.001
45. Stephanie H. Kneeshaw-Price, Brian E. Saelens, James F. Sallis, et al. Neighborhood Crime-Related Safety and Its Relation to Children's Physical Activity. *Journal of Urban Health: Bulletin of the New York Academy of Medicine* 2015;92(3) doi: 10.1007/s11524-015-9949-0
46. Zhu X, Lee C. Walkability and Safety Around Elementary Schools Economic and Ethnic Disparities *American journal of preventive medicine* 2008;34(4) doi: 10.1016/j.amepre.2008.01.024
47. Alton D, Adab P, Roberts L, et al. Relationship between walking levels and perceptions of the local neighbourhood environment. *BMJ* 2007;92(1): 29–33. doi: http://doi.org/10.1136/adc.2006.100826
48. Noonan R, Boddy L, Knowles Z, et al. Cross-sectional associations between high-deprivation home and neighbourhood environments, and health-related variables among Liverpool children. *BMJ Open* 2016;6(e008693)
49. Page AS, Cooper AR, Griew P, et al. Independent mobility, perceptions of the built environment and children's participation in play, active travel and structured exercise and sport: the PEACH Project. *International Journal of Behavioral Nutrition and Physical Activity* 2010;7(1):17. doi: 10.1186/1479-5868-7-17
50. Nguyen A, Borghese MM, Janssen I. Pedestrian traffic safety and outdoor active play among 10–13 year olds living in a mid-sized city. *Preventive Medicine Reports* 2018;10:304-09. doi: https://doi.org/10.1016/j.pmedr.2018.04.010
51. Timperio A, Crawford D, Telford A, et al. Perceptions about the local neighborhood and walking and cycling among children. *Preventive Medicine* 2004;38(1):39-47. doi: https://doi.org/10.1016/j.ypmed.2003.09.026
52. Villanueva K, Giles-Corti B, Bulsara M, et al. How far do children travel from their homes? Exploring children's activity spaces in their neighborhood. *Health & Place* 2012;18(2):263-73. doi: https://doi.org/10.1016/j.healthplace.2011.09.019
53. Oliver M, Mavoa S, Badland H, et al. Associations Between the Neighbourhood Built Environment and Out of School Physical Activity and Active Travel: An Examination from the Kids in the City Study. *Health & Place* 2015;36(2015):57–64. doi: http://dx.doi.org/10.1016/j.healthplace.2015.09.005
54. Lin E-Y, Witten K, Oliver M, et al. Social and built-environment factors related to children's independent mobility: The importance of neighbourhood cohesion and connectedness. *Health & place* 2017;107(113)
55. van den Berg P, Waygood EOD, van de Craats I, et al. Factors affecting parental safety perception, satisfaction with school travel and mood in primary school children in the Netherlands. *Journal of Transport & Health* 2020;16:100837. doi: https://doi.org/10.1016/j.jth.2020.100837
56. Santos MP, Pizarro AN, Mota J, et al. Parental physical activity, safety perceptions and children's independent mobility. *BMC public health* 2013;13:584-84. doi: 10.1186/1471-2458-13-584
57. Stark J, Frühwirth J, Aschauer F. Exploring independent and active mobility in primary school children in Vienna. *Journal of Transport Geography* 2018;68:31-41. doi: https://doi.org/10.1016/j.jtrangeo.2018.02.007
58. Davis A, Jones L. Environmental constraints on health: listening to children's views. *Health Education Journal* 1996;55(4):363-74.
59. Oliver M, McPhee J, Carroll P, et al. Neighbourhoods for Active Kids: study protocol for a cross-sectional examination of neighbourhood features and children's physical activity, active travel,

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independent mobility and body size. *BMJ Open* 2016;6(e013377) doi: 10.1136/bmjopen-2016-013377
60. Ding D, Sallis JF, Kerr J, et al. Neighborhood Environment and Physical Activity Among Youth: A Review. *American Journal of Preventive Medicine* 2011;41(4):442-55. doi: https://doi.org/10.1016/j.amepre.2011.06.036
61. SALLIS JF, PROCHASKA JJ, TAYLOR WC. A review of correlates of physical activity of children and adolescents. *Medicine & Science in Sports & Exercise* 2000;32(5):963-75.
62. Pabayo R, Gauvin L, Barnett TA. Longitudinal Changes in Active Transportation to School in Canadian Youth Aged 6 Through 16 Years. *Pediatrics* 2011;128(2):e404-e13. doi: 10.1542/peds.2010-1612
63. Liu W, Liu W, Lin R, et al. Socioeconomic determinants of childhood obesity among primary school children in Guangzhou, China. . *BMC public health* 2016; 16(1):482. doi: 10.1186/s12889-016-3171-1
64. Datar A, Nicosia N, Shier V. Parent perceptions of neighborhood safety and children's physical activity, sedentary behavior, and obesity: evidence from a national longitudinal study. *Am J Epidemiol* 2013;177(10):1065-73. doi: 10.1093/aje/kws353 [published Online First: 2013/04/11]
65. Kelly P, Krenn P, Titze S, et al. Quantifying the Difference Between Self-Reported and Global Positioning Systems-Measured Journey Durations: A Systematic Review. *Transport Reviews* 2013;33(4):443-59. doi: 10.1080/01441647.2013.815288
66. WHO. Equity, Social Determinants and Public Health Programmes: World Health Organization, 2010:303.
67. Salvy S-J, Roemmich JN, Bowker JC, et al. Effect of Peers and Friends on Youth Physical Activity and Motivation to be Physically Active. *Journal of Pediatric Psychology* 2008;34(2):217-25. doi: 10.1093/jpepsy/jsn071
68. Lee C, Moudon AV. Physical Activity and Environment Reserach in the Health Field: IMplication for Urban and Transportation Planning and Reserch. *Journal of Planning Literature* 2004;19(2):147-81.
69. Chataway ML, Hart TC. (Re)Assessing contemporary "fear of crime" measures within an Australian context. *Journal of Environmental Psychology* 2016;47:195-203. doi: https://doi.org/10.1016/j.jenvp.2016.06.004
70. Lorenc T, Petticrew M, Whitehead M, et al. Fear of crime and the environment: systematic review of UK qualitative evidence. *BMC Public Health* 2013;13(1):496. doi: 10.1186/1471-2458-13-496
71. Solymosi R, Bowers K, Fujiyama T. Mapping fear of crime as a context-dependent everyday experience that varies in space and time. *Legal and Criminological Psychology* 2015;20(2):193. doi: http://dx.doi.org/10.1111/lcrp.12076
72. Boessen A, Hipp JR, Butts CT, et al. Social fabric and fear of crime: Considering spatial location and time of day. *Social Networks* 2017;51:60-72.

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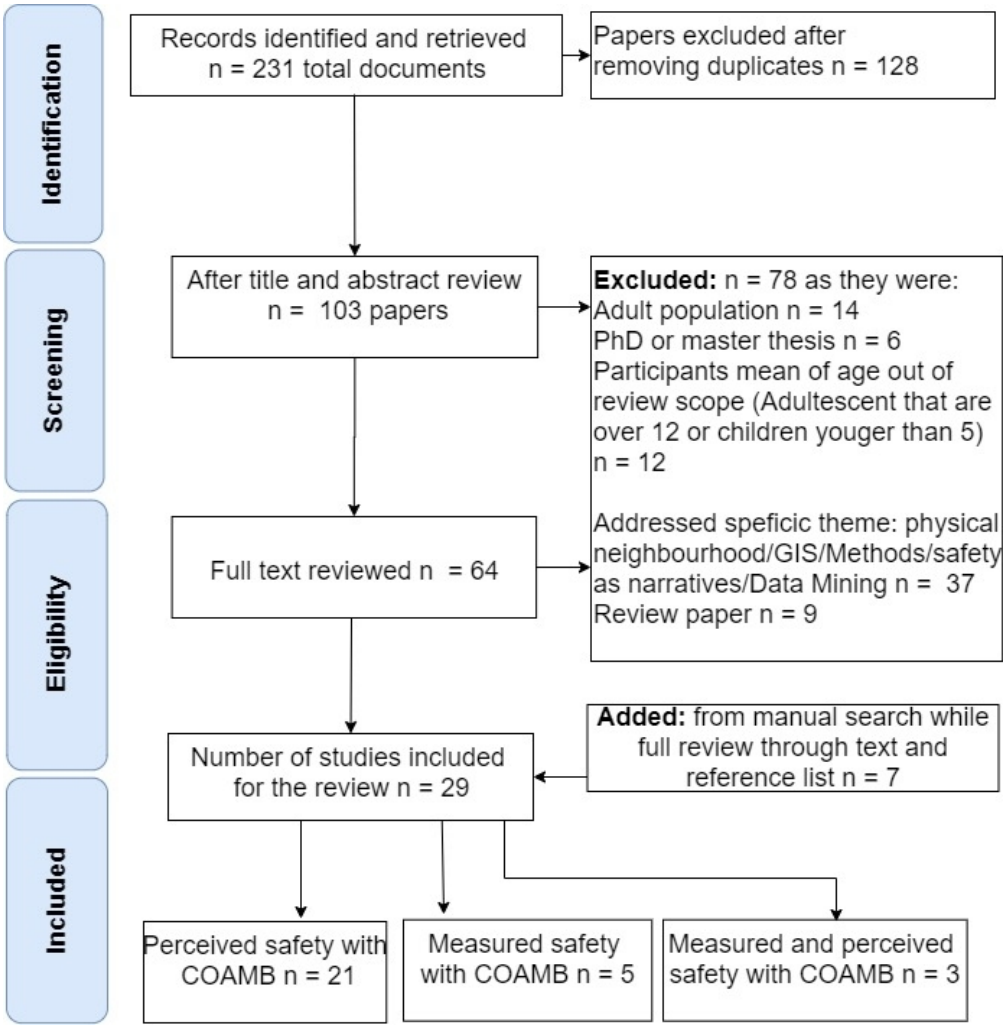


Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses, PRISMA flow diagram for review of evidence on neighbourhood safety correlates to COAMB.

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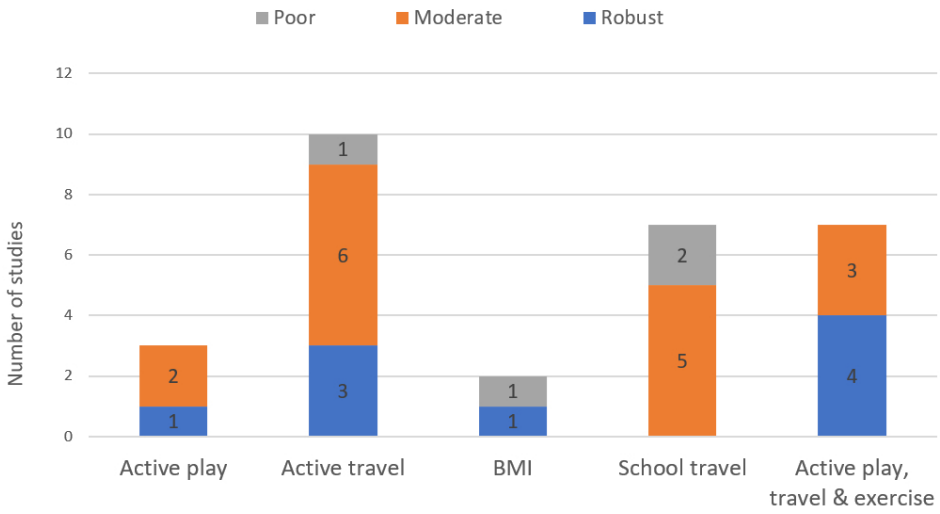


Figure 2. Studies grouped by the type of active behaviour indicating studies' level of methodological quality assessment. The x-axis represents studies grouped by type of active behaviour examined in each study. y-axis denotes the number of studies giving the different levels of evidence 'strength' when total methodological review score was summed. Strength of evidence was robust when summing total score was ≥ 66.66% moderate sum was ≥ 50–< 66.6% and poor when an accumulated score was < 50%. Adapted from previous review by Schoeppe et al.26

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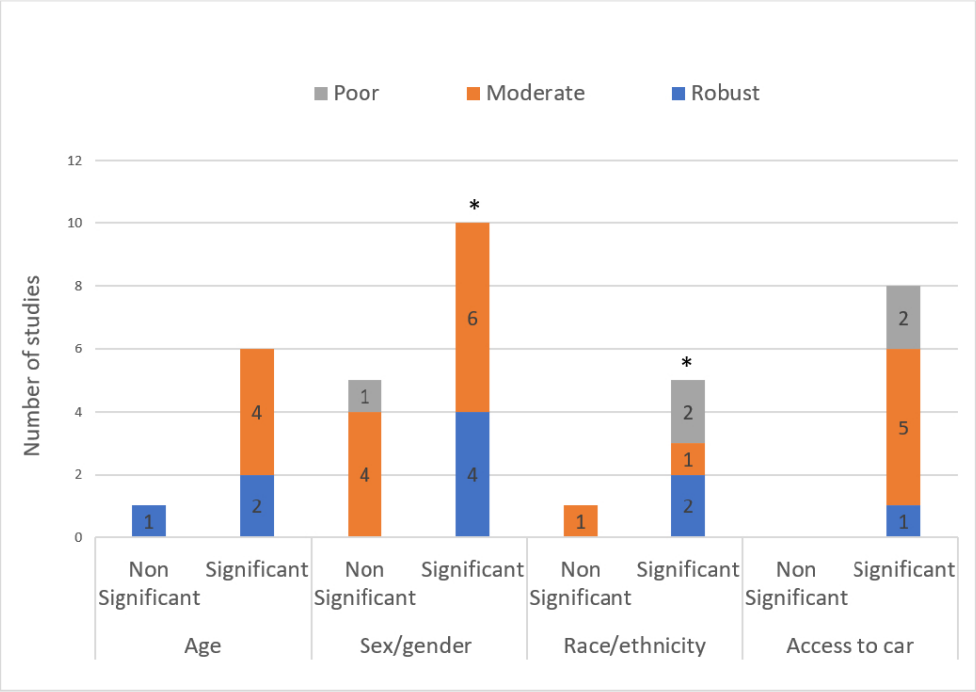


Figure 3. Correlation between individual (child) and family characteristics with children's active mobility behaviour. The x-axis represents variables that accumulated the minimum number of studies to synthesise evidence of correlates between COAMB and sex/gender, age, access to a car, and ethnicity. y-axis denotes the number of studies that examined the correlation showing accumulated significance or non-significance. Note: * = in the sex/gender variable, we saw variability by temporal characteristics (i.e. changes between weekdays and weekends or time of the day (before and after school)) in three studies and reduced COAMB by distance travelled in one study. * = In Race/ethnicity, one study with a significant correlate depicted temporal change (i.e. weekend/weekdays).

90x90mm (300 x 300 DPI)

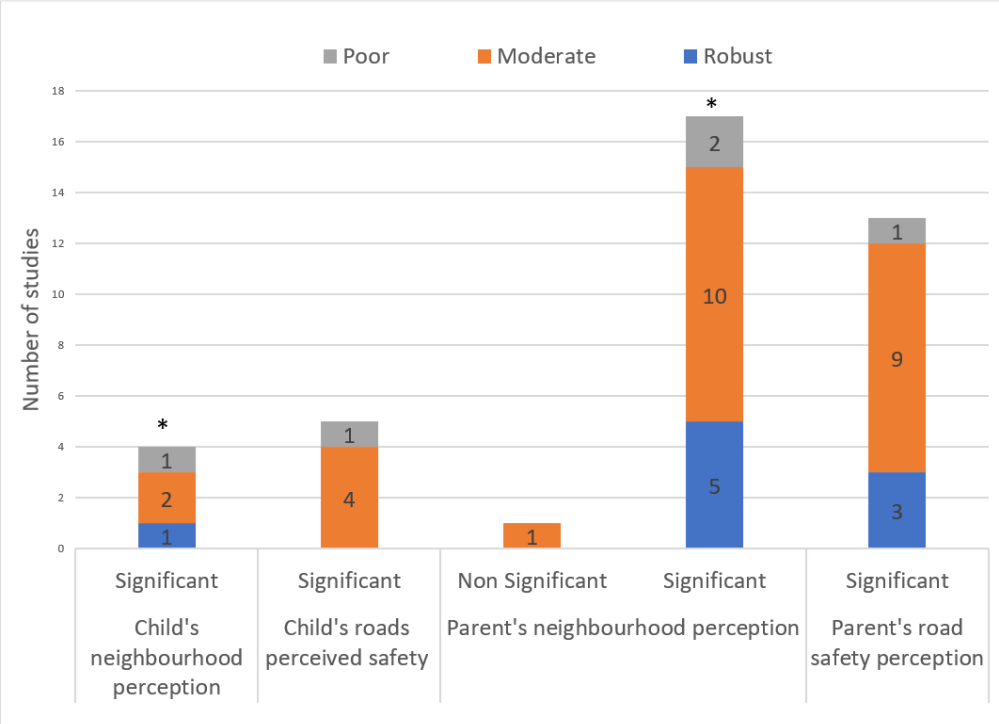


Figure 4. Perceived safety (parents and children) of personal and road danger correlates to reported and measured children's active mobility behaviour. The x-axis represents examined variables that accumulated a minimum number of studies to synthesise evidence for perceived personal and road safety by parents or children. y-axis denotes the number of studies that accumulated significant or non-significance. Note:* = Parents' perceived neighbourhood safety showing significant correlates included three studies depicting temporal characteristics (weekdays/weekends or before and after school), three studies varied by sex/gender and one study by age with one study in perceived road safety varied by gender. * = For children, perceived personal safety varied by gender in one study, but for perceived road safety, one study varied by SES and gender, and two studies depicted variation by gender

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1 of manuscript
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P2 of manuscript
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P3 of manuscript
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P3 of manuscript
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P4 of manuscript
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P4 of manuscript
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	P4 of manuscript
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P4 of manuscript
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P4 of manuscript
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Table 1 supplementary file 2
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Table 1 supplementary file 2
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P5 of manuscript
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P6 of manuscript
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P6 of manuscript
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P6 of manuscript

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Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P6 of manuscript
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P4 of manuscript
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	P5 of manuscript
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P6 of manuscript
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P6 of manuscript
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	P6 of manuscript
Study characteristics	17	Cite each included study and present its characteristics.	Table 2 supplementary file 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 3 supplementary file 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 4 supplementary file 2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P6 of manuscript
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Table 4 supplementary file 2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P7 of manuscript
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Table 2supplementary file 2,
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P7 of manuscript
DISCUSSION			

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P8 of manuscript
	23b	Discuss any limitations of the evidence included in the review.	P8 of manuscript
	23c	Discuss any limitations of the review processes used.	P8 of manuscript
	23d	Discuss implications of the results for practice, policy, and future research.	P8 of manuscript
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	N/A
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	12 of manuscript
Competing interests	26	Declare any competing interests of review authors.	of manuscript
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	of manuscript

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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Search Terms Used

(1) **target population** using terms such as "Children" "youth" or "kids" or "primary school aged children", (2) **exposures of social neighbourhood** "Perceived safety from crime", "Social neighbourhood", "Neighbourhood safety", "Measured crime", "Fear of crime", or "Perception of crime", "road safety", "traffic safety". (3) **moderators and mediators** of "Parents' perception of safety from crime" "parents' and children's perception of safety from crime" or "children perception" "age 5-11" "gender", "ethnicity". (4) The **spatiotemporal** aspects: "spatiotemporal measures, Geographic Information Systems GIS, time geography", "space and time geography" "Objective measures" or "GPS". (5) the **outcome** such as "Active Mobility", "Free play", Medium to Vigorous Physical Activity". "Outside school hours". A combination of at least three keywords was alternated to find the possible variety of papers related that could be scanned and retrieved.

Search strategy in Scopus

Search by document type (ALL), Search in (Article title, Abstract, Keywords), Access Type (ALL), Limit is (Published all years). The source type is articles. Language is in English. Keywords (e.g., "Children" OR "primary school aged children" OR "youth", OR "Active mobility" OR "Active behaviour" OR "Physical activity" OR "Active play" OR "active travel") AND ("Neighbourhood") AND ("Safety" OR "Parents Perceived Safety" OR "Children Perceived Safety" OR "Road safety" OR "Crime") AND ("Outside school hours)" AND ("Spatial data" OR "GIS" Or "GPS" OR "Spatiotemporal").

Search strategy in Google scholar

Search by document type (Article), Search alternate keywords (e.g., "Children" OR "primary school aged children" OR "youth", OR "Active mobility" OR "Active behaviour" OR "Physical activity" OR "Active play" OR "active travel") AND ("Neighbourhood") AND ("Safety" OR "Parents Perceived Safety" OR "Children Perceived Safety" OR "Road safety" OR "Crime") AND ("Outside school hours)" AND ("Spatial data" OR "GIS" Or "GPS" OR "Spatiotemporal").

Search strategy in PubMed

Search Terms: Include related keywords terms of (e.g., "Children" OR "primary school aged children" OR "youth", OR "Active mobility" OR "Active behaviour" OR "Physical activity" OR "Active play" OR "active travel") AND ("Neighbourhood") AND ("Safety" OR "Parents Perceived Safety" OR "Children Perceived Safety" OR "Road safety" OR "Crime") AND ("Outside school hours)" AND ("Spatial data" OR "GIS" Or "GPS" OR "Spatiotemporal"). Select English language, Publication Year No limit

Search strategy in Science Direct

Search for peer-reviewed journal articles (including open access content)

Find articles with these keywords' terms of (e.g., "Children" OR "primary school aged children" OR "youth", OR "Active mobility" OR "Active behaviour" OR "Physical activity" OR "Active play" OR "active travel") AND ("Neighbourhood") AND ("Safety" OR "Parents Perceived Safety" OR "Children

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Perceived Safety" OR "Road safety" OR "Crime") AND ("Outside school hours)" AND ("Spatial data" OR "GIS" Or "GPS" OR "Spatiotemporal"). No limit for Publication Autor(s) or Year(s)

Search strategy in ProQuest

Use command line to use search terms keywords such as (e.g., "Children" OR "primary school aged children" OR "youth") AND ("Active mobility" Or "Active behaviour" OR "Physical activity" OR "active play" OR "active travel") in Anywhere AND ("Neighbourhood Safety" OR "Parents Perceived Safety" OR "Children Perceived Safety" OR "Road safety" OR "Crime") in Anywhere. Add a row of AND ("Outside School Hours") in Anywhere. Add a row of AND ("Spatial Data" OR "GIS" Or "GPS" Or "Spatiotemporal"). Limited to (Peer reviewed), Publication date (All dates), Language (English).

Search strategy in Web of Science

Basic Search: Alternate search terms keywords such as (e.g., "Children" OR "primary school aged children" OR "youth", OR "Active mobility" OR "Active behaviour" OR "Physical activity" OR "Active play" OR "active travel") AND ("Neighbourhood") AND ("Safety" OR "Parents Perceived Safety" OR "Children Perceived Safety" OR "Road safety" OR "Crime") AND ("Outside school hours)" AND ("Spatial data" OR "GIS" Or "GPS" OR "Spatiotemporal"). Search in Topic. Timespan (all Years). Default Number of Search Fields to Display

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Table 1. Characteristics of included studies and findings of evidence of influences on children outdoor active mobility behaviour

#	Study's author(s)/year of data collection n = Participants number/ (sex/gender: number/% of M/F)/ Age (or mean age)/Grade	Country/ Study Design/ [Project]	Measures Safety (Perceived by)/ Measured/ Level of study	Active Behaviour Outcome / Measures	Definition of neighbourhood	Study accounted cofounders and other examined variables	Findings of Associations
1	Alton, Adab, Roberts, and Barrett (2007)/-/ n= 473 (250 M and 60 F)/ 9-11 years old/	United Kingdom Cross-section	Children/questionnaire on the perception of local environment + preferred travel method/ In Six primary schools in a range of socio-economic classification	Level of walking (high walkers, low walkers)/ Self-report Past 7-days	Local area	Sex/gender, race/ethnicity (Asian, black, Chinese, mix, others, white), family characteristics (cars ownership and number of rooms in primary residence) as a proxy for SES)	Child from ethnic minority walks significantly less. Perception of high parental concerns over road safety and heavy traffic associated with a higher walking level. The authors explained this unexpected correlate as children who are high walker may be often warned by their parents about road danger, and therefore children perceive roads as dangerous. Child perception of lack of suitable leisure space in the neighbourhood and worry about a stranger is associated with less walking levels. Child perceptions of the local environment correlate with walking level.
2	(Carver, Timperio, & Crawford, 2008)/between July and December 2004/ n = 188 (44% boys) 8-9 years old	Australia Cross-section from [CLAN] longitudinal study	Parents survey for personal characteristics +frequency of children walking and cycling to 15 destinations + Measured road safety[1]/ From 19 state primary schools of varying socio-economic states across Melbourne.	*MVPA Outside school hours Accelerometer for 8-days	800 m around participant home	Sex/gender, age + Active transport +MVPA + measured road environment	No association between children likelihood of making at least seven walking/cycling trips per week to neighbourhood destination and roads environment
3	Carver, Timperio, Hesketh, and Crawford (2010)/2001 n = 170 (51% M)/ 10-11 years old 15-17 years old (excluded) 2001-2005 longitudinal study	Australia Cross-section from [CLAN] longitudinal study	Parents questionnaire Indices for (avoidance and defensive behaviour, and perceived risk) + active transportation to 15 destination/ In 10 high and ten low socio-economic areas across Melbourne	*Active transport/MVPA outside school hours/ Accelerometer	Local area	Sex/gender + Parents (avoidance, defensive and perceived risk)	The lower level of active transport and lower level of MVPA in a neighbourhood outside school hours associated with constrained behaviour exhibited by parents on both boys and girls. Reduced MVPA associated with constrained behaviour on weekends for boys. Higher constrained behaviour associated with higher MVPA for girls (limitation of study)
4	(Carver, Panter, Jones, & van Sluijs, 2014)/T1 (April-July 2007) n= 1121 (9-10) years old (43%M) T2 (April-July 2008) n= 491 (39%M)	United Kingdom Longitudinal from [SPEEDY] study	Parents perception questionnaire (social/physical environment + rules regarding their children physical activity + perception of traffic safety concerns)/At 1600m of their school in urban/rural areas	Independent mobility to school/ from children questionnaire	Within 800 m pedestrian network buffer around the home (10 min walk)	Sex/gender, sociodemographic (siblings, cars ownership, parents' education) + environmental characteristics around the home and (within 100 m buffer of the shortest route to school	Car access is associated longitudinally with boys decreased odd of walking/cycling independently to school. The proportion of main roads in the neighbourhood and parental encouragements of walking/cycling associated longitudinally with girls walking/cycling to independently to school. Land use mix is associated positively with girls walking cycling independently to school. Boys that are allowed by their parents to play outside have higher odd to walk/cycle independently to school.
5	(Davis & Jones, 1996)/-/ n = 492 of children (not reported gender)/ 9-11-years-old 13-14 years old (excluded)	United Kingdom Cross-section	Children (focussed group discussion)./ From four schools in broadly working-class areas.	Independent mobility/ discussion	Local destination	Sex/gender, age + stranger danger and traffic danger	In children's view, traffic and stranger danger, social and cultural factors create barriers on children being active for both genders, especially for girls. Car escort journey developing unhealthy habits of sedentary living with its associated risks of increased cardiovascular illness.
6	(Fagerholm & Broberg, 2011)/ Sep.-Oct. 2009 n = 35 children (18M/17F)/ 10-11 years old	Finland Cross-section	Parents questionnaire (children and parent) mobility patterns + mobility licences, perceived safety./From two residential areas	*Active route (home to destinations)/ GPS + Travel diary for 7-days	Buffer 500-m from home	Sex/gender + Land use types	Gender difference shows in the distance and speed children travel actively to the boys' advantages. Land-use type associated with different mobility patterns (increase in urban areas). A high perception of safety from parents in residential areas resulted in a high level of independent mobility.

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							Weekdays and weekends have different mobility patterns in term of proximity (near the home on weekdays) with more time out on weekends.
7	Faulkner, Mitra, Buliung, Fusco, and Stone (2015)/ April2010 – May2011 n = 736 (47% M and 52%F) included in the analysis/ (10-12 years old), grade 5-6	Canada Cross-section from [BEAT] project	Parents questionnaire (child outdoor active play + parents' perception of personal and road the neighbourhood safety)/ From 16 elementary primary/intermediate schools in the city of Toronto	*Outdoor playing time MVPA/ GPS + Accelerometer for 7-days	Neighbourhood	Sex/gender, age, SES of the neighbourhood (neighbourhood income) + neighbourhood perception (roads, personal safety, accessibility of facility)	Time spent outdoor was significantly associated with (MVPA) and with sex/gender, but age didn't play a role in this relation. Parental safety perception plays an inverse relation with the duration of outdoor play on weekdays. Parental concerns play a barrier role for children outdoor play. Association of outdoor play duration with the perception of safety (stranger danger and traffic safety), differ between weekdays and weekends.
8	(Janet E. Loebach & Jason A. Gilliland, 2016)/ During April and May of 2010 and 2011. n = 143 (49M/94F) two groups of 9- 11 and 12-13 years-old)/grade 5 - 8	Canada Cross-section from [STEAM] project	Parents & Children Questionnaires/ From seven schools in London	Neighbourhood Activity Space (NAS)/ GPS for 7-days	NAS found within 400, 800 m of home, the second set those found within 1,600 m.	Sex/gender, age, car availability + parents and children environmental perception + neighbourhood types + parental IM licenses to children	IM awarded to a child is associated with parents' perception of neighbourhood safety and a strong predictor for distance travelled. Distance to school predicts active travel. Gender or age was not associated with NAS size. No association between parents' perception of neighbourhood and child spend their time closer to home. Children perception correlate with time spend closer to home but were not as strongly predictive as parental mobility restriction. Neighbourhood types (Residential) was a predictor of time spend out and (commercial) to distance travelled (over 800 m). Smaller home to school distance facilitate active travel and more frequent and distant neighbourhood travel
9	(Helbich et al., 2016)/between December 2008 – April 2009/ n = 97 (60%F) /Aged 6 – 11 years	The Netherlands, Part of [SPACE] project	Road traffic Safety exposures within 100 m buffer width around using GIS/ From six elementary schools located in five neighbourhood	Active trips to school and transport mode of choice GPS for 8-days	Home to school trip	Sex/gender, age + active trips to school + built environmental variables (land use mix, density, closeness and in between indices) + traffic safety control variables (major roads availability, distance, the proportion of cycling path and road accidents) +street density) weather	Gender is significantly associated with AST. Age is significantly associated with AST. Negative association with the distance. Negative weak association of AST with major road. Traffic safety (exposure to major roads/highways) is negatively correlated with AST. Cycling path availability positively corollate with AST.
10	Lin et al. (2017)/between 2011 and 2012 n = 254 (100M, 133F) and 239 parents/ 8–13 years (mean age of 10.5)	New Zealand from [KITC]	CATI-Parents Survey (perception of neighbourhood cohesion using social cohesion scale, + neighbourhood social connection using intergenerational closure scale and parents' concerns of places that will not let their children go alone)/ From nine schools across Auckland	*Independent mobility /Travel diary for 7-days	Neighbourhood	Sex/gender, age, race/ethnicity, household profile + built environment (street connectivity, destination accessibility)	Shorter distance to school accounted for the effect of neighbourhood cohesion and connection associated with higher children independent mobility (IM). IM associated with having an older sibling and limited access to cars. Parent's perception is not associated with children independent mobility though concerns of safety which differed by ethnicity (pacific/Asian).
11	Mehdizadeh, Mamdoohi, and Nordfjaern (2017)/-/ 735 parents of (364M, 371F)/ 7-9-year-old in 9 schools of 735	Iran Cross-section	Parents questionnaire via their children/from nine schools	Perceived Walking Time to school (PWTS)/ questionnaires	Environment around school	Sex/gender, demographics, household characteristics (father/mother driving licence, car owner) SES	Perceived walking time to school (10 min) is the maximum threshold where the proportion of active mode started to decrease. Certain demographics: parental age, household income, accessibility to public transport, type of school (public or private), school service status and psychological factors (parents attitude towards walking in dirty, vandalised and unsafe streets), SES of children and household variables were significant predictors of a PWST to eligible school. Sex didn't have a consistent and definite role in active travel.
12	(Nguyen, Borghese, & Janssen, 2018)/ Jan. 2015 and Dec. 2016/ n= 458 (230M/228F)/ aged 10-12	Canada Cross Section	Parent's questionnaire (perceived pedestrian safety) + Objective measures of Pedestrian safety/ Recruited through social media and study flyer.	Average of minutes per day of active outdoor play/ Accelerometer + GPS in a smartwatch for 7-days	1 km buffer zone around participants home	Sex/gender Race (white, non-white) Family characteristics (single or dual parents' household, number of siblings, household income, parental education, parents' value of outdoor	No association between pedestrian safety measures and outdoor active play. No association between objective and perceived pedestrian measures. Perceived measures of traffic volume, traffic calming and pedestrian infrastructure were not associated with outdoor active play.

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						and income) + Pedestrian safety (traffic volume, traffic speed, traffic calming and pedestrian infrastructure + season	Parents perceived moderate to high traffic speeds had higher outdoor active play value than children whose parents perceived low traffic speed. Objective measure of traffic volume, traffic calming and pedestrian infrastructure but not traffic speed was associated with outdoor active play.
13	Noonan, Boddy, Knowles, and Fairclough (2016), n = 194 children/ 9-10 years old Gender not reported	United Kingdom Cross-section	Parents questionnaire on environment perception using NEWS_Y + Children self-reported PA using PAQ-C. From 10 primary schools in Liverpool	IM/ Self-Reported PA	High and low deprived areas	Home environment, Area deprivation, parent's perception walkability Index	Home environment for HD provides more opportunities for sedentary behaviour and less opportunity for PA, less access to bedroom media equipment, and greater independent mobility were *strongly associated* with higher PA in HD and MD children, respectively.
14	Oliver et al. (2015)/ Between 2011 and 2012/ n = 236 (104 M/132) for weekday analyses, n = 210 (91M/119F) children for weekend days analyses. /age 9-13 Age mean 9.8 from 9 schools/grade 5-8	New Zealand Cross-section from [KITC] project	CATI-Parents questionnaire (Perceived parents safety) + measured roads connectivity/ From nine schools in Auckland	*% MVPA/ accelerometer + GPS for 7-days	Buffer of 800-1000 m around school	Sex/gender, age, race/ethnicity (New Zealand European, Maori, Pacific Island, Indian/Asian/Other Ethnicity), SES+ Neighbourhood street connectivity, street space, destination accessibility+ distance to schools + ratio of high-speed roads around schools + street connectivity. .	On weekdays: Females & access to car accumulated less %MVPA than males. On weekends: Female and ethnicity (Indian, Asian or "other" ethnic group) made less %MVPA. Street connectivity and distance to school were related to the proportion of active trips on both weekends and weekdays. The ratio of high-speed roads associated with %MVPA (weekdays after school). Improved streetscape for active travel was related to %MVP on weekdays. Ethnicity and %MVPA differ by day type. Age, access to cars was negatively associated with %MVPA. Inconclusive evidence of socio-economic association. Parent's perceptions of neighbourhood safety is positively associated with the proportion of active trips on weekdays.
15	(Oluyomi et al., 2014)/-/ n= 830 and their parents of 4th grade (412M/418F)	United States Cross-section from [T-COPPE] longitudinal project	Parents' questionnaire adapted from several surveys including the National Centre for Safe Routes to School Parents Survey, SPAN, (UH-PEAK), NEWS, and EnVivo) Personal safety + Traffic Safety/ From 81 elementary school across Texas	Walking to school (WTS) /parents' questionnaire of National Safe Route to School. Captured safety of en-route to school, home neighbourhood, and school environment)	Within walking distance of 3.2 km to school from the residential address of students	Race/ethnicity (the majority were Hispanic) car ownership, public assistance) Examined two environments (home neighbourhood and en-route environment to school)	Parents Perception of road safety (higher sidewalk availability, well-maintained sidewalks and safe road crossing) is associated with students more likely to walk to school. On route to school: parents' perception of sidewalk, Speed and amount of traffic and intersection along school route, also associated with WTS Parents reported safe walking to a school associated with reported higher children WTS. On Personal safety, parents concern about general neighbourhoods' safety, stray or dangerous animals and the availability of adults with whom child can walk associated with lower en-route to school.
16	(Page, Cooper, Griew, & Jago, 2010)/ between 2006 and 2008 n = 1307 (639 M, 661 F)/ 10-11 years old from 23 schools	United Kingdom Cross-section from [PEACH] longitudinal study	(computerised) children questionnaire perception of the environment (Aesthetics, Safety, Social Norms, Nuisance, constraints, accessibility, minutes of daylight from 3 pm till sunset)/ From 23 schools.	Frequency of participation in outdoor play, exercise and active travel home to school/ questionnaire	School-home	Age, sex/gender, race/ethnicity (white, non-white, but not accounted in analysis), + perception of the environment + the level of deprivation (using Index of Multiple Deprivation (IMD) and derived from seven categories of deprivation + daylight + pubertal status	Boys had a more positive perception of the environment than girls (personal safety and traffic safety) for Local-IM and Area-IM. No gender differences in school travel Boys scores significantly lower in the Constrained scale than girls Physical activity, structured sport and active travel to school weakly significantly correlate to each other. Boys who have greater independent mobility scores had more time of playing out than boys who played less. Girls with a higher positive perception of neighbourhood correlate with playing outdoor often. For boys and girls, and increased likelihood of walking or cycling to school was associated with higher levels of local-IM. Distance from home to school is a predictor for both boys and girls
17	Roberts, Knight, Ray, and Saelens (2016)/Sep.-Dec.2014/ n = 144(72M,72F)/7-12 years old (mean age of 9.7 children)/	United States Cross-section	Parents' questionnaire (perception of the environment)/ From nine counties and cities.	Active children (met the 60 min daily PA)/	Metropolitan area	Sex, demographics race/ethnicity (Hispanic/Latino, African American, American, Indian/Alaska Native, Asian American, White, Other) + perceived parents' safety	Parents' better perception of neighbourhood associated with active kids yet reported a high crime rate and being a victim of a crime of their neighbourhood. Parental perception of street barriers associates with children physical activity. Closer proximity to play areas was significantly associated with greater odd of children meeting the 60 min/day play. Gender (male more active) and race disparities associated with active behaviour.

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18	(Santos, Pizarro, Mota, & Marques, 2013)/ 2010/2011 n= 354 (156M) of grade 6th. (mean age 11.63) and their parents	Portugal Cross-section from [SALTA] longitudinal study	Parents questionnaire (parental physical activity, and perception (adapted from NEWS and previous studies) + Children questionnaire to derive mobility style/ From nine middle schools	IM/ questionnaire of previous week physical activity based on IPAQ	Local destinations	Age, sex/gender, family demography (parents age, education, parental PA, parents' perception of neighbourhood safety (sidewalk, street safety, fear from strangers, crime and traffic safety).	Parental perception of sidewalks and street safety is associated with IM Parents physical activity was significantly associated with more active children. Perception of fear from a stranger, crime and traffic safety was not significantly associated with child IM.
19	(Shokoohi, Hanif, & Dali, 2012)/ Jan.-Feb. 2009 /-/ Grades 3- 5 (48.8%M,51%F)	Iran Cross-section	Parents + children questionnaire on the perception of environmental factors that prevent children from walking to school/From 18 school sites	Walking to school from parent's survey. Differed the trips from home to school and from school to home.	Home-school	Socio-economic status (three income groups of parents)	Parents and children with a negative perception of neighbourhood safety tended to use motor vehicles or to escort their children while walking to and from school.
20	Stark, Frühwirth, and Aschauer (2018)/-/ n = 190 (49%F) from two public schools/ 6-9(10) years old	Austria Cross-section	Parents questionnaire. From two schools of different location in the outskirt of the city of Vienna	AIM/ One weekday and one weekend Travel Diary using (KONTIV-format)	School active travel	Parental attitude, Parents and household characteristics	The most influential variables on IM are: Shorter trip distance to school, higher age and parents perceived social safety and traffics. Working status of parents. The type of school (all-day/half-day primary school) is relevant. Parental attitude (Promoters, Pragmatists and Protectors) strongly influence the degree of AIM.
21	Stephanie H. Kneeshaw-Price et al. (2015)/between Sep. 2007 and Jan 2009 n = 145 (71M, 74F) / 6–11-years-old	United States Cohort study [NIK]	Police report crime + parents' prior crime victimisation survey + parents' perception (stranger danger, general crime, and disorder)	MVPA/ accelerometer for 7-days	Census block	Age, race/ethnicity (non-Hispanic white, Hispanic, and non-Hispanic non-white), Household income + NEWS + collective efficacy	Lower children's physical activity was associated with more neighbourhood crime but not with parents' perception of stranger danger, disorder and neighbourhood safety. Lower MVPA didn't account for race/ethnicity as the majority was non-Hispanic (white).
22	Suminski, Robson, May, Blair, and Orsega-Smith (2018)/ T1 n = 2108/50.5%F/ 5-11 years-old	United States longitudinal study	Measured crime using CRI Index (a higher number than 100 indicate a greater chance of crime) for each zip code in an urban neighbourhood	Body Mass Index/(BMI) score/ measured at baseline and three years later	Zipcode	(BMIz scores, Sex/gender, family characteristics (income, education Race/Ethnicity White, African American, Hispanic, Asian)	Actual, neighbourhood-level crime predicts changes in BMIz scores in white children This relationship varies as a function of race/ethnicity (i.e. in White children but not African American, Hispanic or Asian American)
23	Timperio, Crawford, Telford, and Salmon (2004)/ n = 291 (150M/141F)/ Aged 5–6 years n = 919 (424M/495F)/ aged 10–12 years from 19 primary schools	Australia Cross-section	Parents & Children Survey Compared parents to own children (aged 10-12) view	Frequency of walking and cycling/ parents survey	Local destinations	Sex, Family background: the first language spoken, marital status, parents' education (SES), cars' ownerships, siblings	Parents' perception of the neighbourhood is *associated* with a lower likelihood of walking or cycling (girls in particular). Children perceive parents' perception of their local neighbourhood safety more negative than their own. Children are reporting less concern about heavy traffic, stranger dangers and road safety and lack of parks or sports ground than parents. Age & SES is associated positively with the frequency of trips made to destinations. Sex is not associated with the frequency of active trips for the young group, but Boys are more still active than girls in the 10 – 12 years old.
24	(Tung, Ng, Chin, & Mohd Taib, 2016) n = 256 (42%M/58%F and their parents aged 9-12 years	Malaysia Cross-section	Parents perception using NEWS Constrained behaviour of parents	Children PA level 7-days questionnaire	-	Sociodemographic (age, parents' gender and ethnicity (Malay, Chinese, and were Indian), highest education level, parent's occupation and monthly income) + parents' perception+ children PA	Sex/gender associated with the level of PA (boys accumulate more than girls). Correlates found between parental perception of neighbourhood safety and constrained behaviour of children active play. Land use mix (access) was correlated positively with higher PA level
25	(van den Berg, Waygood, van de Craats, & Kemperman, 2020)/ Fall of 2018/(315M/341F)/ n= 660 children and parents/ aged 7 – 12/ grade 5-8 (mean age 9.5)	The Netherland Cross-section	Parents survey for perceived pedestrian safety Objective measures of Pedestrian safety/ From 14 primary school in the Netherland	Travel mode to school/ children survey (at school)	Participants were of Home-school Within 1 km distance	Age, Sex/gender Household (income, car ownership), weather, street connectivity	Parental perceptions are related to the child's age, income, perception of neighbourhood infrastructure, travel distance and social cohesion. Parents that are less concerned let their children travel actively
26	Vonderwalde, Cox, Williams, Borghese, and Ian Janssena (2019)/ between Jan. 2015 and Dec.2016/	Canada	Measured crime (person, Property) 24 months before measures of active behaviour	Active transportation/ GIS loggers	Crime in 1 km distance to	Sex/gender, age, race (85% white)	Children living in neighbourhoods in the highest neighbourhood crime rate quartile engaged in significantly more active transportation than children living in neighbourhoods in the lowest neighbourhood crime

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	n = 387(185M, 182F)/ 10–13 years old (mean age 11.5)	Longitudinal from [Active Play Study]			participants home	family profile (single or dual-parent, parent income education), walkability index+ Season change	rate quartile and persisted after adjustment for several individual, family, and environmental covariates. Association of AS and crime against a person but not against a property.
27	Villanueva et al. (2012)/ July – December 2007/ n = 926 (463M,463F)/ 10-12 years old included in the analysis.	Australia Cross-section from [TREK]	Parent and Child Questionnaires/ Schools in neighbourhood within three socio-economic state (low, medium, high)	Activity Space/ steps count Pedometer + mapping activity for 7-days	within 800 and 1600 meter of child's home high and low walkable areas	Age, sex/gender, maternal education + School-specific walkability level	Children in high walkable neighbourhood's schools had larger AS. Girls had smaller AS for if parents perceived living on a busy road. Utilitarian destinations were associated with a smaller AS. Sex is associated with AS areas and daily pedometer. Boys' and girls' own confidence in travelling independently was positively associated with covering larger AS areas. Despite some environmental barriers, individual and social-cultural factors likely to encourage children to roam in their local environments.
28	(Waygood & Susilo, 2015)/ 2006 survey/ n = Roughly 31,000 households' participants (52% boys) aged 10-11 years	Scotland Cross-section	Parents survey/ from 2006 Household Survey	Walking to school	Home to school	Sex/gender, Family background +built environment, deprivation index	Car's availability is negatively associate with walking to school Distance to school was strong determinate for children walking to school. Parental perception Good local shops are positively associated with children active travel to school and slow traffic or safe was negatively associated variable,
29	Zhu and Lee (2008)/-/ Children from 73 elementary schools/-/	United States Cross-section	Measured Crime Geocoded 8 major crimes rate/ From 73 public elementary schools	Potential walkers/ neighbourhood-level walkability index	School attendance areas	Race/Ethnicity (Hispanic, non-Hispanic, and White), Poverty + Neighbourhood-level walkability + Neighbourhood-level safety SWI	Poverty associated with negative conditions and low perceived safety. Ethnicity associated with increased crime, traffic danger and poor safety. Unsafe neighbourhoods and poor street conditions may influence not only children's school travels but also their play activities and the overall physical activities of all residents.

Notes: /-/ = data was not reported, M = males, F = Female, BMI = Body Mass Index, IM = independent mobility, AST = active school transport, PA = physical activity, AS = activity space, SES = socioeconomic status, HD = high deprivation, MD = medium deprivation, WTS = walking to school, NAS = Neighbourhood Activity Space, CLAN = Children Living in Active Neighbourhoods, SES = Socioeconomic status, MVPA = Medium-to vigorous Physical Activity, SPEEDY = Sport, Physical activity and Easting Behaviour Environmental Determinants in Young People, BEAT = Built Environment and Active Transport, GPS = Global Positioning System, STEAM = Spatio-Temporal Exposure and Activity Monitoring, KITC = kids in the City, PAQ-C = Physical Activity Questionnaire for Older Children, NIK = Neighbourhood Impact on Kids, CRI = Crime Risk Index (measured crime using actual crime statistics) TREK = Travel Environment and Kids Project, CATI = Computer-aided Telephone Interview, KIC = Kinds in the City, T-COPPE survey = Texas Childhood Obesity Prevention Policy Education project, IMD = Index of Multiple Deprivation is a composite score based on seven categories of deprivation (income, employment, health and disability, education skills and training, housing and geographical access to service), PEACH = Personal and Environmental Association With Child's Health, SPAN = School Physical Activity and Nutrition, UH-PEAK = Urban Hispanic Perceptions of Environment and Activity Among Kids, En Vivo = TV reduction intervention study, NEWS = Neighbourhood Environment Walkability Scale, NEWS-Y = Neighbourhood Environment Walkability Scale for Youth used to assess parental perceptions of neighbourhood design, SALTA = Environmental Support for Leisure and Active Transport, KONTIV = format of travel diary survey for non-home activity patterns, GIS = geographic information systems, IPAQ questionnaire = International Physical Activity Questionnaire, Local-IM = destinations of best friend's house, school, local shops and park or playground, Area-IM = destinations of swimming pool, library, cinema, arcade, bus stop, sports and shopping centre, SWI = School Walkability Index derived from network connectivity and traffic volume, Neighbourhood-level walkability index derived from an (estimate of potential walkers, pedestrian facilities, residential density, street connectivity, land use mix), Neighbourhood-level safety = derived from (traffic danger and the crime rate in a year), TREK = Travel Environment and Kids

Studies denoted with * = Study measures and analysis accounted for temporal changes (weekend/weekdays or outside school, i.e., before and after school hours).

[1] = Objective road measures using GIS were: street network total length, local road index, No of intersections, the total length of walking track, no of speed humps, no of traffic /pedestrian lights, no of barriers

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Table 2. Methodological measure quality assessments and output per study.

Study citation	Study Objectives, design, target population, random sampling (0.25 each)	study participants, inclusion/exclusion, study population, participants recruitments (0.25 each, total 1)	Response rate, data collection, data sources, missing data. (0.25 each, total 1)	Did the active behaviour data collection was objectively measured? (0.25 for each method, total 1 point,)	Did the measure of active behaviour account for temporal characteristics? (0.5 point)	Has the perceived safety measured the temporal characteristics? Or has measured safety used geocoded data in actual crime for personal safety or in road safety? (0.5 point)	Did the study delineate the exposure area "neighbourhood" objectively (0.5 point)	Did the study evidence Accounted For spatiotemporal behaviour in output (1 point)	Was it clearly described the statistical method and to assess significance association, or did the study describe the method of spatial analysis? (0.5)	Did the study account for the four cofounders (age, sex, ethnicity and family characteristics) (0.25 each, total 1 point)	Quality score total	%	overall rating of evidence Quality
(Alton et al., 2007)	1	0.75	0.5	0.25	0	0	0	0	0.5	1	4	50.0	Moderate
(Carver et al., 2010)	1	1	0.75	0.5	0.5	0.5	0	1	0.5	0.5	6.25	78.1	Robust
(Carver et al., 2014)	1	1	0.75	0.25	0	0	0.5	0	0.5	0.75	4.75	59.4	Moderate
(Carver et al., 2008)	1	1	0.75	0.25	0	0	0.5	0	0.5	0.75	4.75	59.4	Moderate
(Davis & Jones, 1996)	1	0.5	0.25	0.25	0	0	0	0	N/A	0.5	2.5	35.7	Poor
(Fagerholm & Broberg, 2011)	1	0.75	1	1	0.5	0	0.5	0.5	0.5	0.5	6.25	78.1	Robust
(Faulkner et al., 2015)	1	1	1	0.5	0.5	0	0	0.5	0.5	0.75	5.75	71.9	Robust
(Lin et al., 2017)	1	0.5	0.75	0.25	0	0	0	0	0.5	1	4	50.0	Moderate
(Janet E Loebach & Jason A Gilliland, 2016)	1	1	1	0.5	0.5	0	0.5	0	0.5	0.75	5.75	71.9	Robust
(Helbich et al., 2016)	1	0.75	0.5	0.25	0	0.5	0.5	0.5	0.5	0.5	5	62.5	Moderate
(Mehdizadeh et al., 2017)	1	1	1	0.25	0	0	0	0	0.5	0.75	4.5	56.3	Moderate
(Noonan et al., 2016)	1	1	0.75	0.25	0	0	0	0	0.5	0.25	3.75	46.9	Poor
(Nguyen et al., 2018)	1	1	0.75	1	0	0.5	0.5	0	0.5	1	6.25	78.1	Robust
(Oliver et al., 2015)	1	1	1	1	0.5	0.5	0.5	1	0.5	1	8	100.0	Robust
(Oluyomi et al., 2014)	1	1	1	0.25	0	0	0.25	0	0.5	0.5	4.5	56.3	Moderate
(Page et al., 2010)	1	1	0.75	0.25	0	0	0	0	0.5	0.75	4.25	53.1	Moderate
(Roberts et al., 2016)	1	1	0.75	0.25	0	0	0.25	0	0.5	0.75	4.5	56.3	Moderate
(Stephanie H. Kneeshaw-Price et al., 2015)	1	1	0.5	0.5	0	0.5	0	0	0.5	0.25	4.25	53.1	Moderate
(Santos et al., 2013)	1	1	0.75	0.25	0	0	0	0	0.5	0.5	4	50.0	Moderate
(Shokoohi et al., 2012)	1	0.5	0.5	0.25	0	0	0.25	0	0.25	0	2.75	34.4	Poor
(Stark et al., 2018)	1	1	0.75	0.25	0	0	0	0	0.5	0.5	4	50.0	Moderate
(Stephanie H. Kneeshaw-Price et al., 2015)	1	1	0.5	0.5	0	0.5	0	0	0.5	0.25	4.25	53.1	Moderate

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(Suminski et al., 2018)	1	1	1	N/A	N/A	0.5	0.5	1	0.5	1	6.5	108.3	Robust
(Timperio et al., 2004)	1	1	0.75	0.25	0	0	0	0	0.5	0.75	4.25	53.1	Moderate
(Tung et al., 2016)	1	1	0.5	0	0	0	0	0	0.5	1	4	50.0	Moderate
(van den Berg et al., 2020)	1	0.75	1	0.25	0	0	0.25	0	0.5	0.5	4.25	53.1	Moderate
(Villanueva et al., 2012)	1	1	0.75	1	0	0	0.5	0	0.5	0.75	5.5	68.8	Moderate
(Vonderwalde, Cox, Williams, Borghese, & Ian Janssena, 2019)	1	1	1	1	0	0.5	0.5	0.5	0.5	1	7	87.5	Robust
(Waygood & Susilo, 2015)	1	0.75	0.5	0	0	0	0	0	0.5	0.5	3.25	40.6	Poor
(Zhu & Lee, 2008)	1	N/A	N/A	0	0	0.5	0.5	0	0.5	0.5	3	50.0	Moderate

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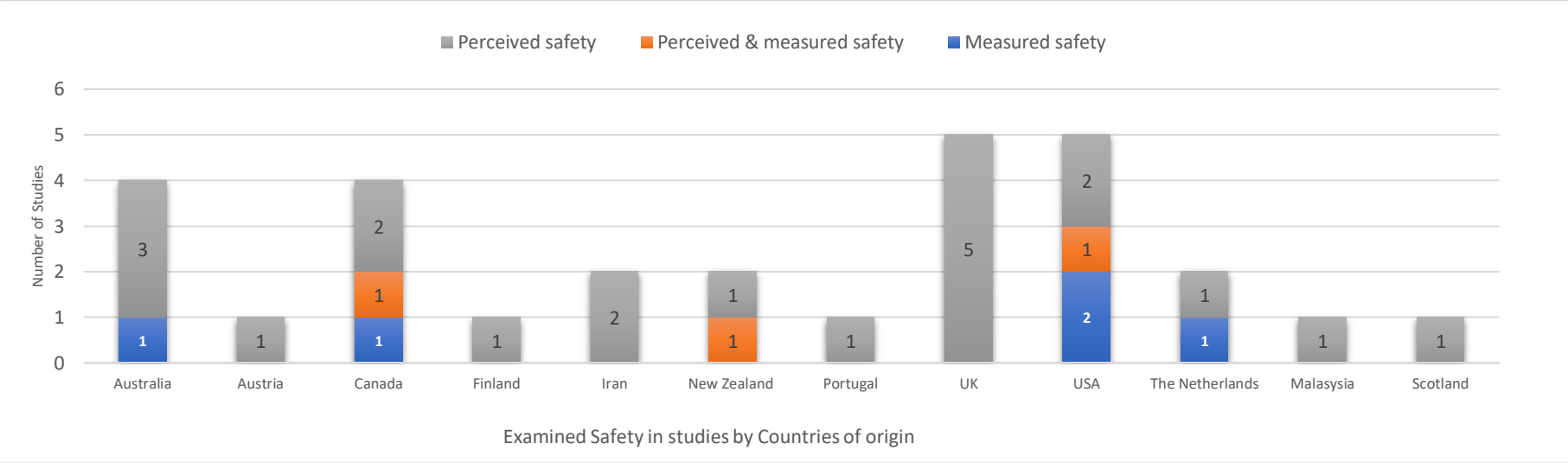


Diagram 1. Country of origin of the studies included in this review including examined neighbourhood safety types.

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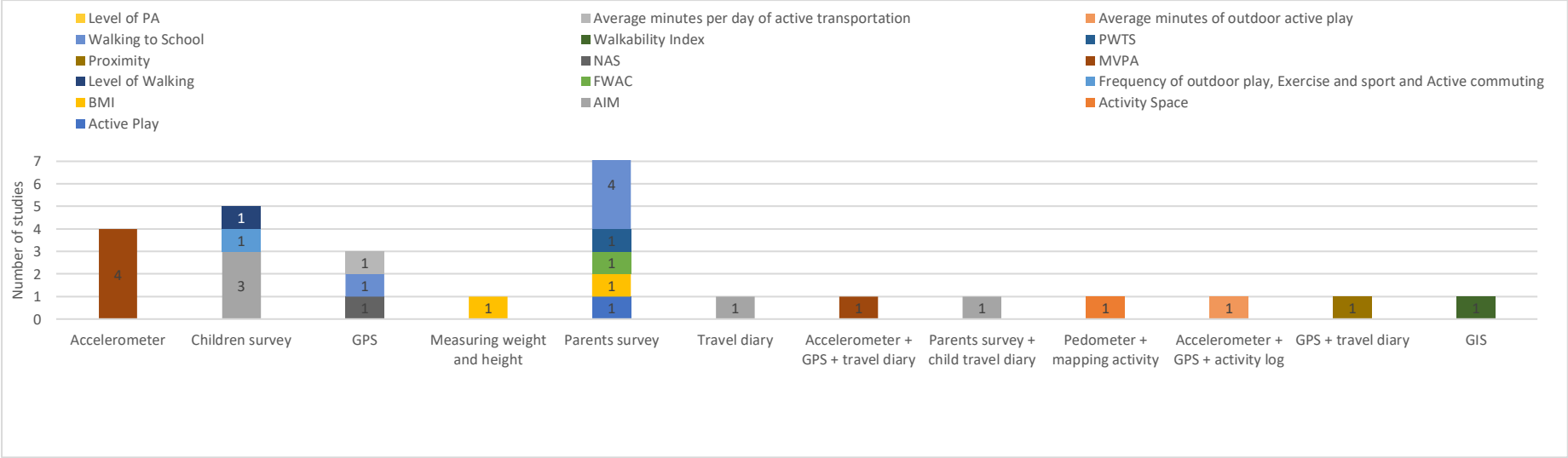


Diagram 2. Variances in measurement methods of children active mobility behaviour and the outcome of measures in 29 studies. Abbreviations: PA = physical activity, AIM = active independent mobility, BMI = body mass index, FWAC = frequent walking and cycling, MVPA = moderate-to-vigorous physical activity, NAS = neighbourhood activity space, PWTS = perceived walking to school, GPS = global positioning system, GIS = geographic information system

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Diagram 3, 4 and 5 illustrated evidence of modified child active behaviour synthesised by individual and family, and neighbourhood safety.

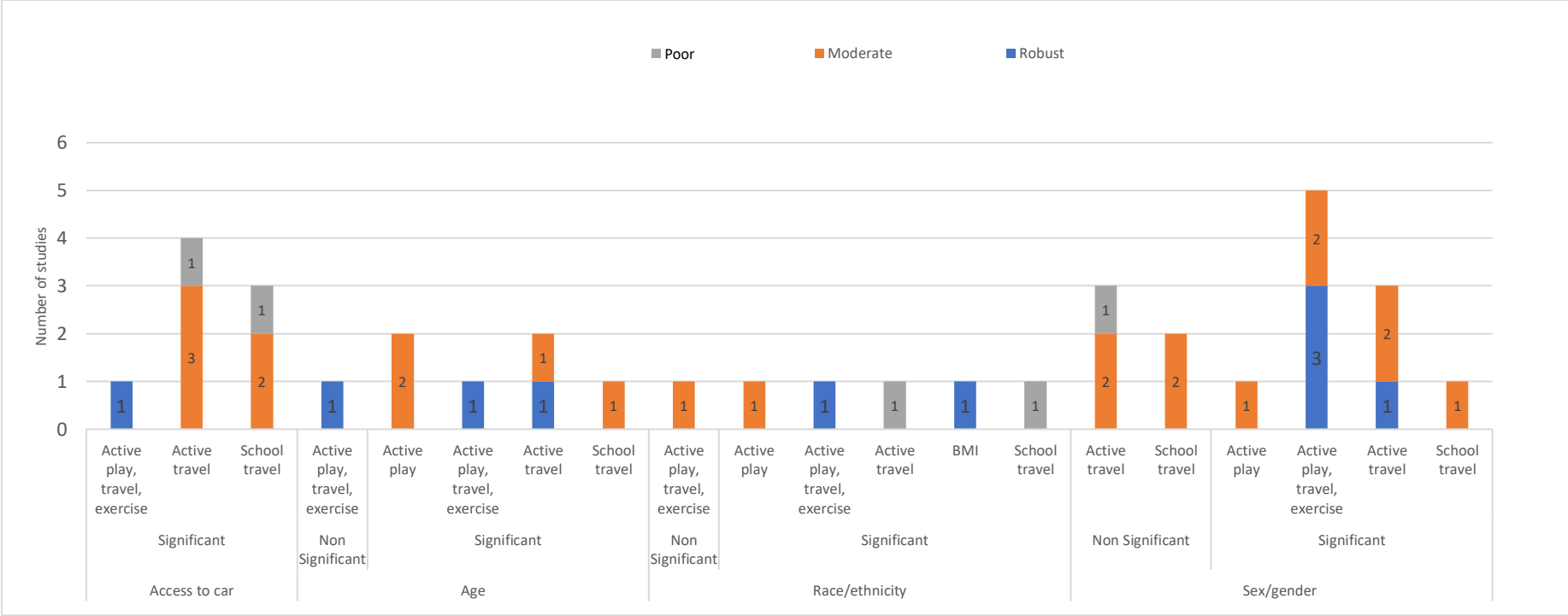


Diagram 3. Findings of Individual and family layers of association to children various type of active behaviour in studies examined neighbourhood safety. Studies were re-grouped each by the examined active behaviour into four categories of (Active travel, active play, school travel and active play, travel, and exercise) and as appraised by the methodological quality assessment. x-Axis depicted variable addressed at the individual and family level and the examined type of child active behaviour. y-Axis represents the number of studies accumulated.

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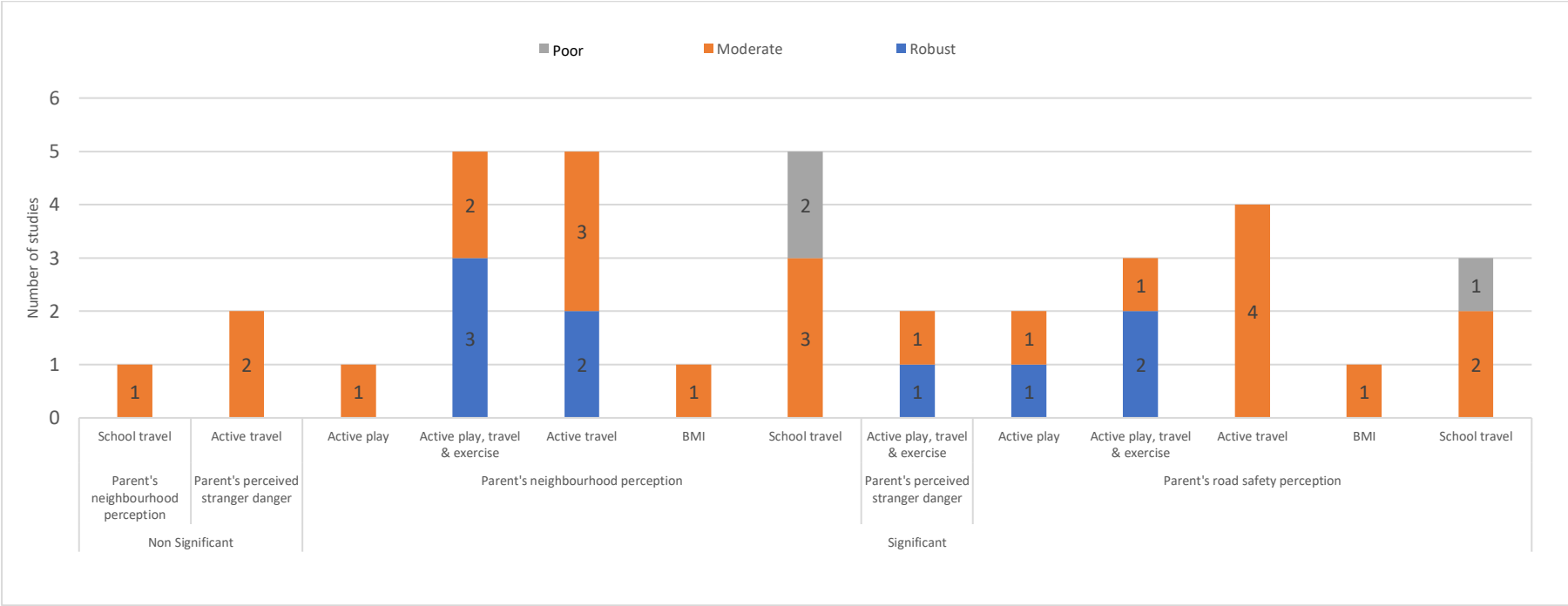


Diagram 4. Findings of parents perceived safety layers of association to children various type of active behaviour. Studies were re-grouped by active behaviour into four categories of (Active travel, active play, school travel and active play, travel, and exercise) and as appraised by the methodological quality assessment. x-Axis depicted variable addressed at the neighbourhood perceived level and the examined type of child active behaviour. y-Axis represents the number of studies.

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Table 3. Studies re-grouped output measure of active behaviour to enable synthesis of evidence.

#	citation	Neighbourhood Safety	Active Type	Re-Grouped Active Mobility Behaviour
1	(Alton et al., 2007)	Perceived	Level of Walking	Active travel
2	(Carver et al., 2010)	Perceived	MVPA	Active play, travel & exercise
3	(Carver et al., 2014)	Perceived	AIM	School travel
4	(Carver et al., 2008)	Perceived & measured	MVPA	Active play, travel & exercise
5	(Davis & Jones, 1996)	Perceived	AIM	Active travel
6	(Fagerholm & Broberg, 2011)	Perceived	Proximity	Active travel
7	(Faulkner et al., 2015)	Perceived	MVPA	Active play, travel & exercise
8	(Helbich et al., 2016; Oliver et al., 2015)	Perceived	MVPA	Active play, travel & exercise
9	(Janet E Loebach & Jason A Gilliland, 2016)	Perceived	NAS	Active travel
10	(Lin et al., 2017)	Perceived	AIM	Active travel
11	(Mehdizadeh et al., 2017)	Perceived	PWTS	School travel
12	(Noonan et al., 2016)	Perceived	BMI	BMI
13	(Nguyen et al., 2018)	Perceived & measured	Average minutes of active outdoor play	Active play
14	(Oliver et al., 2015)	Perceived & measured	MVPA	Active play, travel & exercise
15	(Oluyomi et al., 2014)	Perceived	Walking to School	School travel
16	(Page, Cooper, Griew, Davis, & Hillsdon, 2009)	Perceived	Frequency of outdoor play, Exercise and sport and Active commuting	Active play, travel & exercise
17	(Stephanie H. Kneeshaw-Price et al., 2015)	Perceived & measured	MVPA	Active play, travel & exercise
18	(Roberts et al., 2016)	Perceived	Active Play	Active play
19	(Stark et al., 2018)	Perceived	AIM	Active travel
20	(Suminski et al., 2018)	Measured	BMI	BMI
21	(Shokoohi et al., 2012)	Perceived	Walking to school	School travel
22	(Santos et al., 2013)	Perceived	AIM	Active travel
23	(Timperio et al., 2004)	Perceived	FWAC	Active travel
24	(Tung et al., 2016)	Perceived	MVPA	Active play, travel & exercise
25	(van den Berg et al., 2020)	Perceived	Walking to School	School travel
26	(Villanueva et al., 2012)	Perceived	Activity Space	Active play, travel & exercise
27	(Vonderwalde, Cox, Williams, Borghese, & Ian Janssena, 2019)	Measured	Average minutes per day of active transportation	Active travel
28	(Waygood & Susilo, 2015)	Perceived	Walking to school	School travel
29	(Zhu & Lee, 2008)	Measured	Walkability Index	Active travel

Notes: we categorised each study as per the active behaviour examined in each study: If a study addressed total MVPA it is grouped under "Active travel, play and exercise". If walking to school, the study grouped under "Active school travel". If the addressed active behaviour is the frequency of walking or cycling or independent mobility travel (IM), the study categorised it as "active travel", and when addressed active playing behaviour, it is "active play". Assessed impact on BMI was left as "BMI". Then we grouped studies to synthesise evidence of influences on modifying types of COAMB.

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Table 4. Findings of correlates across the socio-ecological levels and modified child types of outdoor active mobility behaviour.

Effective domains		**+ - 0**	**Correlation with children active mobility**
Individual (child)			
P_Survey Sex: Female	Carver et al. (2010)	+	avoidance behaviour[1]
	(Oliver et al., 2015)	–	%MVPA (F) weekdays
	(Timperio et al., 2004)	–	walking/cycling, after dark (F) of (10-12 YOLD)
		0	5- 6-year-old age group
	(Faulkner et al., 2015)	–	time outdoor (weekends and weekdays)
	(Villanueva et al., 2012)	–	activity Space (F)
	(Carver et al., 2014) greater land use mix (residence/retail outlets, sports facility) at T2	+	walking and cycling to school (F)
Sex: Male	Carver et al. (2010)	+	made more active trips (M)
	(Roberts et al., 2016)	+	active behaviour
	(Fagerholm & Broberg, 2011)	+	Distance travelled, and speed children travel
	(Villanueva et al., 2012)	+	walking/cycling, after dark (M) of (10-12 YOLD) Steps count
	(Mehdizadeh et al., 2017)	0	Perceived walking to school
	(Helbich et al., 2016)	0	Active school travel
	(Carver et al., 2014) being allowed by parents to play outside	+	odd of walking/cycling independently to school
	(Tung et al., 2016)	+	PA (reported)
C_Survey Gender	(Alton et al., 2007)	0	walking trips
	(Page et al., 2010) playing and taking part in structured sport	+	local-IM: (M) more than F
	(Page et al., 2010)	0	Active commuting to school
Age Adult Survey	(J. Loebach & J. Gilliland, 2016)	+	time spend outdoor, distance & IM (older).
	(Stark et al., 2018)	+	IM with older children
	(Timperio et al., 2004) of age 5-6 compared to 10-12	+	walking or cycling with age in particular for (M).
	(Oliver et al., 2016)	+	%MVPA weekdays
	(Faulkner et al., 2015)	0	MVPA
	(Helbich et al., 2016)	+	ATS
	(van den Berg et al., 2020)	+	active school travel
Parents/Family background characteristics			
SES	(Timperio et al., 2004) (boys) 10 -12	+	Frequency of walking or cycling to public transport (active trips) in high SES more than in medium SES.
	(Timperio et al., 2004) 10- 12 F	+	walked and cycled more to school in high SES.
	(Timperio et al., 2004) age 5-6 years old) M	+	in high SES
	(Timperio et al., 2004) age 5-6 years old) F	0	association to SES
	(Tung et al., 2016) high SES	+	PA
	(Timperio et al., 2004)	0	concerns of stranger danger (F)
	(Mehdizadeh et al., 2017) mothers with driving licence		Reduced active travel to school
	(Oliver et al., 2016)	–	(Inconclusive)
	(Noonan et al., 2016) in high deprivation	+	BMI level
		+	self-reported PA
C_Survey And SES	(Timperio et al., 2004) concerns of traffic in (M)	x	found in children in low SES more than children in high SES
Poverty	(Zhu & Lee, 2008)	–	safety & walking
Household income P_survey	Mehdizadeh et al., 2017	+	perceived walking to school
	(van den Berg et al., 2020)	x	increased income & increased perception of safety
Mother driving licence	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	–	active travel
Parental age	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	+	perceived walking to school
Employments Status (mother)	(Stark et al., 2018)	+	IM licence
Home environment (access to Media)	(Noonan et al., 2016)	x	in High Deprivation
Having older siblings	(Lin et al., 2017)	+	IM
	(Carver et al., 2014)	+	IM
P_Survey	(Alton et al., 2007)	–	level of walking

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Cars ownership/ access	(Davis & Jones, 1996)	–	IM
	(Lin et al., 2017)	–	IM
	(Mehdizadeh et al., 2017)	–	active travel
	(Oliver et al., 2015)	–	# of active trips (weekdays)
	(Timperio et al., 2004) Parent of 5-6 years old (F) who owned more than one car associated with	–	less likely of walking or cycling to destinations
	(Waygood & Susilo, 2015)	–	Walking to school
	(Carver et al., 2014)	–	odd of boys walking and cycling to school at T2
C_Survey rooms in the house	(Alton et al., 2007)	–	(significant only for 2.1%)
Parental attitude towards walking	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	+	short perceived walking time to school
Parental Physical Activity	(Santos et al., 2013) Parental physical activity	+	independent mobility
Ethnicity	(Alton et al., 2007)	–	level of walking in minority
	*(Oliver et al., 2016)	+	%MVA Pacific, European and Maori accumulates (weekdays only). %MVA Indian/Asian/others (weekend days) than their counterparts/Pacific on weekdays
	(Suminski et al., 2018) crime risk index	–	BMI (white children)
	(Roberts et al., 2016)	+	in minority
	(Zhu & Lee, 2008) (Hispanic)	+	in crime & traffics
Access to public transport	Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	–	active travel
Attending Public School	(Mehdizadeh, Mamdoohi, & Nordfjaern, 2017)	–	perceived walking to school
Neighbourhood, Perceived (Personal and Road Safety)			
P_Survey low personal safety perception	(Fagerholm & Broberg, 2011)	–	Independent mobility after dark and by distance (only F)
	(Stark et al., 2018)	–	IM
	(Timperio et al., 2004)	–	frequency of walking
	(Mehdizadeh et al., 2017)	–	activity space after 10 min walking
	(J. Loebach & J. Gilliland, 2016)	–	time outdoor and travelled distance
	(Faulkner et al., 2015)	–	outdoor playtime (outdoor plat and MVPA)
	(Roberts et al., 2016)	+	active children
	(Villanueva et al., 2012)	–	Activity space F
	(Oliver et al., 2016)	–	active trips (weekdays)
	(Noonan et al., 2016)	–	outdoor play in MD
	(Carver et al., 2010)	–	active transport (M/F) in 10 -11 years old
	Constrained behaviour (avoidance and Defensive behaviours)	–	MVPA (M) weekends
	(Lin et al., 2017)	0	IM (though parents had concern of safety)
	(Oluyomi et al., 2014)	–	walking to school general neighbourhoods' safety, stray or dangerous animals and availability of adults
	(Carver et al., 2014)	–	walking and cycling to school at T1
	(Shokoohi et al., 2012)	–	walking to school
	(van den Berg et al., 2020)	–	children travel actively
	(Waygood & Susilo, 2015)	–	School active travel
	(Tung et al., 2016)	–	PA
C_Survey Low perceived safety	(Villanueva et al., 2012)	-	activity space
	(Janet E. Loebach & Jason A. Gilliland, 2016)	-	time spend close to home
	(Shokoohi et al., 2012)	–	walking to school
	(Page et al., 2010) positive perception	-	playing out for girls
	(Page et al., 2010) positive perception of the environment	-	greater IM
	(Page et al., 2010) exercising or doing sport every day associated	-	local-IM (boys)
P_Survey Perceived Stranger danger	(Faulkner et al., 2015)	+	on weekdays
	(Lin et al., 2017)	0	though parents have safety concerns
	(Stephanie H. Kneeshaw-Price et al., 2015)	–	MVPA
	(Santos et al., 2013)	0	IM
C-Survey Perceived Stranger danger	(Alton et al., 2007)	–	less walking
	(Davis & Jones, 1996)	–	in particular (F)

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	(Timperio et al., 2004)	–	low SES (F) than high SES
P_Survey Increased social Cohesion/Social Norms	(Lin et al., 2017)	+	IM
	(van den Berg et al., 2020) increase perception of social cohesion	+	parents perceived safety
C-Survey Neighbourhood high aesthetics / Nuisance Social Norms	(Noonan et al., 2016) Neighbourhood aesthetics	0	self-reported PA
	(Noonan et al., 2016) Neighbourhood aesthetics	–	BMI
	(Page et al., 2010) higher score of social norms	+	frequency of outdoor play
P_Survey Perception of Road Safety Low perceived road safety	(Noonan et al., 2016)	–	independent mobility
	(Stark et al., 2018)	–	independent mobility (general traffic safety)
	(Timperio et al., 2004) Parents believe of (heavy traffic) in (5-6 years old) boys	+	Frequency of walking (sidewalks or bike lanes, safe crossing
	(Timperio et al., 2004) Parents believe (for M) of 10 – 12 years old of no lights or crossing.	–	walking and cycling
	(Timperio et al., 2004) The parental belief of F needs to cross many roads to reach play area 10 - 12	–	likelihood of walking or cycling
	(Villanueva et al., 2012) safe neighbourhood crossing	–	activity space in particular girls,
	(Faulkner et al., 2015) Fast drivers (weekdays),	–	MVPA (active play)
	(Roberts et al., 2016) if parents perceived a lack of sidewalk and signals on a busy street.	–	active children
	(Santos et al., 2013) perceived sidewalk and street safety	–	independent mobility
	(Oluyomi et al., 2014) road safety in the home environment of higher sidewalk availability, Speed, amount of traffic, intersection safety, road crossing problem and availability of crossing guard.	–	Likelihood of walking to school
	(Carver et al., 2014) concern about traffic.	-	likely for boys and girls to cycle independently
	(Tung et al., 2016) traffic hazards (perceived_)	–	PA
	(Waygood & Susilo, 2015) slow or safe traffic	+	Walking to school
	(Davis & Jones, 1996) traffic	–	Independent mobility
	(Nguyen et al., 2018) parents' perception of high or moderate traffic speed higher	+	outdoor active play
Perceived increased street connectivity	(van den Berg et al., 2020) increased connectivity	+	parental perception of safety
Destination accessibility	(Oliver et al., 2016)	+ –	the proportion of trips in active mode (weekdays) %MVPA (weekend)
Improved streetscape (measured)	(Villanueva et al., 2012)	+	Activity space
C_Survey Road safety	(Alton et al., 2007)	–	road safety and heavy traffic and level of walking
	(Villanueva et al., 2012)	–	activity space
	(Davis & Jones, 1996)	–	IM in particular F (traffic danger)
	(Timperio et al., 2004)	–	low SES (M) than in high SES
	(Page et al., 2010)	+	local and Area-IM with traffic safety (for girls) positive perception of road safety (+ IM) for girls
Neighbourhood's Safety –Perception of other elements related to safety in the Physical environment			
P_Survey Perceived **lack of leisure facility**, Parks and sport ground	(Timperio et al., 2004)	–	likely to walking &cycling three times a week (10-12) F
	(Alton et al., 2007)	–	with a crossing of no light or crossings
	(Faulkner et al., 2015)	0	duration of playout
Land use diversity: Residential/ Commercial/ Industrial/	(Fagerholm & Broberg, 2011)	+	mobility mainly in urban structure (residential, commercial and traffic areas)
	(Villanueva et al., 2012)	–	AS in utilitarian destination (within 800 m)
	(Janet E. Loebach & Jason A. Gilliland, 2016)	+	Time spend close to home (residential) and (+) distance travelled and time spends in commercial (beyond 800m).
		–	mobility in agricultural and industrial (on 400-800m buffer)
	(Faulkner et al., 2015)	+	time playing outside (Residential &Commercial)

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	*(Oliver et al., 2016)	0	%MVPA
	(Helbich et al., 2016) urban environment and well-connected street and lights	+	Active school travel
	(Carver et al., 2014) (land use mix of residential, retail shops, sports centre)	+	walking/cycling independently to school (F)
Land use mix (access)	(Tung et al., 2016)	+	PA
P_Survey proximity to destinations	(Stark et al., 2018)	–	AIM
	*(Fagerholm & Broberg, 2011)	–	trajectory (with increased distance),
	*(Oliver et al., 2016) distance to school	–	the proportion of trips made in active mode, + distance to school
	(Lin et al., 2017)	–	AIM
	(Mehdizadeh et al., 2017) perception of more than 10 min walking	–	perceived walking to school distance and active travel
	(Roberts et al., 2016)	+	to play area, active behaviour
	(Janet E. Loebach & Jason A. Gilliland, 2016)	–	majority of time spent closer to home (400 m buffer around the home)
		–	school travel and neighbourhood travel
	(Helbich et al., 2016)	–	AST
	(Waygood & Susilo, 2015)	–	Walking to school
	(van den Berg et al., 2020)	–	school active travel
C_Survey proximity	(Janet E. Loebach & Jason A. Gilliland, 2016)	+	distance travelled and activity space near home
	(Page et al., 2010)	–	longer route and active commuting to school (for both boys and girls Active commuting
P_Survey perception of limited access to public transportation	(Timperio et al., 2004)	– –	likely of walking or cycling F of 5-6-year-old. likely of walking or cycling F of 10-12-year-old.
P_Survey Availability of Neighbourhood Amenities	(J. Loebach & J. Gilliland, 2016)	+	neighbourhood activity space
P_Survey High walkable neighbourhood	(Villanueva et al., 2012)	+	activity space
P_Survey High Neighbourhood aesthetics	(Roberts et al., 2016)	+	active children
	(Noonan et al., 2016)	–	BMI z-score and waist circumference
	Mehdizadeh et al., 2017	+	Perceived walking to school
P_Survey Owning a dog	(Timperio et al., 2004)	+	frequency of walking
C-Survey Accessibility to destination	(Timperio et al., 2004) child believe of no parks access to parks (F/M)	–	walking and cycling in 10-12 years old (M)
	(Page et al., 2010) easy access to range pf destination (Page et al., 2010) greater perceived accessibility	+ +	taking part in structured exercise/sport every day Active commuting to school
Neighbourhood– Measured safety			
Measured safety from crime (high level of actual crime)	(Stephanie H. Kneeshaw-Price et al., 2015)	–	MVPA
	(Vonderwalde, Cox, Williams, Borghese, & Janssen, 2019)	+	active transportation in High crime areas
	(Zhu & Lee, 2008)	–	walkability
	(Suminski et al., 2018)	+	BMI z-score
Road safety	(Nguyen et al., 2018) children from the highest traffic volume	+	outdoor active play
	(Carver et al., 2014) the proportion of main roads	–	odd of walking and cycling (F) walking and cycling to school independently
	*(Oliver et al., 2015) ratio of high-speed roads around schools (weekdays)	–	%MVPA
	street connectivity	+	the proportion of trips made in active mode (on weekend and weekdays)
	(Helbich et al., 2016) exposures to major roads/highway Availability of cyclist path	– +	AST AST
	(Carver et al., 2008)* measured road environment	0	Likelihood of making at least 7 walking and cycling active trips per week

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Perceived and measured road safety	(Nguyen et al., 2018) perceived road safety	x	measured road safety
Perceived and measured personal safety	(Stephanie H. Kneeshaw-Price et al., 2015) Measured crime	0	with parents' perception
Parents to children perceived safety	(Timperio et al., 2004)		parent of 10 – 12 perception was more negative than their own children

Note: Association identified between safety and active mobility behaviour: (+) positive association, (-) negative association, (0) Non-Associated/Not Significant difference, x = association not to active behaviour. Abbreviation: P_Survey = studies examined the perception of safety among parents; C_Survey = studies examined the perception of safety among children. M = male (boys), F = female (girls), IM = independent mobility, MBI – body mass index, SES = socioeconomic status, HD = high depreciation, MD = Medium Deprived, CRI = Crime Risk Index, BMI Body Mass Index, SES = Socioeconomic level, * = behaviour show significant correlate on Weekdays, ** = Behaviour show on weekend, MVPA = Moderate to vigorous physical activity. [1] "Avoidance behaviour is where no further engagement in habits and activities due to perceived risk, e.g. parents driving children to school instead of walking or cycling., [2] Defensive behaviour is where habits are altered in an attempt to reduce perceived risk, e.g. parental accompaniment to children while walking to school", Ferraro, 1995.

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